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Extensions to
Hierarchical Bradley–Terry Models
with Dynamic Abilities

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Abstract

Many professional sporting competitions exhibit a home-ground advantage. When scheduling constraints stipulate an imbalanced schedule — such as team divisions or uneven travel requirements — a natural question arises regarding impacts on the competitors’ home effects. We discuss extensions to Hierarchical Bradley–Terry models, a novel approach to quantifying these complex order effects in head-to-head competitions. The history and developments of paired comparison models are surveyed, with coverage of existing implementations and their limitations. A revised modelling framework is then proposed to allow for generalised, user-defined home advantage structures between the competitors; furthermore, we consider a simple dynamic extension to account for the fluctuation of team strengths over time. Key implementation details are also outlined. Applications feature twenty seasons of match data from the Australian Football League (AFL) and National Rugby League (NRL), two competitions which have received relatively little attention in the paired comparison model literature. For the AFL, significant differences were found when grouping teams based on their home state, with evidence to suggest home effects are team-specific. Conversely, the NRL data is best explained assuming home advantage is instead common to all teams. Various additional case studies are included to demonstrate the flexibility of the framework.

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Chapter 1: Introduction

In this chapter, we give a short introduction to the field of sports analytics and why paired comparison models provide an enticing methodical framework for such. The focus will be on the history and developments of the Bradley–Terry model, particularly for dynamic extensions, though other prevalent models are included. We examine key theoretical and computational advancements in the literature, as well as survey some noteworthy applications. Following this, existing implementations of the models are reviewed in 1.3 and previous work is summarised then critiqued in 1.4 and 1.5. Finally, the aims and structure of this thesis are outlined in 1.6.

1.1 Sports Analytics: A Brief Overview

Sport is a worldwide phenomenon. With a global market size forecasted to exceed \$600 billion USD by 2030 (Kearney, 2025), sports culture has undeniable socioeconomic importance. The advent of “Big Data”¹ has brought about possibilities to improve performance and decision-making in the world of sports. As such, **sports analytics** has developed into a “fully-fledged field of scientific inquiry ... within sports science and data science” (Glickman & Jones, 2025). It is comprehensively defined by Morgulev et al. (2018) as

the management of structured historical data, the application of predictive analytic models that use these data, and the utilization of information systems, in order to inform decision makers and enable them to assist their organizations in gaining a competitive advantage on the field of play.

Baseball is commonly associated with the origin of formal statistical analyses in sport, where the literature began developing in the late 20th century alongside computational capabilities.² The popularisation of the term “sports analytics” itself is typically attributed to *Moneyball* (2011). The film³ follows the true story of Oakland Athletics (the A’s) — a Californian MLB

¹A buzzword typically used to describe the collection of very large datasets and their application to analytical problems. It is usually used in the context of the increasingly large volume of raw, unstructured data endlessly collected by various devices, such as wireless sensor networks and cameras. See Lohr (2013) on the etymological origins of the term, which has largely been superseded by “AI” jargon.

²“Sabermetrics” was the term coined to describe baseball-specific analytics, so named for the Society for American Baseball Research (SABR) founded in 1971.

³based on the 2003 book *Moneyball: The Art of Winning an Unfair Game* by Michael Lewis

team — under general manager Billy Beane, who used player statistics to field a competitive baseball team with a tight payroll. Despite not winning the World Series, in 2002 the team famously won twenty consecutive games. The unprecedented success of Beane’s data-driven strategy prompted a paradigm shift in the usage of statistics in sports. This is what Szymanski (2020) calls “the Moneyball moment” — “the acceptance that data analytics could make a difference to team performance in baseball.” Mainstream adoption followed. As one example, the Boston Red Sox hired Theo Epstein, who used similar data-based strategies to end an 86-year long championship drought⁴ in 2004. Most major sports leagues now have dedicated analytics departments, where statisticians work alongside scouts. Morgulev et al. (2018) and Watanabe et al. (2021) highlight the diversity of sports analytics applications beyond improving team performances, including on-field player behaviour, business management, economics, and social science.

The modern landscape of sports analytics is not without its challenges. Recent advancements in data collection and statistical modelling — particularly ML/AI — have undoubtedly bolstered the field. The era of “Big Data” has allowed for increasingly accessible, granular, and voluminous datasets. However, quantity alone does not necessarily lead to better insights; “today we are rich in data but ... poor in theory” (Szymanski, 2020). In reality, the analyst is faced with great technical efforts: endogeneity problems, the integration of quantitative and qualitative sources, the “monumental task ... to move from a culture of data silos to a centralized system” (Morgulev et al., 2018). Further, Szymanski (2020) critiques the erosion of scientific principles in modern sports analytics, likening it to alchemy due to increasing “secretiveness and obscurantism.” For the area to progress amicably, research must reconcile with the ruthless pursuit of competitive advantage. This motivates a pivot toward models on the fundamental structure of competition, as opposed to complexity for its own sake.

A key interest to us is the subdomain of methods for estimating team and player strengths to predict game outcomes. This is logical: to predict outcomes well, we need to model and make inferences on team abilities. While the literature offers a wide variety of modelling options — including predicting point scores in both high-scoring (NFL) and low-scoring (soccer) sports, analysing player geospatial data, and using rank models for multiplayer sports — we wish focus on head-to-head competitions in this thesis. These **paired comparison models** ask only about the binary outcome of matches, the “win-or-lose”. Despite seeming reductive in comparison to the aforementioned models, such models tend to be more robust and less susceptible to bias because of their simpler specification. Notably, Szymanski (2020) substan-

⁴The drought was nicknamed the “Curse of the Bambino” after Babe Ruth, widely considered the greatest baseball player of all time, who departed the Red Sox in 1920.

tiates concerns with overly complex models by showing similar predictive performance from bookmakers, SOTA models⁵, and simple logit models. Furthermore, paired comparison models are practically appealing for their clear applications to betting markets and historical analyses. Thus, these models offer an alluring approach to analysing competitor strengths.

1.2 A Review of Paired Comparison Models

There is a sizeable literature on the analysis of paired comparisons. We will hence introduce the fundamental model formulations and results, then comprehensively examine order effects and dynamic extensions due to their relevance in this thesis. Following this, we peruse a variety of related advanced topics. There are several overviews of paired comparison models available; for the interested reader, we suggest any of (Agresti, 2013, Ch. 11), (Cattelan, 2012), (Hamilton et al., 2025), (Glickman & Jones, 2025), and (Fang et al., 2026). More antiquated — but nevertheless instructive — summaries are given by (Bradley, 1984) and (David, 1988).

The foundational framework for analysing head-to-head comparisons is the **linear paired comparison model** (Cattelan, 2012; David, 1988). Consider a set of I items⁶ in head-to-head competition with one another and denote θ_i as the *strength*⁷ of competitor $i = 1, \dots, I$. The model hence describes $\binom{I}{2}$ probabilities with I parameters which can be used to determine an overall ranking of the items. Let $\{i \succ j\}$ denote the event that i beats j , and define the binary outcome of a match by

$$Y_{ij} = \mathbb{I}(i \succ j) = \begin{cases} 1 & i \text{ beats } j \\ 0 & j \text{ beats } i \end{cases} \quad (1.1)$$

The linear paired comparison model assumes that

$$\pi_{ij} := \mathbb{P}(Y_{ij} = 1) = F(\theta_i - \theta_j) \quad (1.2)$$

where $F : \mathbb{R} \rightarrow [0, 1]$ is a monotonically increasing function such that $F(-x) = 1 - F(x)$, i.e., a cumulative distribution function (CDF) of a zero-symmetric random variable.

1.2.1 Historical development

The most common choices for F are the CDFs of the logistic and normal distributions. The former describes the **Bradley–Terry (BT) model** (Bradley & Terry, 1952), the main model

⁵state-of-the-art models, namely, Nate Silver’s *FiveThirtyEight* (now defunct)

⁶read: competitors, teams, players, objects, etc.

⁷read: ability, merit, worth, etc.

of interest in this work, which has the form

$$F(\theta_i - \theta_j) = \text{expit}(\theta_i - \theta_j) = \frac{e^{\theta_i}}{e^{\theta_i} + e^{\theta_j}} \quad (1.3)$$

where $\text{expit}(x) = (1 + \exp(-x))^{-1}$ is the logistic (or *expit*) function.⁸ The latter choice corresponds to the **Thurstone–Mosteller (TM) model** (Mosteller, 1951) and has the form

$$F(\theta_i - \theta_j) = \Phi(\theta_i - \theta_j) \quad (1.4)$$

The models were popularised throughout the early 1950’s; however, the existence of the BT and TM models in the literature can be traced to the late 1920’s with Zermelo (1929) and Thurstone (1927), respectively. Theoretical developments continued steadily alongside increasingly broad applications in the following years (Bradley, 1984).

A crucial thing to note for these models is that — because they are defined as the function of a difference of parameters — the strengths $\boldsymbol{\theta}$ are *invariant* up to a constant. It is easy to see that $\boldsymbol{\theta}^* := \boldsymbol{\theta} + c\mathbf{1}$ leads to the same probability distribution for any $c \in \mathbb{R}$. There are a few ways to handle this. Typically, a linear constraint is imposed on $\boldsymbol{\theta}$, such as the sum constraint $\sum_{i=1}^n \theta_i = 0$ or by setting $\theta_k = 0$ for some chosen reference item k .

There are also some assumptions on the *connectivity* of the comparisons. This requires appealing to the language of graph theory, in particular by viewing the comparison data as a directed graph with non-negative edge weights. The main assumption is that of *irreducibility*: in any partition of the items into two (nonempty) sets, there must be at least one comparison of some pair across these sets. This is equivalent to having strong connectivity of the graph. Rather importantly, this implies the existence of a unique maximiser of the log-likelihood. See (Hunter, 2004) and Section 3.2 for more on these assumptions.⁹

Historically, the BT model has been defined on the probability scale rather than the log-odds scale. For instance, the original works of Zermelo (1929) and Bradley and Terry (1952) presented the model as a ratio of the ‘true ratings’ $\lambda_1, \dots, \lambda_I$,

$$\mathbb{P}(Y_{ij} = 1) = \frac{\lambda_i}{\lambda_i + \lambda_j}$$

Of course, the difference with (1.3) is simply the exponential transformation $\theta_i = \log \lambda_i$. The log-odds parameterisation has become favoured in the modern literature, perhaps for the intuitive, additive expression on the log-odds scale

$$\text{logit}(\pi_{ij}) = \log \frac{\pi_{ij}}{1 - \pi_{ij}} = \theta_i - \theta_j \quad (1.5)$$

⁸See A.1 for more on this function and its properties.

⁹crucially, on how the irreducibility assumption is specified when extensions like order effects are included

and so we will continue to use this.

This parameterisation also reveals an important connection between the linear paired comparison model and generalised linear models (GLMs) (McCullagh & Nelder, 1989). In particular, (1.2) can be viewed as a GLM, with the choice of logit and probit link functions corresponding to the BT and TM models, respectively. More formally, first recognise that each $Y_{ij} \sim \text{Ber}(\pi_{ij})$. Let n_{ij} be the number of sampled comparisons between i and j , where $n_{ij} = n_{ji}$, and w_{ij} be the number of times that i was preferred to j , with $n_{ij} = w_{ij} + w_{ji}$. Therefore, assuming that the comparisons are independent, we have $W_{ij} \sim \text{Bin}(n_{ij}, \pi_{ij})$. In this view, the BT model is thus also as a form of logistic regression.

1.2.2 Common extensions: ties and covariates

In many situations, a paired comparison may not result in a clear win or loss, but rather a **tie** or ‘non-selection’ judgement. For example, this may occur due to equal scores or evaluations, time constraints, stalemates, mutual forfeits, etc. In some sports competitions, a binary outcome is forced via rules for overtime or secondary win conditions; on the other hand, rules may be in place to prevent perpetual games by defaulting to a draw in some specific situation. Indeed, the original application of Zermelo (1929) was player rankings in chess tournaments, where draws are an exceedingly common outcome.¹⁰

Formal extensions address this by treating ties as an explicit outcome, typically written as $Y_{ij} \in \{-1, 0, 1\}$. Well-known treatments for the BT model are given by Rao and Kupper (1967) and Davidson (1970), where another parameter is introduced that governs the frequency of ties relative to the competitor strengths. A key property of these extensions is that the original model (1.3) can be recovered as a special case.¹¹ More recently, Glickman (2025b) proposed a model for player strength-dependent ties with applications to high-level chess.

Alternatively, a more common approach in practice is to ignore the trinary model and instead just allocate half of a win to the respective competitors. Turner and Firth (2012) describe this as a “crude but simple device,” nevertheless finding in some tests that the results are “very similar ... to the more sophisticated analyses that have appeared in the literature.” However, the justifiability of such an approach depends heavily on the context. In those scenarios with demonstrably higher probabilities of ties, the formal approaches are necessary to avoid unreliable strength estimates (Glickman, 2025b). Having a principled model to explicitly

¹⁰about half of top-level games are drawn (Zhou, 2018), with some data indicating up to 95% in certain formats (Glickman, 2025b)

¹¹The (Davidson, 1970) model furthermore preserves Luce’s choice axiom or the “independence of irrelevant alternatives” (IIA).

estimate the probability of a draw is also desirable when analysing such competitions.

Another useful generalisation is to account for explanatory variables or **covariates**. Originally proposed by Springall (1973), the latent strengths are expressed as a linear combination of competitor-specific features

$$\theta_i = \mathbf{x}_i^T \boldsymbol{\beta} \implies F(\theta_i - \theta_j) = F((\mathbf{x}_i - \mathbf{x}_j)^T \boldsymbol{\beta}) \quad (1.6)$$

where $\mathbf{x}_i$ is a vector of covariates intrinsic to competitor i 's ability and $\boldsymbol{\beta}$ the corresponding vector of coefficients. An example application is given by Dittrich et al. (1998), who use categorical subject-specific covariates to analyse university rankings. Alternatively, one could have the strength included alongside some covariates — such as $\theta_i - \theta_j + (\mathbf{x}_i - \mathbf{x}_j)^T \boldsymbol{\beta}$ — to examine how specific features affect baseline strengths (S. E. Stern, 2011). The latent strengths here could then be interpreted as competitor-specific intercepts β_{i0} .

Environmental factors, such as weather conditions or stadium attendance, are considered comparison-specific covariates. The model is much the same as the competitor-specific version, but these terms instead characterise factors for the comparison itself rather than its participants. These approaches for subject-specific, comparison-specific, and interaction-based covariates are discussed throughout the literature; see (Cattelan, 2012) and (Schauberger & Tutz, 2019) for more comprehensive reviews.

1.2.3 Order effects

An extension of particular interest to us is the incorporation of **order effects**, which captures how the presentation or use of one item before another can effect the outcome of the comparison. For example, this could account for biased judgements in an analysis of preferences, such as the first of two foods sampled by a subject; another example would be to note which player used the white pieces in a chess game. In the context of professional league sports, one interpretation is a **home-ground advantage** (HGA).¹²

The conventional approach is to add a single parameter to the model, such as

$$\mathbb{P}(Y_{ij} = 1) = F(\theta_i - \theta_j + \alpha) \quad (1.7)$$

where α quantifies the advantage enjoyed by the home team, in this case i . This can be viewed as a type of comparison-specific covariate as discussed before. For two evenly matched teams, i.e., $\theta_i = \theta_j$, the probability of the home team winning is $(1 + \exp(-\alpha))^{-1}$. This of course reduces back to a coin flip when there is no home advantage ($\alpha = 0$).

¹²The ‘home’ terminology really depends on the context. A home ground can be referred to as a home field, court, ice, venue, etc. A home team may alternatively be called the respective venue’s tenant or defender.

The approach was developed by Davidson and Beaver (1977) for the Bradley–Terry model, which also naturally combines with the two common tie extensions mentioned before. It seems that this paper actually describes within-pair order effects α_{ij} for each (i, j) with the constraint $\alpha_{ij} = \alpha_{ji}$. The model hence assumes that each unique pair of competitors has a specific addition to the log-odds, but the constraint implies that the change is bidirectionally equal. The monograph by David (1988) combines the extension for ties and order effects into one model, which recovers (Davidson & Beaver, 1977) as a special case when ignoring ties.

Applications are plentiful throughout the literature; most major head-to-head sports have at least some amount of home-ground investigation using paired comparison models. For example, Agresti (2013) provides an instructive analysis of American League (Eastern Division) baseball games with a common HGA. Koehler and Ridpath (1982) investigate NBA results and find evidence of a common home-court effect that is notably stronger than baseball. Test match cricket is analysed by Dewart and Gillard (2019) with a common HGA, as well as individual player ratings and prediction. The combination of both order effects and ties in chess are examined, for instance, by Glickman (2025a) using a novel model.

In Chapter 2, we outline our models for increasingly complex parameterisations of HGA, such as team-specific and hierarchical home-ground effects. Considerations are also made for neutral venue scenarios, which act as another comparison-specific covariate to nullify the HGA parameter when neither team has a particular home advantage. Section 1.5 summarises previous contributions to our work, which initially proposed these (seemingly novel) models.

1.2.4 Time-varying models

The consideration of **time variation** when modelling competitor strengths over multiple time blocks is certainly important. Indeed, “it is difficult to imagine sports settings in which game outcomes are collected over a span of time, but in which competitor strengths do not correspondingly evolve” (Glickman & Jones, 2025). The idea is to now index the strength of each competitor i at some time t as θ_{it} . Hence, write $Y_{ijt} = \mathbb{I}(i \succ j \text{ at time } t)$ such that

$$\mathbb{P}(Y_{ijt} = 1) = F(\theta_{it} - \theta_{jt}) \tag{1.8}$$

Approaches for this extension can be generally categorised as non-stochastic and stochastic.

Non-stochastic methods

The **non-stochastic** or **deterministic** approach assumes that the strength of competitor i is modelled by some smooth function f_i such that

$$\theta_{it} = f_i(t \mid \boldsymbol{\gamma}) \quad (1.9)$$

where $\boldsymbol{\gamma}$ is a common parameter vector.

Many choices of f_i have been explored in the literature. Perhaps the most obvious option is to use a polynomial or spline. For instance, Araki et al. (2019) use piecewise quadratic splines in the TM model to investigate age effects in sumo wrestling, with emphasis on predicting match results. Their model essentially describes two half parabolas conjoined at the turning point,

$$f_i(t \mid a_i, b_i, \delta_i, \mu_i) = \begin{cases} \mu_i - 2a_i\delta_i t + a_i t^2 & , \quad t \leq \delta_i \\ \mu_i + (b_i - a_i)\delta_i^2 - 2b_i\delta_i t + b_i t^2 & , \quad t > \delta_i \end{cases} \quad (1.10)$$

where $\boldsymbol{\gamma} = (\mathbf{a}^T, \mathbf{b}^T, \boldsymbol{\delta}^T, \boldsymbol{\mu}^T)^T$ are the parameters to be estimated subject to $a_i, b_i \leq 0$ and $\delta_i \geq 0$. Note that the time index t refers to the age of the wrestler and not the time block that the match occurred in.

McHale and Morton (2011) analyse tennis matches (ATP tour) using a local likelihood method where older observations are downweighted according to exponential decay. Specifically, if match k was played at time t_k , then the likelihood contribution of match k at time t is weighted by $\exp(\epsilon(t - t_k))$, where the ‘half-life’ parameter ϵ controls the rate of decay. Their model also includes a covariate for surface type, which has been known to influence player performance. Later papers (Baker & McHale, 2014, 2017) look at Grand Slam tournament data to determine the strongest tennis players, instead using barycentric rational interpolants (BRIs). In this model, the strength of i at time t is given by

$$f_i(t \mid \mathbf{w}_i, \boldsymbol{\lambda}_i) = \frac{\sum_{k=1}^{m_i} w_{ik} \lambda_{ik} / (t - t_{ik})}{\sum_{k=1}^{m_i} w_{ik} / (t - t_{ik})} \quad (1.11)$$

where m_i is the number of nodes, λ_{ik} is the strength (estimate) of i at t_{ik} , and the w_{ik} are weight parameters. This is typically simplified by taking a constant number of nodes $m_i = m$ for each competitor; if the nodes are also uniformly spaced in time then $t_{ik} = t_k$ and $w_{ik} = (-1)^k$. The problem of selecting m and knot locations is largely the same as in spline interpolation, however Baker and McHale (2014) show that BRIs have a simpler formulation and perform better.

BRIs are touched on again by Krese and Štrumbelj (2021), who additionally consider a Bayesian formulation — where the spline coefficients have priors — as well as Gaussian Process (GP) priors for the latent strengths. The latter two methods would fall under state-space models,

which are discussed shortly. They find that these Bayesian approaches often surpass MLE-based BRIs, particularly regarding robustness to overfitting and the flexibility of GPs to discontinuities. Indeed, Baker and McHale (2017) also modified their previous BRI model to use empirical Bayes for similar reasons. Fitting for the Bayesian models appears to be done with a variant of the Hamiltonian Monte Carlo algorithm. These claims are supported by analysis of ATP tennis and NBA games, including a common HGA for the latter.

Some non-parametric models have been proposed too. Bong et al. (2020) estimate strengths with kernel smoothing, a pre-processing approach to smooth the paired comparison data given some kernel function W_h with bandwidth h that controls smoothness over time. Given the dataset $\mathcal{D}$, this may be written

$$f_i(t \mid h, \mathcal{D}) = \sum_{k \in \mathcal{D}} W_h(t, t_k) Y_{ik} \quad (1.12)$$

This is effectively a type of weighted moving average. The authors apply this to NFL data using a standard Gaussian kernel and LOOCV¹³ to tune the bandwidth parameter h .

Stochastic methods

A natural suggestion is to define a **stochastic process** on the strength parameters, also referred to as **state-space modelling**. In this case, a Markovian model typically governs the evolution of the parameters from one time block to the next (Glickman, 2001). For example, consider a first-order autoregressive (AR-1) model

$$\theta_{i,t+1} = \rho \theta_{it} + \varepsilon_{i,t+1} \quad (1.13)$$

where $|\rho| < 1$ and $\varepsilon_{i,t+1} \sim N(0, \sigma^2)$ represents the innovation with unknown variance σ^2 . Enforcing $\rho = 1$ simply gives a random walk. The main appeal of a stochastic model is that the strengths can experience random innovations over time (also called shocks or jumps), rather than some smooth trajectory in a deterministic model.

Glickman and Jones (2025) explain that the stochastic approach can be taken to two extremes: the recovery of constant strengths across time and the treatments of strengths as if there is no connection between them over time. Here we see the classic bias-variance trade-off in effect. The former is the original static model (1.2), where all data is used with no consideration for possible time variation. A constant model assuming no time dependence — e.g., $\sigma^2 = 0$ and $\rho = 1$ in (1.13) — will miss important dynamic effects and hence have high bias. The latter case means independently analysing (fewer) matches per time block, then attempting to summarise

¹³leave-one-out cross-validation, a hyperparameter tuning method

parameter estimates. A model assuming no time dependence — e.g., $\sigma^2 \rightarrow \infty$ in (1.13) — would obviously have a high variance.

A stochastic model like in (1.13) is a compromise between these extremes. The AR-1 model assumes that θ_{it} and $\theta_{i,t+1}$ are distinct but related, and the innovation variance σ^2 captures the magnitude of the difference between these parameters over time. The inference approach in this case is usually Bayesian. A benefit of such is that the posterior can be iteratively updated with new information (i.e., the next time block of data) rather than re-performing the whole analysis each time (Glickman, 2001).

These state-space models can be interpreted as instances of dynamic generalised linear models (DGLMs) (West et al., 1985) with a binomial link. Analyses of dynamic paired comparisons seemed to emerge in the 1990’s, with Glickman (1993) developing a Markov chain Monte Carlo (MCMC) method for posterior inference of chess player strengths. Fahrmeir and Tutz (1994) present a more flexible state-space model based on a Gaussian random walk that can account for ties and order effects. This is accompanied by a generalised Kalman filter algorithm for posterior mode estimation and application to the German National soccer league¹⁴. A very similar approach to Fahrmeir and Tutz (1994) is taken by Knorr-Held (2000) with an additional hyperparameter for “temporal smoothness of the abilities of each team.” This is also fitted using MCMC methods and applied to both German soccer and NBA datasets.

Glickman (2001) extends (1.13) by allowing for competitor-specific innovation variances that themselves vary stochastically, written σ_{it}^2 for competitor i at time t . The model, defined based on the random walk, is then

$$\theta_{i,t+1} \mid \theta_{it}, \sigma_{i,t+1}^2 \sim N(\theta_{it}, \sigma_{i,t+1}^2) \quad (1.14)$$

$$\log \sigma_{i,t+1}^2 \mid \sigma_{it}^2, \tau^2 \sim N(\log \sigma_{it}^2, \tau^2) \quad (1.15)$$

with suitable choice of priors for τ^2 and identifiability discussed further in the paper. It is easily extended to handle covariates and an AR parameter ρ . Results are given for an application to NFL games, including an AR parameter to represent any draft lottery effect and a common HGA parameter. The stochastic treatment of the innovation variance allows for abrupt changes over time, with examples like sudden shifts in cognitive development and financial markets. A similar random walk is described by Gorgi et al. (2019) for tennis, including covariates for court surfaces and player-specific features.

An exponentially-weighted moving average (EWMA) model is applied by Cattelan et al. (2013) to the analysis of Italian soccer (Serie A) and NBA. Though typically a smoothing technique,

¹⁴called *Bundesliga*

it is used in this context as a quasi-state-space model with recursive updates given by

$$\theta_{i,t+1} = (1 - \lambda)\theta_{it} + \lambda\mu_i(t) \quad (1.16)$$

where $\lambda \in [0, 1]$ is a smoothing parameter and $\mu_i(t)$ is a local strength estimate based only on the most recent match of i at time t . The choice of μ_i is specific to the competition of interest, such as for capturing HGA. Additionally, some important points are made about how teams may join and/or leave the competition over time, which affects identifiability constraints and parameter initialisation.¹⁵

Finally, Maystre et al. (2019) seem the first to propose defining the latent strengths with continuous-time GPs. Each competitor i has strength modelled as

$$\theta_{it} \sim \mathcal{GP}(0, \kappa(t, t')) \quad (1.17)$$

where $\kappa(\cdot, \cdot)$ is the covariance function (or *kernel*) and the mean function is taken to be zero. The Gaussian Process framework is rather powerful, thanks in part to its Bayesian formulation and the flexibility of kernel options. The authors furthermore develop an efficient Bayesian inference algorithm for fitting such. The choice of kernel is a critical user decision, requiring some level of domain knowledge; general discussion can be found in Duvenaud (2014) and Rasmussen and Williams (2005). Maystre et al. (2019) analyse several different competitions — such as NBA, ATP tennis, a large online chess dataset, and competitive *StarCraft*¹⁶ — with specially designed kernels. They find good predictive performance relative to even SOTA rating models and demonstrate handling extensions like comparison-specific covariates and intransitivity. As mentioned before, Krese and Štrumbelj (2021) also empirically supported the advantages of GPs in hindcasting.

1.2.5 Further topics

Connections to the Bradley–Terry model

Though many of the extensions discussed above were developed in the context of the BT model, they can nonetheless be applied to the TM model or any model satisfying (1.2). For example, a Thurstonian framing of a stochastic extension (the AR-1 model) was given by Böckenholt (2002). As mentioned previously, the BT and TM models are very similar, essentially a difference only of choice of logit or probit link function.¹⁷ In this vein, H. Stern (1992) investigates

¹⁵Some sports, such as major soccer leagues, are subject to *promotion and regulation* processes every season. In Serie A, for example, the last three teams are demoted and the best three teams in the lower league are promoted for the next season.

¹⁶an RTS video game popular in esports, particular in South Korea

¹⁷Some other link functions have been proposed, such as a two-parameter gamma (H. Stern, 1990).

the practical differences in link choice for (1.2), and finds “the evidence is strong that for samples of size usually encountered in practice all linear paired comparison models are virtually equivalent.” Furthermore, Glickman and Jones (2025) compare the BT and TM models on NBA data for the sake of illustration and find “nearly identical” rankings of teams.¹⁸

Luce’s Choice Axiom (Luce, 1959), also called the “independence of irrelevant alternatives” (IIA), provides a theoretical basis for analysis of paired comparisons. It states that the ratio of the probabilities of choosing one item over another is unaffected by the existence of other items in the set. The BT model is hence occasionally referred to as the Bradley–Terry–Luce model in the literature. This axiom leads to an important generalisation of the BT(L) model called the **Plackett–Luce (PL) model** (Plackett, 1975),

$$\mathbb{P}(i_1 \succ \dots \succ i_N) = \prod_{j=1}^{N-1} \frac{\exp(\theta_{i_j})}{\sum_{k=j}^N \exp(\theta_{i_k})} \quad (1.18)$$

where the observations are instead rankings of $N \leq I$ items $\{i_1, \dots, i_N\}$ rather than paired comparisons. Applications are competitions involving multiple simultaneous participants, such as horse racing or swimming. As the choice axiom is satisfied, the comparison of two items $\{i, j\}$ in the PL model reduces to the standard BT probabilities in (1.3). Furthermore, the BT model itself is naturally the special case where $N = 2$.¹⁹ See (Fang et al., 2026) for more on multiple comparisons and their properties.

The BT model can be approached from many different theoretical angles. The original paper (Bradley & Terry, 1952) proposed the model in the analysis of incomplete block designs. Similarly, Agresti (2013) emphasises the related concepts of square contingency tables and quasi-symmetry models, where BT is a logit formulation of the latter. Hamilton et al. (2025) provides an accessible yet extensive review of the many different motivations for the BT model, including some novel connections.

Computational developments

A natural question is about the procedures for estimating the model parameters, and doing so efficiently. The standard approach is maximum likelihood estimation (MLE). Assuming a dataset $\mathcal{D} = \{(i, j, k) \mid i \succ j \text{ in comparison } k\}$ consists of independent comparisons, then the MLE is given by

$$\hat{\theta}_{\text{MLE}} := \arg \max_{\theta} \sum_{(i,j,k) \in \mathcal{D}} \log F(\theta_i - \theta_j) \quad (1.19)$$

¹⁸Comparisons are done with LOOCV; BT (with HGA) only slightly outperforms by log-loss.

¹⁹Interestingly, in their original paper, Bradley and Terry (1952) describe an extension for triplet comparisons, i.e., the PL case for $N = 3$. Bradley (1984, Sec. 4.4) also reviews these methods.

subject to some constraint such as $\mathbf{1}^T \boldsymbol{\theta} = 0$ for identifiability. The constraint does not guarantee the existence of a unique maximiser; rather, strong connectivity of the comparison graph ensures the negative objective above is convex. Practically, this assumption means that there cannot be a team which wins every match because its strength estimate will diverge to infinity. In situations where strong connectivity is violated, one can instead impose regularisation of the parameters (e.g., ridge or lasso) (Varin & Firth, 2024).

The original work of Zermelo (1929) proposed an iterative algorithm for computing MLEs, though it converges slowly (Hunter, 2004); efficiency improvements have been suggested (Newman, 2023). The powerful class of minorisation-majorisation (MM) algorithms²⁰ and gradient descent are also widely used (Hunter, 2004; Vojnovic et al., 2020). Spectral methods like Rank Centrality have been developed (Negahban et al., 2015), in particular for dynamic settings (Tian et al., 2024). See Fang et al. (2026) for a survey of recent advancements of algorithms for paired and multiple comparison models.

The Bayesian approach has received increasing attention in paired comparison analysis (Whelan, 2017), though contributions date back to (Davidson & Solomon, 1973). A significant appeal is that the specification of priors induce implicit regularisation, and hence allow parameter estimation even when strong connectivity is not satisfied. Another aspect is of course convenient uncertainty quantification via the posterior. The posterior distribution $p(\cdot)$ of the strength vector $\boldsymbol{\theta}$ given the comparison data $\mathcal{D}$ has the following relationship:

$$p(\boldsymbol{\theta} \mid \mathcal{D}) \propto g(\boldsymbol{\theta}) \times \prod_{(i,j,k) \in \mathcal{D}} F(\theta_i - \theta_j) \quad (1.20)$$

where $g(\cdot)$ is the prior distribution of $\boldsymbol{\theta}$. A point estimate for $\boldsymbol{\theta}$ is either the posterior mean

$$\hat{\boldsymbol{\theta}}_{\text{PM}} := \mathbb{E}[\boldsymbol{\theta} \mid \mathcal{D}] \quad (1.21)$$

or maximum a posteriori (MAP) estimate

$$\hat{\boldsymbol{\theta}}_{\text{MAP}} := \arg \max_{\boldsymbol{\theta}} p(\boldsymbol{\theta} \mid \mathcal{D}) \quad (1.22)$$

We have seen that Bayesian methods appear naturally in stochastic extensions (Glickman, 2001; Krese & Štrumbelj, 2021; Maystre et al., 2019). Techniques have been developed for both posterior mean and MAP estimation, such as MCMC and EM algorithms, respectively (Caron & Doucet, 2012; Fang et al., 2026).

Theory developments

While the efficient computation of parameter estimates is important as datasets grow larger, so too is the study of the statistical theory. The existence and uniqueness of MLEs, as well as

²⁰of which expectation-maximisation (EM) algorithms are a special case

their properties of consistency and asymptotic normality, has received considerable attention (Bong et al., 2020; Hunter, 2004; Yan, 2025). Uniform consistency and asymptotic normality of the MLE for the BT model as $I \rightarrow \infty$ is established in a seminal paper by Simons and Yao (1999), and later extended to minimal conditions in sparse settings by Han et al. (2020). Most of these papers seem to use the graph theoretic approach. Again, Fang et al. (2026) succinctly summarise these theoretical contributions and more. Regarding the likelihood expression (1.19), it is easy to show that the win counts w_{ij} for each pair (i, j) are sufficient statistics through the GLM connection.

Recall that the models discussed so far assume independence of the paired comparisons; however, real-world data often exhibits dependence that necessitates some care from the modeller. Cattelan (2012) provides an extensive review of methods to handle dependent data, categorised based on structural assumptions. An intuitive concern is when *transitivity* is violated. The standard BT model has strong stochastic transitivity, yet some competitions naturally contain intransitive choices. For example, a *circular triad* is when we have a relationship $i \succ j \succ k \succ i$ between competitors i, j, k , such as in rock-paper-scissors or the famous Fire/Water/Grass type matchup in *Pokémon*. These situations can be handled with higher-dimensional BT models at the cost of a final ranking.

Beyond structural violations, dependency can arise in experimental design, such as when an individual judge provides multiple preferences. Cattelan (2012) discusses various ways to model correlation between observations, more so for Thurstonian models. For instance, Dittrich et al. (2008) consider dynamic effects in longitudinal paired comparisons, where a fixed set of judges make repeated comparisons in over time. Inference becomes quite complex for these extensions.

Lastly, there is an extension for when the outcome of a paired comparison is ordinal rather than binomial. Consider a preference survey where the respondent may express varying levels of agreement toward one of two options. If the options were strong preference, mild preference, or indifference for either item, then there would be $H = 5$ responses on this scale. These models hence describe outcomes $Y_{ij} \in \{1, \dots, H\}$ and cumulative probabilities

$$\mathbb{P}(Y_{ij} \leq h) = F(\tau_h - (\theta_i - \theta_j)) \quad (1.23)$$

with category thresholds $\tau_1 < \dots < \tau_{H-1}$ and where F is now a cumulative link. When $H = 3$, we have the aforementioned win-tie-loss scale with $\tau_1 = -\tau_2$ necessarily.²¹ Agresti (2013) and Cattelan (2012) describe these in more detail. Furthermore, S. E. Stern (2011) suggests a ‘moderated’ paired comparison model when the ‘size’ of a preference — i.e., margin of victory — is known. Application is given to cricket by defining ‘size’ as a function of team scores

²¹Specifically, this is the formulation of Rao and Kupper (1967).

and modelling outcomes with a beta distribution, including exponential downweighting of past matches and a common HGA.

Rating systems

Traditional likelihood-based approaches quickly become intractable when the number of items is large. These massive competitions — like international and online chess, amateur sports leagues, and online gaming — can have thousands if not millions of participants and matches played around-the-clock. Attempting global inference is infeasible in these competitions as the comparison data is very sparse and continuously updated with new results. Rating systems provide ‘local’ parameter updates immediately after each comparison and hence time complexity scales linearly in I .

The Elo rating system, named for its creator Elo (1978), is arguably the most well-known and foundational, with users as diverse as chess (FIDE), soccer (FIFA), competitive *League of Legends* (LoL), and many more. The Glicko system was developed in the 1990’s and offers some benefits over the Elo system thanks to its Bayesian approach, such as a reliability factor and uncertainties. More recently, novel ideas like Microsoft’s *TrueSkill* have been tailored to handle the complexities of multi-player and team-based online video games, such as FPS games and MMOs. A review of rating systems, including their application to NBA data and the evolution toward modern algorithms, is provided by Glickman and Jones (2025).

1.3 Review of Existing Implementations

Here we survey some notable implementations of paired comparison models. A majority of these are R packages (R Core Team, 2026); however, Python was also considered due to its rising popularity. We will briefly examine key features and drawbacks to motivate our own implementation. Another survey of R packages for both classical and more specialised paired comparison models is given in (Cattelan, 2012).

1.3.1 R packages

Perhaps the most popular implementation is `BradleyTerry2` (Turner & Firth, 2012), which supplants the earlier `BradleyTerry` package (Firth, 2005). It seems that tournament analysis was the main use case for these models. The package supports fitting of some of basic BT models with logit, probit, and cauchit link choices. Some emphasis is given regarding explanatory variables, as in (1.6), including the case when covariate data is missing, but only a constant

order effect (i.e., common HGA) is mentioned. Models can be fitted in three ways, from the standard likelihood methods to penalised and bias-reduced forms; in the latter scheme, finite parameter estimates can be obtained when the strong connectivity assumption is false. The authors note the lack of support for explicit handling of ties and contest-specific random effects.

The authors went on to develop the more generalised **PlackettLuce** package (Turner et al., 2019) for analysis of ranking data using the model in (1.18). This package hence supports Bradley–Terry models as a special case when rankings are done in pairs and adds many more features. Of note is the accommodation for ties and pseudo-comparisons to ensure strong connectivity. However, the authors indicate that future work is needed to “formally [account] for spatio-temporal heterogeneity in worth parameters,” among other efficiency improvements.

Another package is **prefmod** (Hatzinger & Dittrich, 2012), which allows for ordinal comparison data. Both the standard BT model and a pattern model approach are possible, with the latter handling situations where responses may be missing not at random (MNAR). The **BTLLasso** package (Schauberger & Tutz, 2019) provides various tools for subject-specific, comparison-specific, and interaction-based covariates. Notably, both common and team-specific HGAs can be modelled and matches at neutral venues can be flagged. Lastly, the **bpcs** package (Issa Mattos & Martins Silva Ramos, 2022) supports the Bayesian approach to paired comparisons and their various extensions by utilising Stan. The package seems quite feature-rich and fits using the efficient No-U-Turn sampler (NUTS) algorithm²².

1.3.2 Python packages

As one of the most ubiquitous programming languages, Python warrants some examination of its packages too. The **choix** package (Maystre, 2025) seems to be the most popular²³ implementation of ranking models. It supports both the BT and PL models, as well as ordinal responses and network choice models. Multiple algorithms are available, such as the MM algorithm of Hunter (2004) and a Bayesian framework for regularisation. However, **choix** appears to be missing some important features such as handling of ties, covariates, and inference beyond parameter estimation (e.g., standard errors).

Other Python implementations, to the best of our knowledge, are rather scarce. Most seem to be supplementary code for models presented in corresponding papers, rather than fully-featured packages in their own right. For example, Jerdee and Newman (2024) provide the **pairwise-ranking** package for the methods in their paper on randomness in social compe-

²²which was used in (Krese & Štrumbelj, 2021)

²³based on <https://pypistats.org/packages/choix>

titions that can lead to upsets. They attempt to quantify the ideas of ‘luck’ and ‘depth of hierarchy’ in competitions by modifying the form of the logit link in (1.3). Similarly, supplementary code is provided by Bong et al. (2020) for their kernel smoothing approach to a dynamic BT model. While these latter two implementations are useful for reproduction of results and transparency, they are not general-purpose. Overall, it is clear that there is a lack of feature-rich, out-of-the-box models available for paired comparison analysis, especially for dynamic extensions.

1.4 Review of Relevant Applications

Here we survey some relevant research, or lack thereof, of head-to-head competitions that considers home-ground and dynamic effects. In general, of course, the potential applications for paired comparison models are both numerous and diverse. Besides sports analytics, the models have found extensive use in other domains like the social sciences (David, 1988) and, recently, machine learning (Jiang et al., 2023; Ouyang et al., 2022). The suggested review papers mentioned earlier in 1.2 list various applications across a range of scientific disciplines (Cattelan, 2012, Sec. 2.2) (Fang et al., 2026, Sec. 6) and sports contexts (Glickman & Jones, 2025). While paired comparison models are the focus here, we include some papers on HGA and time variation in sport that use other models or qualitative approaches.

It is common knowledge that sports teams are volatile. Certainly, many sports show dynamic team performances throughout the seasons due to factors like players’ age, strategy developments, and draft systems (Podlog et al., 2015). Despite this, the BT model literature is relatively scarce on historical analyses with time variation. Indeed, it appears that analyses with dynamic effects are usually done to corroborate novel modelling ideas rather than for their own sake, as with most cited in 1.2.4. Studies utilising competition-specific domain knowledge are hence few and far between. As one example, Cattelan et al. (2013) thoroughly discuss their results using EWMA on the NBA and Serie A matches, highlighting individual team performance records and home advantages. This approach is quite relevant as they model the home and away abilities as separate EWMA processes, which essentially allows for team-specific HGAs that vary over time. Notably, they show that one limiting case leads to a model where the home ability is constant in time and common to all teams.

The heterogeneity of home-ground advantage — that is the concept of significantly different team-specific advantages — has been examined with differing conclusions. In the paired comparison model literature, however, this possibility has been seldom considered beyond a generic subject-specific covariate extension. In most papers mentioned thus far, we have seen that any

practical modelling of HGA usually defaults to just a single parameter common to all teams and constant across time. Though some sports have inconclusive evidence of a heterogeneous HGA, others are renowned for their iconic home venues and loyal fan bases, such as the NBA (Ribeiro et al., 2016). For example, Clarke and Norman (1995) analysed match margins in English soccer league with a least squares model and found notable differences in HGA.²⁴

Some interesting scenarios can be explored when looking at home-ground over time. An obvious contemporary example is the impact of the COVID-19 pandemic on sports. Leitner et al. (2023) give a systematic review of the change in home advantage in soccer from a psychological perspective and confirm significant impacts due to so-called “ghost games”²⁵. Further, Liu et al. (2025) look into the effect of COVID-19 on EPL results using a simple dynamic BT model with team-specific HGA. As expected, they find that team performance and HGA fluctuated due to pandemic disruptions compared to the past ten seasons. This also begs the question on whether HGA depends on the venue itself or fan attendance. For instance, Ponzo and Scoppa (2018) investigated if crowd support could be separated from “other mechanisms such as players’ familiarity with the stadium and travel fatigue” and found evidence in favour of such. Clearly, the effect of the home ground can be much more complex than a simple constant parameter would suggest.

The main competitions of interest in this thesis are the Australian Football League (AFL) and National Rugby League (NRL). As such, lastly, we look at previous literature on these two sports — using head-to-head models or not — and whether home or time effects have been covered. Very little research on modelling AFL data with paired comparison models could be found. The preprint by Soffner and Leonelli (2024) seems the most relevant, repeatedly fitting a BT model with a common HGA plus team-specific covariates to each season in order to predict the next. This quasi-dynamic, ‘rolling window’ approach estimates strengths within each season but specifies no dependence structure over time. They find evidence of a variable common home advantage, among other things, and suggest that team strengths in AFL are stationary for no longer than three years.

To our knowledge, no literature exists on applying paired comparison models to the NRL. Other models have been applied to both these competitions, however. Examples include linear score margin models for team-specific HGA (Clarke, 2005); nonparametric score margin models during games (Guan et al., 2022); AR-1 score margin models with common HGA (Leushuis, 2018); ensemble methods (Barry, 2026); classification trees (Woods et al., 2017); Bayesian forecasting (Manderson et al., 2018); modified Elo ratings (Cameron et al., 2025; Ryall &

²⁴Interestingly, the paired HGA was found to be linearly related to distance between the club venues.

²⁵games without supporters, due to lockdown restrictions etc.

Bedford, 2010); and deep learning methods like LSTMs (Yang & Wesemann, 2025) and neural networks (Fahey-Gilmour et al., 2019), just to name a few. Research gaps in these competitions thus present fertile ground for new analysis with the BT model for home-ground and dynamic effects.

1.5 Review of Previous Work

Before we outline the aims of this thesis, it is important to acknowledge previous work — done by Jones (2021) and Zhang (2024) — upon which we build.²⁶ These papers proposed and developed the theory for “Hierarchical” Bradley–Terry (HBT) models, a novel extension which allows for more complex treatments of home-ground advantage. We briefly examine the key ideas and limitations to guide our current work.

Citing a lack of literature and software for modelling HGA effects, the Honours thesis by Jones (2021) introduced the HBT model to analyse pairwise comparisons in **hierarchical** competition structures. In national-scale sports leagues, such as the NBA and NFL, teams are typically partitioned into groups based on geography or historical allocations. The main application of HBT models was to the NBA, where teams are categorised into conferences (Eastern and Western) and then divisions (three per conference). AFL data was also analysed despite having no explicit hierarchical structure, as the HBT model can be applied to any user-defined grouping; for example, AFL teams could be grouped by their state to see if within-state and interstate home-ground effects are significantly different.

The original codebase by Jones (2021) was written in **R** and chiefly utilised the `optim` function for standard ML estimation. The Master’s project report by Zhang (2024) modified and corrected some aspects of the previous implementation, looking to improve user-friendliness. Functionality was added to simulate NBA playoffs based on parameter estimates. The possibility of ‘neutral’ venues — where neither team experiences a HGA — was an additional improvement by Zhang (2024); this simple concept seems mentioned only a couple times throughout the literature, namely in (Schauberger & Tutz, 2019) and (Krese & Štrumbelj, 2021).

There are several improvements for us to consider. First and foremost, Jones (2021) notes that competitions cannot reasonably be assumed to be stationary over many years. While Jones provides some rudimentary analysis of the time dependence in AFL and NBA, the incorporation of a dynamic extension in HBT models is needed. Secondly, various aspects and design decisions in the implementation can be critiqued. For example, documentation and a more object-

²⁶These works are unpublished at time of writing but can be found in their respective GitHub repositories via the bibliography.

oriented approach would significantly benefit the readability of the code; optimisations to parameter estimation and inference could also be explored.

1.6 Aims & Outline

Based on our literature review, the selection of a model for analysing head-to-head match-ups is a non-trivial task. The modeller should have both domain knowledge for the sport in context and an intimate understanding of the relevant extensions, such as whether one must account for ties, HGA, and dynamic effects. We have seen, however, that implementations to readily support these analyses are still rather limited if not absent.

In this thesis, we will further develop the Hierarchical Bradley–Terry model by considering dynamic effects. The competitions of interest are the AFL and NRL, two of the most popular sports in Australia. Applications of paired comparison models to these sports, however, is scarce or not present at all. Our primary focus will hence be on investigating heterogeneous home advantage effects in the two leagues, such as differences across groupings of the teams or changes during events like the COVID-19 pandemic.

The proposed dynamic extension is relatively simple. We wish to now allow the *team strengths to vary freely*, with no specified dependence structure across time blocks, while *constraining HGA effects to be constant*. Recall that this is a special case of the stochastic extension, as in the AR-1 model (1.13) when the innovation variance approaches infinity. In this way, the team strengths θ_{it} are effectively modelled as nuisance parameters. For scope reasons, no extensions for ties nor other covariates (besides HGA) are considered at present. This is acceptable as ties occur relatively infrequently in AFL/NRL; for further justification see 2.7. Furthermore, we assume that the paired comparisons are independent. Other assumptions — like strong connectivity of subgraphs for each time block — are considered as necessary in Chapter 3.

With extensive reviews conducted and our motivations established, we present the following succinct aims for this project:

1. To consolidate previous theory on HBT models and add a nonparametric dynamic extension as described above;
2. To overhaul the implementation of HBT models to improve code clarity, documentation, usage of OOP principles, inference, visualisation tools, and more;
3. To collect and clean data on AFL and NRL matches over several seasons; and
4. To hypothesise and examine various HGA effects in AFL and NRL matches, accounting for time variation.

The remaining parts of the thesis are hence as follows. Chapter 2 formally describes preliminaries and defines the HBT model variants. Chapter 3 explains the codebase design and implementation details for the models. Results for the AFL and NRL applications are presented and analysed in Chapter 4. Chapter 5 summarises the achievements and limitations of the project, then describes directions for improvements and future applications of HBT models.

Chapter 2: Models

For each Bradley–Terry model, we primarily require the **likelihood function**. The log-likelihood (LLH) is the function to maximise for optimal **parameter estimates**. The score statistic (gradient of the LLH) is necessary for gradient-based optimisation methods that we use; the Hessian of the LLH is necessary for computing analytic standard errors (SEs). We first specify some notation in 2.1 before giving the likelihood results for each model. Nested relationships are discussed in 2.6 and further modelling ideas in 2.7.

2.1 Overview

There are several different notations used throughout the literature. Furthermore, there have been some notational discrepancies within previous work as generalisations are made. Naturally, we must also introduce new notation for our dynamic extensions, hence the importance of this section. Here we summarise the likelihood framework for parameter optimisation, then consolidate a consistent set of symbols and their meanings for use throughout this thesis.

2.1.1 A modelling framework

To recap, the original Bradley–Terry model describes the probability of winning (and losing) a paired comparison, that is, the event that one team beats another team. Extensions of the BT model then describe outcomes of paired comparisons under more specific conditions, such as given a certain venue or time of the match. Each model is specified by some parameters $\boldsymbol{\theta}$, where the number of parameters p depends on the model. For a given competition, such as some sport of interest, we have a dataset $\mathcal{D}$ that contains information about all paired comparisons and their conditions. For example, in the simplest BT model, each paired comparison can be described by a 3-tuple: the two competitors and the outcome. Therefore, in general, our goal is to estimate — and make inferences on — the parameters $\boldsymbol{\theta}^*$ based on the data $\mathcal{D}$.

Let us then lay the groundwork for maximum likelihood estimation. We write $\mathcal{L}(\boldsymbol{\theta} \mid \mathcal{D})$ to denote the likelihood function $\mathcal{L} : \mathbb{R}^p \rightarrow \mathbb{R}$ evaluated at the parameters $\boldsymbol{\theta} \in \mathbb{R}^p$ given the data $\mathcal{D}$. The log-likelihood is hence written $\ell(\boldsymbol{\theta} \mid \mathcal{D}) := \log \mathcal{L}(\boldsymbol{\theta} \mid \mathcal{D})$.¹ The likelihood contains the

¹we usually just write $\mathcal{L}(\boldsymbol{\theta})$ and $\ell(\boldsymbol{\theta})$ for convenience

information for all probabilities in a given dataset, and so maximisation chooses parameters that make the observations most probable. In this sense, the data can be thought of as evidence that supports (or refutes) any given set of parameter values. As the logarithm is strictly increasing, ℓ is used for practical reasons like numerical stability.

2.1.2 Notation

We first formally define a **competition** based on the factors of interest outlined in 1.6. In our framework, we describe a competition as consisting of $I \in \mathbb{Z}^+$ competitors over $T \in \mathbb{Z}^+$ time blocks, where $I > 1$. The competitors in each unique pair (i, j) may have a certain **relationship** to one another. Specifically, we let $r_{ij} = k$ denote the relationship between i and j as being of level $k \in \{1, \dots, K\}$, where $K \in \mathbb{Z}^+$ is the number of different levels and $i, j = 1, \dots, I$. These relationships are often induced by certain groupings of the competitors, as in a competition **hierarchy**. What exactly these relationships or ‘hierarchy’ are depend on the context of the competition and modelling choices by the user.

Furthermore, each paired comparison is played at a **venue** which may only be described as either *home* or *away* for a given team. Hence, let us deftly define an **advantage indictor** v_{ij} as

$$v_{ij} = \mathbb{I}(i \text{ is home}) - \mathbb{I}(j \text{ is home}) = \begin{cases} 1 & , \quad i \text{ has home advantage} \\ 0 & , \quad \text{neutral} \\ -1 & , \quad j \text{ has home advantage} \end{cases} \quad (2.1)$$

which explains how home-ground advantage exists when teams i and j play a match at a given venue. It is not so simple to say that if i is home then j is away. The extension to consider neutral venues includes when both teams are away (*neither* team experiences a home-ground advantage) and when both teams are home (the effect of a home-ground advantage is *neutralised*).

Therefore, in general, each paired comparison can be written as the event

$$\{i \succ j \mid v_{ij}, r_{ij}, t\} \quad (2.2)$$

to denote a match where i beats j given some advantage indicator $v_{ij} \in \{-1, 0, 1\}$, relationship level $r_{ij} \in \{1, \dots, K\}$, and time $t \in \{1, \dots, T\}$, for any $i, j \in \{1, \dots, I\}$. Note that if $i = j$ then (2.2) is considered a null event, as we assume no team can play against itself; similarly, we would say r_{ii} is undefined. Altogether, suppose the dataset $\mathcal{D}$ contains N observed paired comparisons from the competition. These are the fundamental factors for the models in this thesis, and we shall consider specifics and border cases as necessary throughout.

The contents of the parameter vector $\boldsymbol{\theta}$ of course depend on the model. As a simple overview, there are only two types of parameters: the **team strengths** θ_{it} and the **order effects** α_k for the given team i , time block t , and hierarchy level k . Recall that we are interested in allowing the team strengths to vary over time while order effects are kept constant (hence the subscript indices). We will see that certain subscripts are dropped depending on HGA complexity and which factors are of interest.

The form of the likelihood function depends on the observed data $\mathcal{D}$. In particular, it can be shown that the pairwise **win counts** are sufficient statistics, in that $\ell(\boldsymbol{\theta})$ can be expressed simply as a function of these win counts. However, these win counts must be partitioned for more complex models as the data is described with more factors, such as venue and time. We make this precise with the following definitions:

- $W_t \in \mathbb{R}^{I \times I}$ is the matrix of win counts in time block $t = 1, \dots, T$, where each element w_{ijt} represents the number of times i has beaten j , for $i, j = 1, \dots, I$.
- $H_t, A_t, U_t \in \mathbb{R}^{I \times I}$ are the matrices of home, away, and neutral venue win counts, respectively, in time block $t = 1, \dots, T$. For example, h_{ijt} represents the number of times i beat j at *home* in time block t .

Furthermore, as teams cannot play themselves, the diagonal terms of each win matrix are zero (such as $w_{iit} = 0$ for any i and t).

With these definitions outlined, we can move to specify our dynamic models where each probability is now conditional on some time block. See 2.6 to easily recover the original static analogues, as in previous work. For each model, we state the number of parameters, then $\ell(\boldsymbol{\theta})$ and its derivative components; only the nonzero Hessian elements are listed for brevity. Furthermore, most derivation details are skipped and can be found in A.1.

2.2 Vanilla Bradley–Terry (VAN)

The vanilla Bradley–Terry model refers to the original model described by Bradley and Terry (1952).² The model is simple in that it only considers the team strengths θ_{it} and no order effects. The number of parameters is hence $p = I \times T$. It simply models the probability of team i beating team j at time t as

$$\pi_{ijt} := \mathbb{P}(i \succ j \mid t) = \text{expit}(\theta_{it} - \theta_{jt}) \quad (2.3)$$

The log-likelihood function is given by

$$\ell(\boldsymbol{\theta} \mid \mathcal{D}) = \sum_{ijt} w_{ijt} \log \pi_{ijt} \quad (2.4)$$

The elements of the score statistic and Hessian are hence

$$\frac{\partial \ell}{\partial \theta_{it}} = \sum_j w_{ijt} - (w_{ijt} + w_{jit})\pi_{ijt} \quad (2.5)$$

and

$$\frac{\partial^2 \ell}{\partial \theta_{it}^2} = - \sum_j (w_{ijt} + w_{jit})\pi_{ijt}(1 - \pi_{ijt}) \quad (2.6)$$

$$\frac{\partial^2 \ell}{\partial \theta_{it} \partial \theta_{jt}} = (w_{ijt} + w_{jit})\pi_{ijt}(1 - \pi_{ijt}) \quad (2.7)$$

2.3 Common Home-ground Advantage (CHA)

We introduce a single home-ground advantage effect α which is common to all teams; this is a typical extension (Agresti, 2013). There is only one additional parameter, so $p = I \times T + 1$. The probability of winning now depends on the venue that the match was played at, with

$$\pi_{ijt}^{(h)} := \mathbb{P}(i \succ j \mid v_{ij} = 1, t) = \text{expit}(\theta_{it} - \theta_{jt} + \alpha) \quad (2.8)$$

$$\pi_{ijt}^{(a)} := \mathbb{P}(i \succ j \mid v_{ij} = -1, t) = \text{expit}(\theta_{it} - \theta_{jt} - \alpha) \quad (2.9)$$

$$\pi_{ijt}^{(u)} := \mathbb{P}(i \succ j \mid v_{ij} = 0, t) = \pi_{ijt} \quad (2.10)$$

as the probability i beats j at time t given i is at a home, away, or neutral venue, respectively.

The log-likelihood function is given by

$$\ell(\boldsymbol{\theta} \mid \mathcal{D}) = \sum_{ijt} h_{ijt} \log \pi_{ijt}^{(h)} + a_{ijt} \log \pi_{ijt}^{(a)} + u_{ijt} \log \pi_{ijt}^{(u)} \quad (2.11)$$

²The terminology ‘vanilla’ comes from Jones (2021).

The elements of the score statistic and Hessian are hence

$$\frac{\partial \ell}{\partial \theta_{it}} = \sum_j h_{ijt} + a_{ijt} + u_{ijt} - (h_{ijt} + a_{jit})\pi_{ijt}^{(h)} - (a_{ijt} + h_{jit})\pi_{ijt}^{(a)} - (u_{ijt} + u_{jit})\pi_{ijt}^{(u)} \quad (2.12)$$

$$\frac{\partial \ell}{\partial \alpha} = \sum_{ijt} h_{ijt}(1 - \pi_{ijt}^{(h)}) - a_{ijt}(1 - \pi_{ijt}^{(a)}) \quad (2.13)$$

and

$$\frac{\partial^2 \ell}{\partial \theta_{it}^2} = - \sum_j (h_{ijt} + a_{jit})\pi_{ijt}^{(h)}(1 - \pi_{ijt}^{(h)}) + (a_{ijt} + h_{jit})\pi_{ijt}^{(a)}(1 - \pi_{ijt}^{(a)}) + (u_{ijt} + u_{jit})\pi_{ijt}^{(u)}(1 - \pi_{ijt}^{(u)}) \quad (2.14)$$

$$\frac{\partial^2 \ell}{\partial \theta_{it} \partial \theta_{jt}} = (h_{ijt} + a_{jit})\pi_{ijt}^{(h)}(1 - \pi_{ijt}^{(h)}) + (a_{ijt} + h_{jit})\pi_{ijt}^{(a)}(1 - \pi_{ijt}^{(a)}) + (u_{ijt} + u_{jit})\pi_{ijt}^{(u)}(1 - \pi_{ijt}^{(u)}) \quad (2.15)$$

$$\frac{\partial^2 \ell}{\partial \alpha^2} = - \sum_{ijt} h_{ijt}\pi_{ijt}^{(h)}(1 - \pi_{ijt}^{(h)}) + a_{ijt}\pi_{ijt}^{(a)}(1 - \pi_{ijt}^{(a)}) \quad (2.16)$$

$$\frac{\partial^2 \ell}{\partial \theta_{it} \partial \alpha} = - \sum_j (h_{ijt} + a_{jit})\pi_{ijt}^{(h)}(1 - \pi_{ijt}^{(h)}) - (a_{ijt} + h_{jit})\pi_{ijt}^{(a)}(1 - \pi_{ijt}^{(a)}) \quad (2.17)$$

N.B. the subtle differences in indices of each sum. The remaining models are essentially increasingly complex generalisations of the home-ground order effects.

2.4 Team-specific Advantage (TSA)

This model extends the common home-ground model by allowing each team to have their own advantage parameter α_i . The number of parameters is $p = I \times (T + 1)$. We have the win probabilities

$$\pi_{ijt}^{(h)} := \mathbb{P}(i \succ j \mid v_{ij} = 1, t) = \text{expit}(\theta_{it} - \theta_{jt} + \alpha_i) \quad (2.18)$$

$$\pi_{ijt}^{(a)} := \mathbb{P}(i \succ j \mid v_{ij} = -1, t) = \text{expit}(\theta_{it} - \theta_{jt} - \alpha_j) \quad (2.19)$$

$$\pi_{ijt}^{(u)} := \mathbb{P}(i \succ j \mid v_{ij} = 0, t) = \pi_{ijt} \quad (2.20)$$

The log-likelihood is given by (2.11). The elements of the score statistic are (2.12) and

$$\frac{\partial \ell}{\partial \alpha_i} = \sum_{jt} h_{ijt} - (h_{ijt} + a_{jit})\pi_{ijt}^{(h)} \quad (2.21)$$

The Hessian elements are (2.14), (2.15), and

$$\frac{\partial^2 \ell}{\partial \alpha_i^2} = - \sum_{jt} (h_{ijt} + a_{jit})\pi_{ijt}^{(h)}(1 - \pi_{ijt}^{(h)}) \quad (2.22)$$

$$\frac{\partial^2 \ell}{\partial \theta_{it} \partial \alpha_i} = - \sum_j (h_{ijt} + a_{jit})\pi_{ijt}^{(h)}(1 - \pi_{ijt}^{(h)}) \quad (2.23)$$

$$\frac{\partial^2 \ell}{\partial \theta_{jt} \partial \alpha_i} = (h_{ijt} + a_{jit})\pi_{ijt}^{(a)}(1 - \pi_{ijt}^{(a)}) \quad (2.24)$$

2.5 Hierarchical Advantage (HIE)

The Hierarchical BT model suggests that home-ground advantage varies depending on the relationship level between teams in a match. Before discussing the model formulae, we need to clarify an important point regarding these relationships r_{ij} . In previous work, HBT models were based on relationships assumed to be symmetrical, $r_{ij} = r_{ji}$. We relax this assumption to allow for possibly different relationships depending on which team is home. Let the first index be the home team and second index be away, so $r_{ij} = 1$ would mean a match with i at home against j uses HGA parameter α_1 . Thus, $r_{ij} = k$ and $r_{ji} = l$ may not necessarily be equal!

To illustrate, consider the NBA. Currently, the National Basketball Association (NBA) consists of 30 teams spread across North America. The league categorises these teams into two *conferences* — Eastern and Western — both further subdivided into three *divisions* consisting of five teams each.³ These groups inherently reflect geographical factors like location and population density, so it would be interesting to analyse differences in home-ground advantage within and across divisions. We continue this example with two specific cases.

A modeller may impose a **symmetric relationship structure** to examine matchup effects. For instance, games between teams in the same division may use HGA parameter α_1 , teams with different divisions but same conference α_2 , and across-conference games α_3 . In this case, the relationship depends on the pair (i, j) such that $r_{ij} = r_{ji}$. Conversely, an **asymmetric relationship structure** could be used when modelling certain constraints on the home-court advantage. For instance, each of the six divisions could be allocated a unique HGA $\alpha_1, \dots, \alpha_6$. Now the relationships are determined only by the home team, so $r_{ij} \neq r_{ji}$ when i and j belong to different divisions. As a side note, recall that we said the ‘self-relationships’ r_{ii} are undefined; we will represent this with the special index $k = 0$ to mean ‘no home-ground parameter’ and fix $\alpha_0 = 0$. See Figure 1 for a visualisation of these two relationship structures.

We discuss these relationships further in 2.6.2. Continuing on, let each relationship level have a unique home-ground advantage α_k for $k = 1, \dots, K$. The model hence has $p = I \times T + K$ parameters. The win probabilities now depend on the relationships r_{ij} , where we have

$$\pi_{ijt}^{(h)} := \mathbb{P}(i \succ j \mid v_{ij} = 1, r_{ij} = k, t) = \text{expit}(\theta_{it} - \theta_{jt} + \alpha_k) \quad (2.25)$$

$$\pi_{ijt}^{(a)} := \mathbb{P}(i \succ j \mid v_{ij} = -1, r_{ji} = l, t) = \text{expit}(\theta_{it} - \theta_{jt} - \alpha_l) \quad (2.26)$$

$$\pi_{ijt}^{(u)} := \mathbb{P}(i \succ j \mid v_{ij} = 0, t) = \pi_{ijt} \quad (2.27)$$

as the probability that i beats j at time t given i is at a home, away, or neutral venue,

³the NBA Wikipedia page provides a nice map and table of the teams’ locations: https://en.wikipedia.org/wiki/National_Basketball_Association#Teams

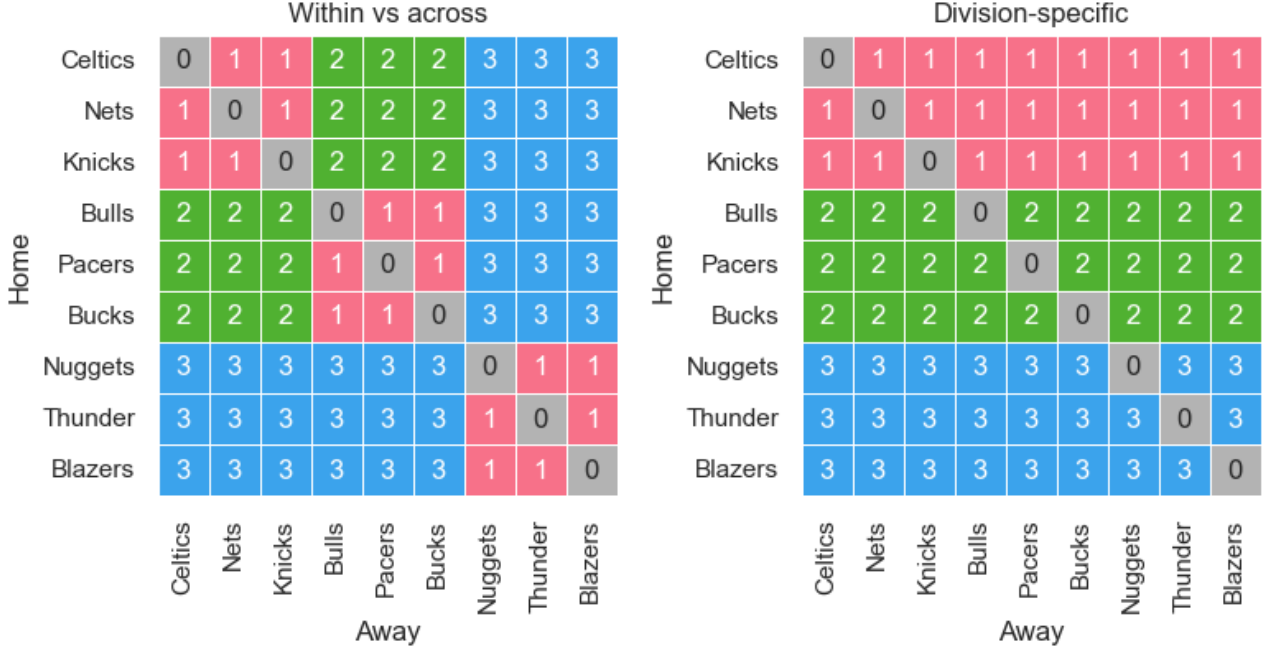


Figure 1: Visualisation of the relationship structure examples for the NBA. Only 9 of the 30 teams are shown for clarity. The teams are: Celtics, Nets, and Knicks (Atlantic division, Eastern conference); Bulls, Pacers, and Bucks (Central division, Eastern Conference); and Nuggets, Thunder, and Blazers (Northwest division, Western conference). We can clearly see how these structures are symmetric (left) or asymmetric (right).

respectively.

The log-likelihood is given by (2.11). The elements of the score statistic are (2.12) and

$$\frac{\partial \ell}{\partial \alpha_k} = \sum_{ijt} [h_{ijt}(1 - \pi_{ijt}^{(h)})] \mathbb{I}(r_{ij} = k) - [a_{ijt}(1 - \pi_{ijt}^{(a)})] \mathbb{I}(r_{ji} = k) \quad (2.28)$$

The Hessian elements are (2.14), (2.15),

$$\frac{\partial^2 \ell}{\partial \alpha_k^2} = - \sum_{ijt} [h_{ijt} \pi_{ijt}^{(h)} (1 - \pi_{ijt}^{(h)})] \mathbb{I}(r_{ij} = k) + [a_{ijt} \pi_{ijt}^{(a)} (1 - \pi_{ijt}^{(a)})] \mathbb{I}(r_{ji} = k) \quad (2.29)$$

$$\frac{\partial^2 \ell}{\partial \theta_{it} \partial \alpha_k} = - \sum_j [(h_{ijt} + a_{jit}) \pi_{ijt}^{(h)} (1 - \pi_{ijt}^{(h)})] \mathbb{I}(r_{ij} = k) + [(a_{ijt} + h_{jit}) \pi_{ijt}^{(a)} (1 - \pi_{ijt}^{(a)})] \mathbb{I}(r_{ji} = k) \quad (2.30)$$

2.6 Connecting and Comparing Models

In this section, we describe the nested relationships of our models and briefly explain how to compare adequacy of fit with hypothesis testing. Typically, a model is called *nested* if it can be expressed as a special case of another model. There are two key ‘axes’ of generalisation in our models to consider: (i) **order effects** and (ii) **time**.

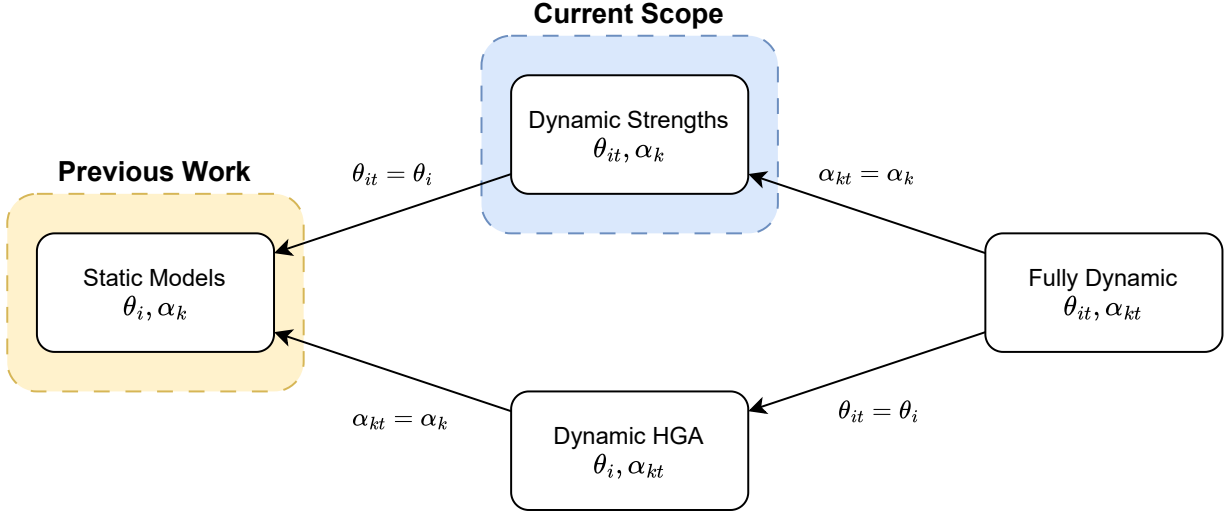


Figure 2: Nested relationships with respect to time.

2.6.1 Static and dynamic models

The nestedness with respect to time is much easier to grasp first. In previous work, all models were **static** since their parameters were constant in time. Of course, we may now let all parameters vary with time to have fully **dynamic** models. In this thesis, however, we are interested in restricting the dynamic models so that only the team strengths vary while the order effects remain constant. Altogether, this presents three distinct levels of temporal modelling, shown in Figure 2. The static models can be recovered by constraining parameters, say $\theta_{it} = \theta_i$ for all $t = 1, \dots, T$ for each $i = 1, \dots, I$.

Additionally, Figure 2 shows how to constrain the parameters using equality to get the special cases; however, certain nested models can alternatively be recovered with **aggregation**. With any static analogues, for instance, one could take $T = 1$ and reformat the data by summing the win matrices over time, such as $w_{ij} = \sum_t w_{ijt}$. The practical benefit of the aggregation approach is that it is usually easier to simply reformat the data than change the implementation itself.

2.6.2 Relationship matrices

On the other hand, we also have nestedness in the different **order effect parameterisations**, i.e., how the home-ground advantage is handled. Before we can explain how these models are nested, however, we need to touch on the relationships r_{ij} once again. Firstly, it is convenient to compile all the r_{ij} into a **relationship matrix** $R \in \mathbb{R}^{I \times I}$ for a given model. Our notions of symmetric and asymmetric relationships naturally also describe their corresponding matrix. With the lens of relationship matrices, we can unify all the models presented thus far.

The key idea is that all home-ground models are special cases of the HIE model with a certain

corresponding relationship matrix. To recall our definition: if teams i and j have the relationship $r_{ij} = k$, then a match played by i at home against j uses HGA α_k . Now, in the CHA model, all home matches of course assume a common advantage α regardless of the specific home team. The CHA model can therefore be viewed as a special case of the HIE model where all relationships are the same, that is $r_{ij} = 1$ for all pairs (i, j) . Similarly, in the TSA model, each team has its own HGA parameter α_i and so we can say $r_{ij} = i$ for any pair (i, j) . The vanilla model would simply imply a constant zero relationship matrix to indicate that no home-ground effect is modelled.

For completeness, a few more connections can be drawn with this generalisation. In previous work, HBT models were initially conceived as only two specific cases of the HIE model we defined in Section 2.5. The key idea was to have a unique HGA parameter for each hierarchy level — as we do — but treated more as hierarchical analogues to the CHA and TSA models. An example of the former — called the *common hierarchical* HGA model by (Jones, 2021) — proposed the symmetric relationship structure we described in the NBA scenario earlier. One could examine the home effect differences based on the matchup: division rivals would use an α_1 , conference rivals α_2 , and teams playing across the two conferences α_3 . The latter — the *team-specific hierarchical* HGA model from (Jones, 2021) — supposed further that each team has a different HGA at each level of the hierarchy, i.e., division rivals use their own α_{1i} and so on. Both these approaches are now special cases of a HIE model with certain R .

Another concept mentioned in previous work was a pair-wise home-ground effect where each matchup (i, j) has a unique HGA α_{ij} . In the NBA, for example, the Celtics may have a unique home-court advantage over the Lakers, being one of a few famous inter-conference rivalries.⁴ This model is the most general form of those listed previously, totalling $p = I \times (T + I - 1)$ parameters. Of course, this is now seen as a HIE model where every r_{ij} is unique. To accompany the two NBA examples in Figure 1, the relationship matrices for these other models are depicted in Figure 3. For the matrices with all 30 NBA teams, refer to Appendix A.2.

As we have seen, the HIE model with arbitrary relationship matrix is the general form of all possible home-ground treatments. The question now becomes how to frame the nestedness of these models, for which one answer is to think of **partitions**. Let us define the set of ordered pairs for a given competition as

$$\mathcal{P} = \{(i, j) \mid i \neq j, \quad i, j = 1, \dots, I\} \quad (2.31)$$

⁴Famous enough for its own Wikipedia article: https://en.wikipedia.org/wiki/Celtics-Lakers_rivalry

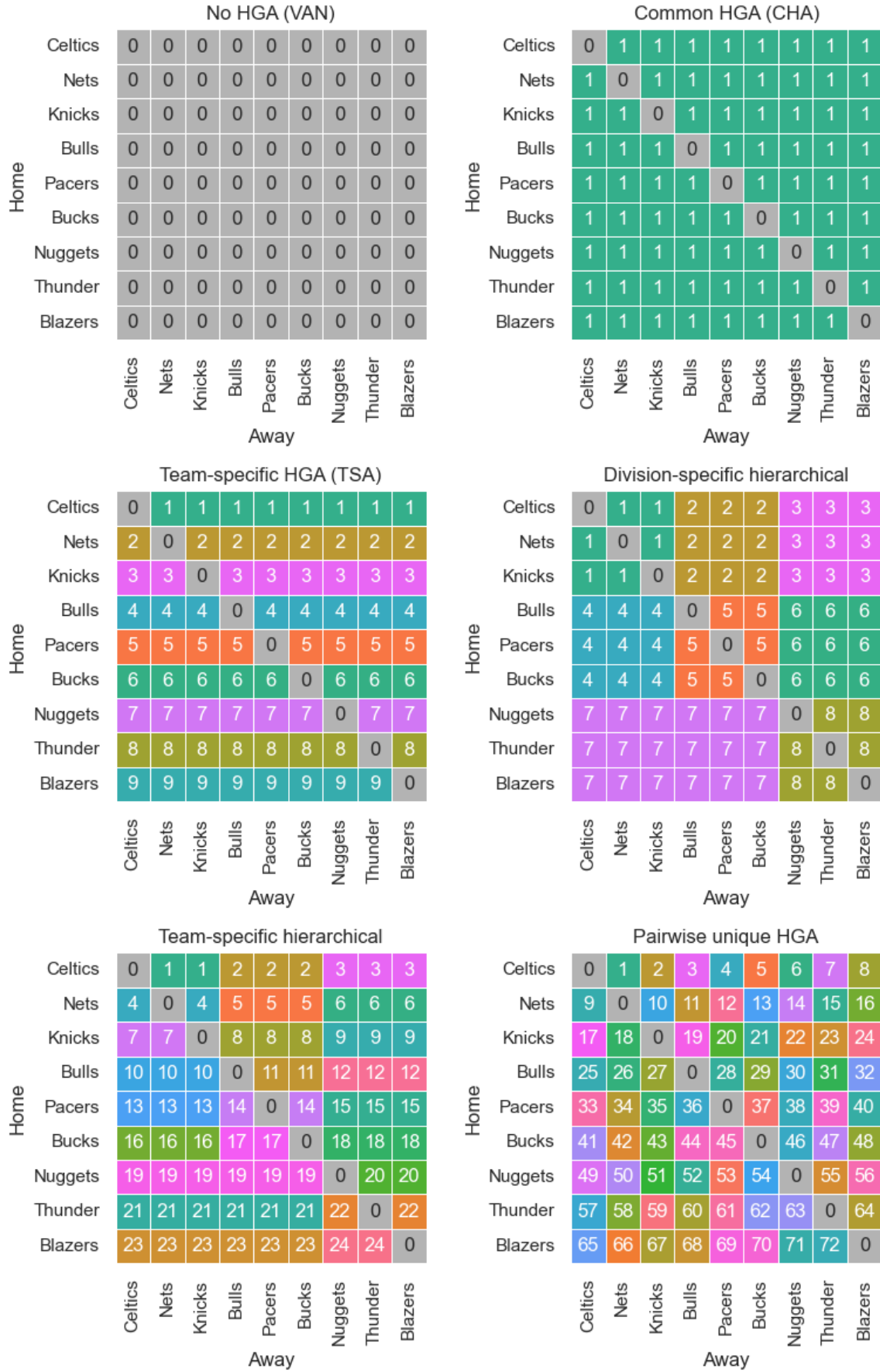


Figure 3: More relationship matrices for the NBA example. Again, this is only a subset of the full roster to demonstrate the many possible relationship structures. This view ties together all possible home-ground advantage parameterisations.

We can then partition $\mathcal{P}$ into K subsets based on a relationship matrix R by

$$P_k = \{(i, j) \mid r_{ij} = k, \quad (i, j) \in \mathcal{P}\} \quad , \quad k = 1, \dots, K \quad (2.32)$$

This leads us to a natural definition of nestedness: one model is nested within another if the partition induced by the former is coarser than the latter. Formally, let models M_1 and M_2 have a relationship matrices $R_1 = [r_{ij}^{(1)}]$ and $R_2 = [r_{ij}^{(2)}]$ with K and L levels, respectively, where $K \leq L$. These relationship matrices hence partition $\mathcal{P}$ as $\{P_k\}_{k=1}^K$ and $\{Q_l\}_{l=1}^L$, respectively. Thus, M_1 is nested in M_2 iff $Q_l \subseteq P_k$ for some $k \in \{1, \dots, K\}$, for all $l = 1, \dots, L$. Equivalently, for all $(i, j), (i', j') \in \mathcal{P}$,

$$r_{ij}^{(2)} = r_{i'j'}^{(2)} \implies r_{ij}^{(1)} = r_{i'j'}^{(1)} \quad (2.33)$$

To solidify with a simple example, consider the nested relationship between the CHA and TSA model. The CHA model yields the trivial partition $P_1 = \mathcal{P}$, while the TSA model partitions into $K = I$ subsets, i.e., the row vectors of R . Clearly then, the CHA model is nested in the TSA model. The nestedness of some important models is shown in Figure 4, which is based on the NBA scenarios discussed so far.

2.6.3 Hypothesis testing

Having established these nested relationships, we move to understand how to compare such models. Consider two models, M_0 and M_1 , where M_0 is nested in M_1 . The likelihood ratio test allows us to determine which model provides a more appropriate fit to some dataset based on the difference in their deviances. Hence, define the test statistic

$$D = -2 \times (\ell(\boldsymbol{\theta}_0) - \ell(\boldsymbol{\theta}_1)) \quad (2.34)$$

where $\boldsymbol{\theta}_0 \in \mathbb{R}^m$ and $\boldsymbol{\theta}_1 \in \mathbb{R}^n$ are the parameter vectors of M_0 and M_1 , respectively. The null hypothesis H_0 is that M_0 is sufficient; the alternative H_1 is that M_1 provides a better fit. We use Wilks' theorem for an asymptotic distribution of D .

Theorem 1 (Wilks' Theorem). *Let model M_0 be nested in M_1 and define D as in (2.34). As the sample size increases, D asymptotically follows a χ^2 distribution under H_0 . The degrees of freedom is the difference in number of parameters between the two models, $n - m$. (Agresti, 2013, p. 11)*

For further reading on inference like comparison of deviances in GLMs, see for instance Agresti (2013, Ch. 4.5). There are some situations where models are not nested and hence we cannot use

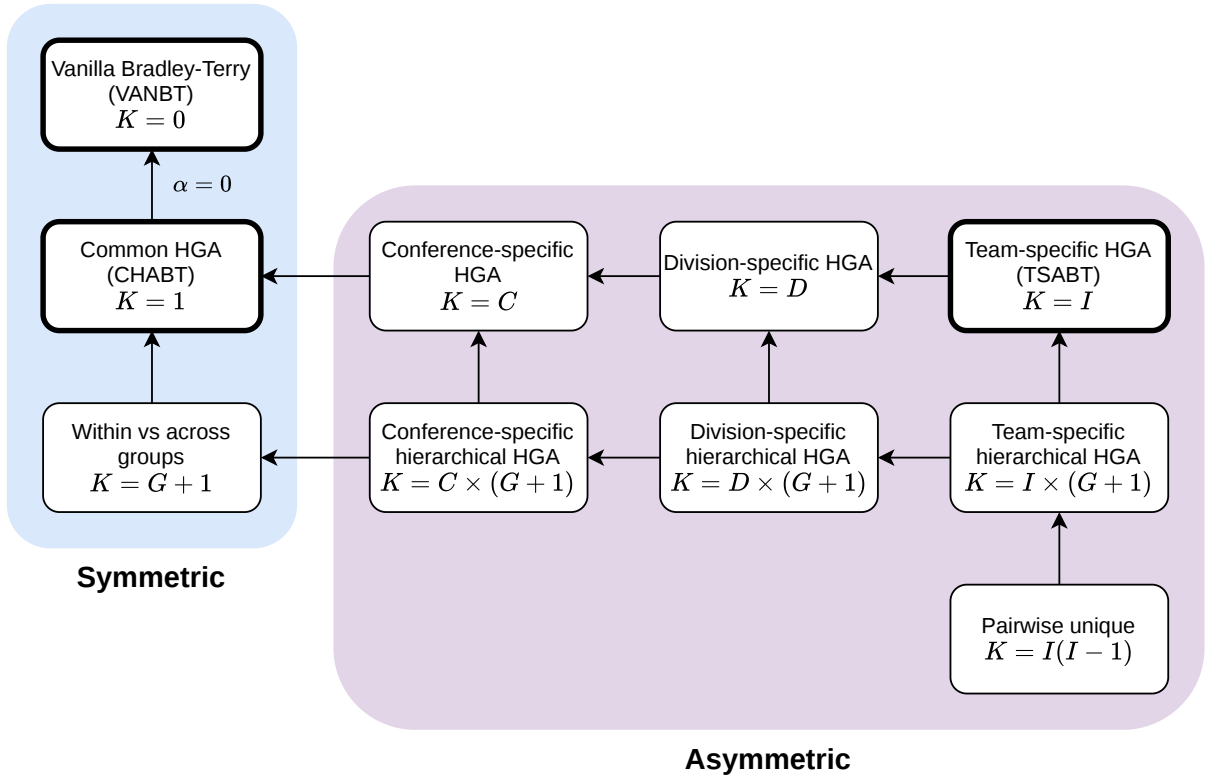


Figure 4: Nested relationships with respect to HGA parameterisation, based on the NBA examples. Models with bolded outlines are those described explicitly in Sections 2.2-2.4, though all models are special cases of the HIE model in Section 2.5. Home-ground parameter counts K for each model are included for ease of reference, where the total number of parameters is $p = TI + K$. In the NBA example, there are $G = 2$ two groupings of the $I = 30$ teams: $C = 2$ conferences split into $D = 6$ divisions total.

this test. For example, consider comparing the team-specific HGA (TSABT) and the ‘within-vs-across’ hierarchical model; Figure 4 tells us that there is no nested relationship between these two models — they are ‘parallel’ — and so we cannot use a deviance test. In these cases, we should examine quantities such as AIC and BIC when making judgements on fit adequacy.

2.7 Further Modelling Options

Having outlined the key results for the models of interest, a natural question is that of further generalisations. Here we briefly discuss more modelling choices for order effects, as well as other possible avenues for extensions. This question will be revisited throughout the thesis.

We have already mentioned the pair-wise unique HGA model, though such a model is likely far from practical in real competitions. Notice in particular that the number of pair-wise advantage terms α_{ij} is quadratic in the number of teams I . Another obvious extension is to let the order effects be time-varying. These are the ‘fully dynamic’ models also mentioned prior. For instance, in the common HGA model, each time block in the competition would have a

distinct home-ground advantage α_t . This is equivalent to just fitting an individual ‘static’ BT model to each time block and treating them as one model.⁵ However, this approach is antithetical to our aims in this report, which presumes that these order effects are constant in time while the team strengths are allowed to fluctuate. This is investigated empirically in Section 4.4. Beyond scope reasons, there are also computational reasons to avoid modelling dynamic order effects due to the size of such models.

As such, one asks what else exactly we can generalise without bloating model complexity. Let us first consider venues. Recall that we have already abstracted the concept of a ‘venue’ out of our dataset; each paired comparison is described with the simple advantage indicator v_{ij} . That is, we do not require additional information about the physical venues themselves because the v_{ij} are sufficient. Should team i change what it considers its ‘home’ venue at some point during the competition — even if *within* a time block — this does not affect the modelling. The models currently only require the *direction* of the advantage $v_{ij} \in \{-1, 0, 1\}$ in a given comparison. Incorporating any venue-specific information would likely require yet another model with different order effect parameterisation⁶; such models are not of primary interest here.

Now consider hierarchies. A comparatively inconsequential extension would be time-varying hierarchy relationships, as in pair (i, j) having relationship level r_{ijt} at time t . For example, this could reflect the rare event when a team may change their location or home city, as has happened a couple dozen times in the history of the NBA.⁷ This is very straightforward to incorporate into our framework; as with v_{ij} , the model only views r_{ij} as a feature of the dataset which determines which HGA parameter to use for a given paired comparison. What would be more troublesome is if the number of hierarchy levels K , or the *interpretation* of such, changes over time. This is not so much a concern with the models’ capabilities, but rather impacts interpretability when analysing the fitted parameters like α_k . Could this still be considered the same competition, or are we best to apply separate models before and after a major change? This is a highly subjective problem to be dealt with on a case-by-case basis.

To summarise, there are many possible ways to extend the models covered in this thesis. The main problems with these generalisations is that they either greatly inflate model complexity or that they are well beyond scope. Some technicalities have been clarified, such as how nuanced interpretations of factors like v_{ij} and r_{ij} between matches does not affect modelling. Additionally, we must be conscious of when certain changes in the competition may be too

⁵In fact, this is the situation for the vanilla model in 2.2 because it has no advantage parameters!

⁶For example, the home team at venue $v = 1, \dots, V$ could have advantage α_v . As an application, perhaps one team has multiple home venues and wishes to know if one offers a substantial benefit.

⁷see https://en.wikipedia.org/wiki/List_of_relocated_NBA_teams

drastic for subsequent parameter interpretations to be meaningful. As yet another example, one could consider adding more comparison-specific covariates such as match attendance, weather, or field conditions. In justifying our choice of relatively simple models when faced with such plentiful extensions, we primarily appeal to the principle of parsimony. Certainly, overfitting is a genuine concern with these more excessively parameterised models, and this is also realised in their computational tractability. Furthermore, we do not wish to stray from our desired scope outlined in 1.6; there are yet implementation details and case studies to cover in this report.

Chapter 3: Implementation

This chapter serves as documentation for the key features of our new implementation of HBT models.¹ We start with an overview of the codebase structure and the benefits of an object-oriented design approach. Some practical details are then considered, such as handling the data, enforcing constraints, and the choice of optimisation algorithm. We touch on some features provided for analysing results, like standard errors and visualisation, before concluding with comparisons to previous work.

3.1 Codebase Architecture

Python was chosen for the implementation. The language is highly approachable and hence one of the most popular for data analysis, from basic visualisations to machine learning. One key benefit of an object-oriented programming approach is the concept of *inheritance*. As there are multiple models of interest, we can define a single parent class — and all of its generic functions — and then construct child classes for specific models as necessary. Our codebase hence consists of this fundamental class and its children, with other modules for features like visualisation, data utilities, and hypothesis testing.

The `BaseBradleyTerry` class is an abstract base class (ABC) for Bradley–Terry models based on our framework. It specifies variables and methods common to all models, and otherwise defines stubs for model-specific methods like the log-likelihood and its derivatives. Besides various getters and setters, the superclass functions include data validation, the fitting algorithm, constraint enforcement, error calculation, and odds and probability calculations. See Appendix B for tabulated descriptions of these variables and functions.

The `BaseHierarchicalBT` class is itself an ABC that extends the `BaseBradleyTerry` class. This base class adds a few things necessary for the hierarchical models, such as variables and functions for interacting with the relationship matrix. There is a child class for each of the models defined in Chapter 2 which inherits from their respective ABC. These classes need only to implement their LLH functions and some getters to ensure the parent functions run correctly.

The standard workflow was designed to be intuitive for users familiar with popular ML pipelines

¹The implementation can be found at the following link: <https://github.com/booms14nger/uqhons-bt>.

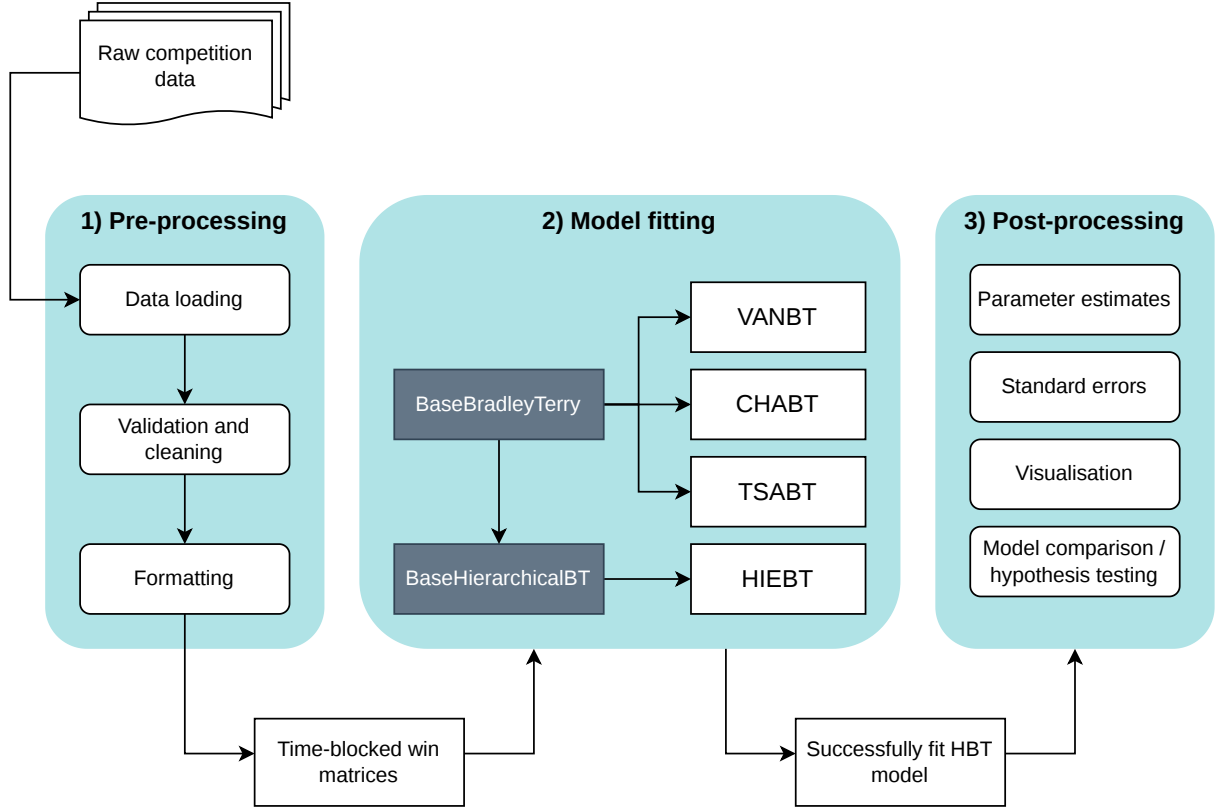


Figure 5: A visual representation of the typical pipeline for HBT model usage. Many of these items are explained in more detail throughout Chapter 3. The middle box shows the inheritance relationships between the classes, with the ABCs in grey.

like `scikit-learn` (Pedregosa et al., 2011). After loading and massaging the data into the required format, one simply needs to instantiate the desired model and pass the data via the `fit()` function. Assuming the model fits successfully, the user can analyse the results using various tools described in Section 3.3. This typical usage process is depicted in Figure 5.

3.2 Practical Details

3.2.1 Data structures

Recall that we defined four different types of win matrices in each time block: the overall wins W_t is partitioned into home wins H_t , away wins A_t , and neutral wins U_t . In the implementation, we use `DataFrames` from the `pandas` package to store the win matrices. These frames are then compiled into a dictionary indexed by time. The vanilla model, for example, only requires the W_t for the time blocks $t = 1, \dots, T$ of interest. All other home-ground models, however, require the partitioned matrices H_t, A_t, U_t and so use a nested dictionary. An illustrative example is given in Figure 6.

The hierarchical models additionally require the relationship matrix, for which a `DataFrame` is

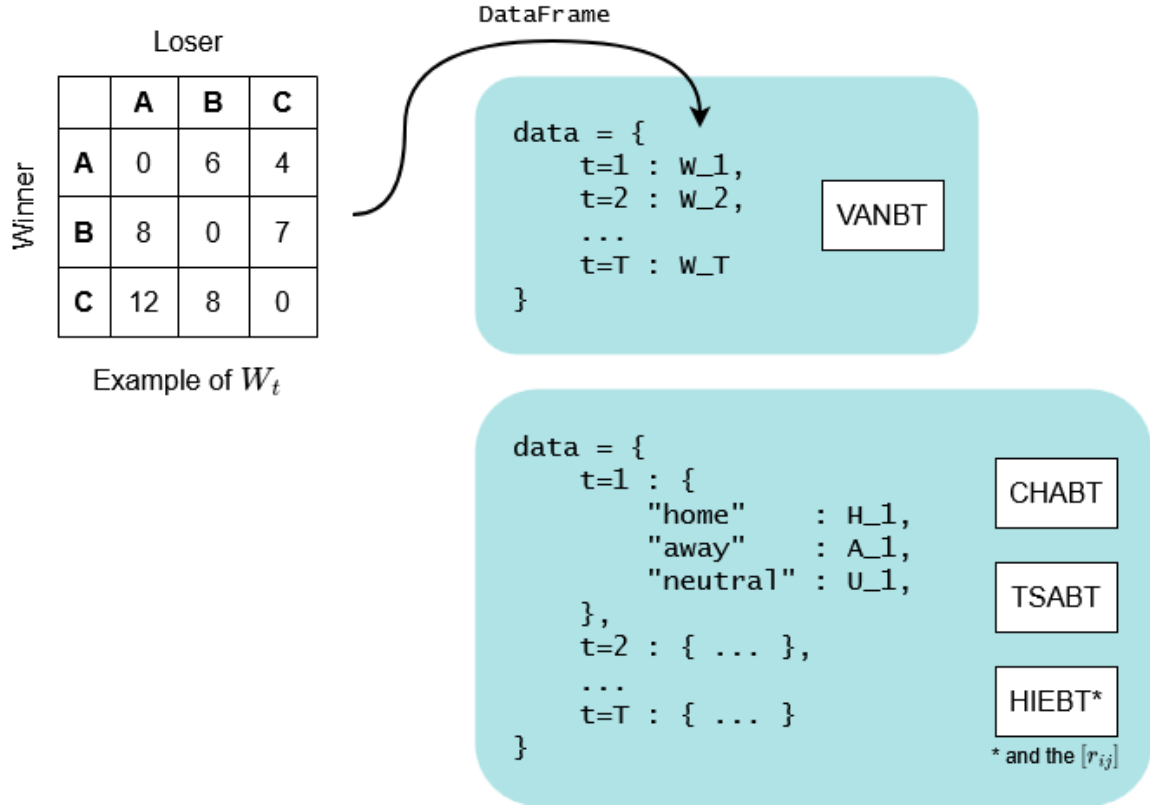


Figure 6: Pseudocode example of the main data structures used in our models. The vanilla model uses a dictionary that maps the time t to the corresponding win matrix W_t . The home-ground models use a similar approach by mapping time t to a nested dictionary, indexed by the venue type, containing H_t, A_t, U_t .

also used. Support functions are provided to aid in converting from typical long-form match results to these data structures. Something to note is that the model parameters are vectorised; that is, all team strengths and HGA parameters are stored in a single, one-dimensional array. To avoid confusion, this vectorisation has been abstracted away from the end user, who can instead get parameters with the dedicated `get_param()` and `get_ranking()` functions.²

3.2.2 Data validation and important edge cases

Our data validation needs to be rather rigorous. The BT model gives us several important assumptions to check and constraints to enforce, particularly in the dynamic case. Not only that, but our data structure requirements are quite strict. As such, we need to ensure the data is formatted correctly lest the model fit poorly or not at all. Currently, the implementation checks the following qualities of the data:

1. The passed object is a dictionary
2. The values of this dictionary are either DataFrames (vanilla) or dictionaries of DataFrames

²These functions are user-friendly. E.g., the former uses the team name and year rather than requiring knowledge of their internally allocated index.

- (home-ground models) with the correct venue keys
3. The win matrices are square
 4. The team names (rows and columns) are consistent across all win matrices
 5. The win counts are all non-negative
 6. There may be teams which are inactive or missing in certain time blocks
 7. There are at least two active teams at all times
 8. The digraph for each time-blocked win matrix is strongly connected (after removing any inactive teams)
 9. There exists at least one **global reference team**, i.e., a team that exists in every year of the competition (over the time period of interest)

Checks (3) and (4) also apply to the relationship matrix in the hierarchical model.

Here we need to consider some edge cases and how they are handled by the code. An obvious situation would be competitions where the digraph is not strongly connected; this could arise in datasets with small sample sizes or in sports with unbalanced schedules. While strong connectivity is often a fundamental assumption for BT models (Hunter, 2004), one can still try fit the parameters in cases where it is violated. If a team has no recorded losses, for instance, then their strength parameter explodes to infinity during ML estimation. As discussed in Chapter 1, there are various ways to compute finite parameter estimates when strong connectivity is violated, such as regularisation or a Bayesian approach.

The consequences of requiring strong connectivity can quickly become meticulous. In the home-ground models — where each win matrix must be split further into venue-specific wins — one may need to additionally ensure strong connectivity of each of these matrices.³ As we still use a simple linear constraint (discussed in the next section), the implementation will print a warning when there are assumption violations and then attempt to fit regardless. This is justifiable for our use case since the competition datasets of interest are indeed strongly connected.

Another important edge case is when a team may not be present in every time block. A real-life example is when analysing the AFL over the 2010’s, as the Gold Coast Suns did not enter the competition until the 2011 season, and the GWS Giants in 2012. Our framework essentially fits a separate BT model to each time block under the constraint of constant HGA(s), so these scenarios should be permissible in theory. In the win matrices where these teams do not play, they are just given a column and row vector of zeroes. These teams are still automatically allocated parameters for the years in which they are missing; such parameters do not get esti-

³As an example, consider a TSA model and a team which never wins at home. There is hence no data to distinguish this team’s strength from its HGA parameter.

mated as there is no data to do so. It is thus necessary to track these **inactive parameters** so they can be explicitly removed from summaries, visualisations, and when calculating statistics like AIC and BIC. While teams are indeed allowed to enter and leave the competition in our models, there must be at least one “global reference” team that has match data in all time blocks to satisfy our constraint.

3.2.3 Parameter estimation and constraints

As mentioned previously, model fitting is done by maximum likelihood estimation. A common optimisation function used in Python applications is `minimize` from `scipy` (Virtanen et al., 2020). This function is used for local minimisation — possibly subject to constraints — of a given multivariate function, with a variety of methods available. At present, we use `minimize` with BFGS,⁴ a standard gradient descent algorithm. This requires the (negative) log-likelihood $-l(\boldsymbol{\theta})$ and its gradient $-\nabla l(\boldsymbol{\theta})$, as well as an initial guess for which we currently just use the zero vector $\boldsymbol{\theta}^{(0)} = \mathbf{0} \in \mathbb{R}^p$.

Recall that the parameters require an identifiability constraint. We employ the set-zero constraint for this purpose; however, there is some nuance when applying it to our framework. Specifically, such a constraint is needed in each time block because the strengths are independent. Our constraint hence becomes

$$\theta_{at} = 0 \quad , \quad t = 1, \dots, T \tag{3.1}$$

where a must be a global reference team. If a was missing from a certain time block, then it would be nonsense to constrain based on an inactive parameter, i.e., that time block would be effectively unconstrained. This constraint was chosen primarily to follow previous work and because it gives a consistent and intuitive baseline for making qualitative comparisons across time (relative to the reference team). It is also easy to translate the strengths by a constant post hoc such that another team becomes the global reference.

In practice, this constraint is currently enforced immediately *after* optimisation with a “rebas-ing” function. Multiple options are provided for the choice of constraint team. By default, the code will pick the first valid team alphabetically; other choices include the team with the first index, the worst estimated average strength, or a user-specified team. In each time block, the rebasing function subtracts the reference team’s strength from all strengths. The inactive parameters are set to `nans` to emphasise that they are not estimated and to distinguish from the zeroes of the reference team.⁵

⁴the Broyden—Fletcher—Goldfarb—Shanno (BFGS) algorithm

⁵`nan` is a special constant provided by `numpy` to represent “Not a Number” (NaN) by IEEE 754 standards.

3.3 Analytical Tools

3.3.1 Outputs and inference

Having outlined the pre-processing and optimisation steps, with all their intricacies, we can now discuss the tools provided for analysing model results. Perhaps the most essential is the summary output, a concise yet effective presentation of all the important variables and results for a model. For this we provide a familiar `summary()` function inspired by those for the `glm` function in R. It prints essential metadata — like number of teams, times, observations, and active parameters — as well as fit quantities like the LLH, AIC, BIC, and fit iterations. After this, all strength parameters and their corresponding standard errors (SEs) are tabulated by time, with any HGA parameters printed separately.⁶ By default, the teams are sorted by average strength.⁷

As mentioned, various model fit statistics and hypothesis testing functions are included in a dedicated module. Besides AIC and BIC, there are functions to carry out a comparison of deviance as described in 2.6. Given two models, the `simple_report()` function will neatly print their fit statistics and hypothesis test results (if nested). It was designed to be similar to functions like `anova` in R. The algorithm for nestedness (2.33) is shown in the snippet below (with docstring removed).

```
1 def is_nested(m1: BaseBradleyTerry, m2: BaseBradleyTerry) -> bool:
2     # Time-based nestedness
3     if m1.n_times > m2.n_times:
4         return False
5
6     mask = ~np.eye(m1.n_teams, dtype=bool)
7     R1 = m1._get_rel_mat()[mask]
8     R2 = m2._get_rel_mat()[mask]
9
10    # Relationship nestedness: check coarser partition
11    for lvl in np.unique(R2):
12        idx = (R2 == lvl)
13        if len(np.unique(R1[idx])) > 1:
14            return False
15
16    return True
```

⁶Because there can often be a lot of strength parameters in our models, there is a option to control the verbosity by not including the errors or printing the strengths at all.

⁷“NaN” is printed for the strengths and SEs of any inactive parameters.

3.3.2 Error calculation

The calculation of standard errors is fairly typical, though the constraints must be handled appropriately. The SEs for the parameters are derived from the estimated covariance matrix, which requires inverting the observed information matrix, i.e., the negative Hessian evaluated at the MLE. For the i -th parameter θ_i of some model, we have

$$\hat{\Sigma} = -H(\hat{\boldsymbol{\theta}})^{-1} \implies \text{SE}(\hat{\theta}_i) = \sqrt{\hat{\Sigma}_{ii}} \quad \forall i = 1, \dots, p \quad (3.2)$$

Due to the presence of inactive parameters and our identifiability constraint, we need to be rather careful to ensure the Hessian is not singular. Fortunately, this simply requires removing the rows and columns corresponding to these inactive and reference team parameters, inverting, and then adding back zero vectors to the covariance matrix to keep the indices consistent.

Note that the BFGS algorithm approximates the Hessian during optimisation; however, we choose to use our analytically derived results for a few reasons. Precision is of course a key factor as the analytic Hessians are exact and hence give more accurate SE values. The approximated Hessians may also be less numerically stable due to convergence termination criteria, particularly in higher-dimensional parameter spaces (as we often have), and thus cannot be guaranteed to be invertible. A very important assumption underlying this approach to SE calculation, however, is the asymptotic normality of the MLE as the sample size increases. This has been covered thoroughly in the literature⁸ and justified for HBT models in previous work (Jones, 2021).

3.3.3 Visualisation

The topic of visualisation is not only highly subjective but also complex. As such, while the implementation provides some generic plotting functions, the experienced user is recommended to generate their own plots. A win matrix can be easily plotted as a digraph using a package like **NetworkX** (Hagberg et al., 2008), though these quickly become impractical for more than a handful of competitors over several time blocks.⁹ More relevant is plotting the parameter estimates, for which we provide two main plots for each model: a static version using just the *averaged* strengths and a dynamic version showing all strength parameters. The static version uses simple but effective ‘dumbbell’ plots, inspired by previous work, but has the obvious downside of excluding the entire time axis. The dynamic versions of course include all the parameters in the model as a line plot, but can easily lose visual clarity when the number of teams is large. Figure 7 provides a simple example of the two basic visualisation options for a common HGA model.

⁸see Section 1.2.5 – Theory developments

⁹This package is still useful as it provides algorithms to check strong connectivity.

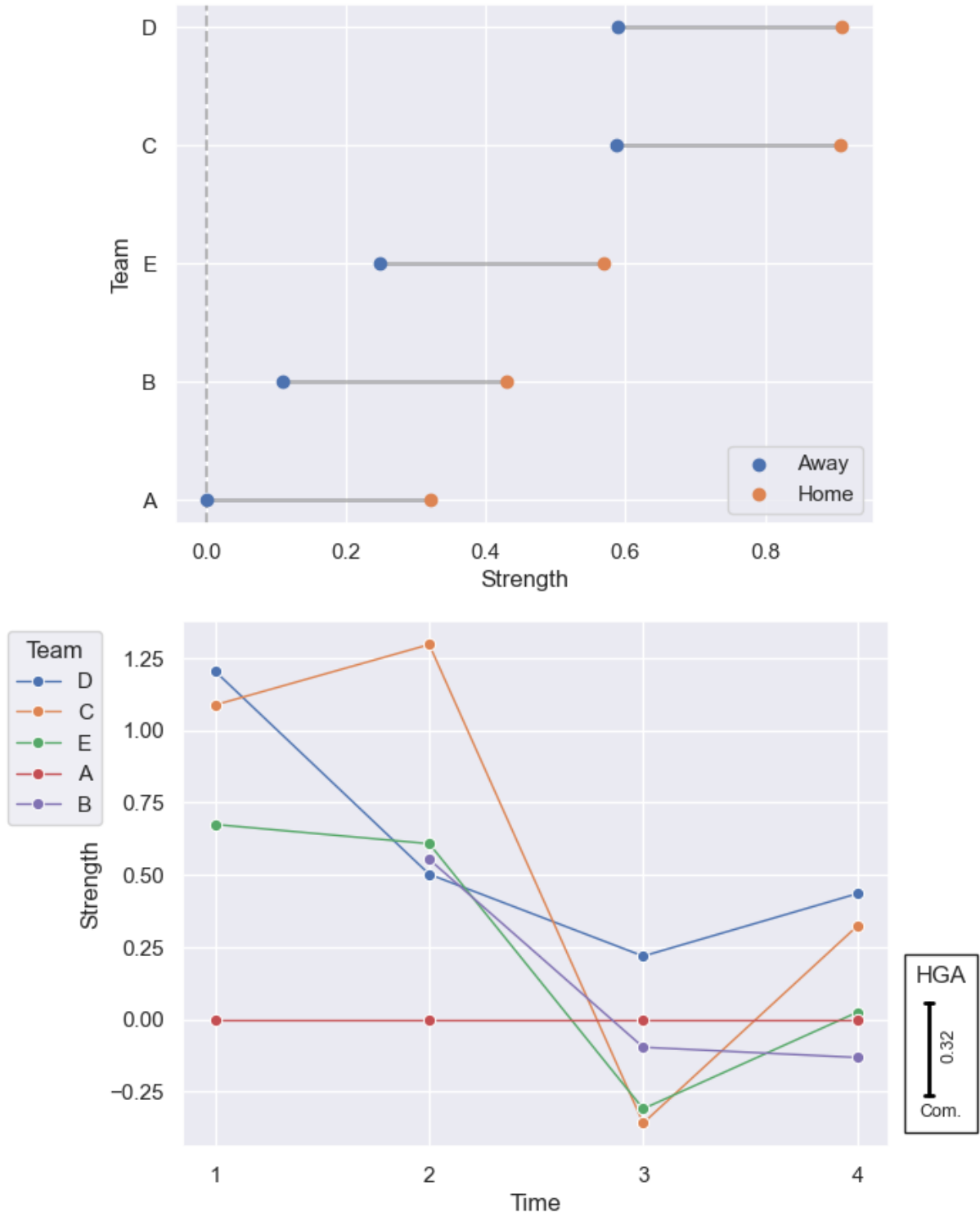


Figure 7: An example of the included visualisation functionality for a CHA model fitted to some dummy competition data with five teams over four time blocks. Team “A” has been set to the reference team by default and Team “B” is not present in the first time block. The static plot (top) is a dumbbell plot showing the average estimated team strengths with and without the common HGA. The dynamic plot (bottom) is a line plot showing the fluctuations in the estimated strengths over time and includes a scale bar for the HGA (which is constant).

A lot of minor details went into the current plotting functionality but are not discussed at length here. The default visualisation options could be improved by using other packages and software. For example, standard errors are currently missing from the plots due to clutter. Though far from the scope of this project, an ideal direction would be interactive visualisations where the user can easily change the reference team and highlight certain trends, among other things. A website, for instance, would be an effective medium for this project in future.

3.4 Discussion of Improvements

3.4.1 Previous implementations

The new implementation offers numerous advantages over the previous. Most important is perhaps our object-oriented approach, as the most recent HBT implementation (Zhang, 2024) suffers from massive code redundancy. By instead defining a common base class, we streamline the development of the desired models and significantly reduce the overhead for future modifications. This design philosophy presents the user with a simple yet familiar workflow for their data analysis and abstracts away the more technical details. Another major change is of course using Python rather than R, unlike all previous implementations. While R is certainly a powerful language, the change to Python arguably makes our code more approachable and opens the way for integration with Python’s rich and modern data visualisation packages.¹⁰

On a more technical level, our implementation tends toward the code standard required for a publicly available Python package. It seems the first to rigorously validate input data and print helpful warning messages for end users. Multiple utility functions, model methods, and visualisation options have also been added. A rather critical feature lacking in previous implementations is descriptive code comments and documentation in general. All our classes and functions are equipped with sufficient docstrings to clearly identify purpose, parameters, outputs, and general usage. Reproducibility is also emphasised, with a requirements file for dependencies and several Jupyter notebooks for interactive tutorials and demonstrations of the standard pipeline. Furthermore, another crucial result of our implementation is that all previous work can be easily recovered as special cases of the models.

3.4.2 Further changes

Despite the major improvements in the new implementation, several aspects could be revisited. Most relevant would be the current method for parameter estimation. One potential problem

¹⁰The author’s familiarity with Python also contributed to this.

with previous work is that optimisation was done unconstrained despite the translational invariance of the BT model. Using BFGS is nice for its efficiency;¹¹ however, in an unconstrained setting, the Hessian would be theoretically singular at the maximum and hence convergence may be poorer than in a constrained scheme. Our approach to enforce the reference team constraint during estimation is simple: implicitly remove the corresponding strengths from the parameter vector and assume them to be zero. As such, our minimisation is done in a reduced space which improves numerical stability. The downside, however, is that fitting tends to take longer than in the unconstrained setting, so more research should be done on what causes this in an effort to regain some fit efficiency.

On the other hand, we could also change the identifiability constraint. The current set-zero constraint (3.1) is suitable for our applications but not best in general. A major downside of this constraint is the need for a global reference team, as choosing different reference teams in each time block sacrifices interpretability across time. The sum-to-zero constraint, $\sum_i \theta_{it} = 0$ for all t , is a solid alternative for our model framework (with a little more implementation effort). This would relax the need for a global reference team while maintaining interpretability as comparisons then become relative to the “average team” in each year. Regularisation is another possible avenue, however, removing an explicit linear constraint hinders interpretability of the parameter values. One could add a small regularisation effect on top of a linear constraint to try prevent singular Hessians and fit when strong connectivity is violated.¹²

Finally, another core element of polished packages is unit testing. Some functions currently exist for checking parts of the models, such as numerical approximations of the gradient and Hessian for a given log-likelihood to ensure the theory has been implemented correctly. Synthetic competition data generation is also demonstrated for testing purposes, but this should be revisited to construct more rigorous unit testing for the models. On a related note is the need for some benchmarking. It would be desirable to compare the performance speed of our implementation as aspects of the data grow, especially compared to previous work and preeminent implementations like `BradleyTerry2` and `PlackettLuce`. Code efficiency was not a major focus in the current implementation, so further options like handling sparse win matrices and LLH vectorisation are possible. Nonetheless, our implementation offers numerous benefits and effectively serves the modelling applications of interest in this thesis.

¹¹ $O(p^2)$ time complexity

¹²the `choix` package does this

Chapter 4: Applications

In this chapter, we apply our models to two sporting competitions: the AFL and NRL. We perform some general analysis of match results over the last 20 years, with an emphasis on home-ground advantage. This is followed by a case study on the effects of the COVID-19 pandemic on the estimated home-ground effects. Finally, we discuss some points regarding the stationarity of these competitions.

4.1 Australian Football League (AFL)

4.1.1 Background

The Australian Football League (AFL) is a professional men’s Australian rules football competition. The league debuted in its inaugural 1897 season as the Victorian Football League (VFL). As the competition expanded to states outside Victoria during the 1980s, the name officially changed to AFL in 1990. There are currently 18 teams from the mainland states of Victoria, New South Wales, Queensland, South Australia, and Western Australia.¹ It is one of the most popular sports in Australia, with the AFL Grand Final often one of the most-watched programs every year. The AFL is also the richest sports league by revenue in Australia.

There are several resources that explain the history and rules of AFL.² Notably, the league dictates the “Laws of Australian Football” and the various amendments made each season.³ At a basic level, each match is played by two teams over 20-minute quarters, where players can score points by sending the ball⁴ through the opposing team’s goalposts for a goal (6 points) or behind (1 point). Ties are a possible outcome, though finals games are played into overtime until a winner is decided.⁵

As our models only require the outcome and advantage indicator for each match, the season structure is more important to cover here. The premiership season, also called the home-and-

¹Tasmania is set to join the league in 2028 as the Devils.

²e.g., the official AFL website: <https://www.afl.com.au/about-afl>

³The 2026 Laws document is 72 pages long, for example, so they can be quite extensive.

⁴football, footy

⁵This change was implemented in 2015, likely after the 2010 grand final between Collingwood and St Kilda was drawn and hence entirely replayed the following week to much dismay.

away season, currently runs for 25 rounds from March to August. The season has several special matches and rounds, such as the Indigenous Round, public holidays like ANZAC Day, and other famous derbies. During the premiership season, teams are ranked in a ladder by points allocated based on match outcomes. A team is granted 4 points for a win, 2 for a draw, and none for a loss or bye.⁶ The top 8 teams play in the finals series throughout September, culminating in the grand final to crown the season premiers.⁷

Many factors contribute to the creation of the fixtures each season. In the 2026 home-and-away season, for example, there are 18 teams to play 23 games each over 25 rounds — plus locations, times, and dates to determine — inherently leading to an **imbalanced schedule**. As explained by Smith (2024), the league has a dedicated scheduling team that prioritises things like maximising viewership, revenue, and fairness. Of course, individual clubs and venues may also have contracts that require a number of games to be played at a given venue each season. The numerous special matches and rounds also add more constraints to the timetabling problem. As only some teams clash more than once per season, the allocation of these extra matches is driven by club rivalries, current finals contenders, and weightings based on the previous season’s ladder. The fixture also attempts to balance the burden of travel time and fatigue for different teams, particularly those outside of Victoria, which affects scheduling of home games. Overall, despite the imbalanced schedule, the win matrices are still always strongly connected in each time block.

4.1.2 Dataset and EDA

To reiterate, we are first interested in doing some generic analysis on AFL match results from the past 20 years. Data was sourced from the “AFL Data Analysis” GitHub repository⁸ which itself scrapes the AFLTables website⁹. The latter is a comprehensive and frankly excellent online repository of AFL and Rugby League data, though there is no convenient method of downloading this data directly. The dataset contains match information from the initial 1897 VFL season to the 2025 AFL season, though the 2025 season was incomplete when this data was collected. In future, it would be better to collect data from an automatically updated resource such as the R package `fitzRoy` (Day et al., 2026). However, as we are analysing the 2005–2024 match results, the current dataset is sufficient. Note that while finals series results

⁶Several tie-breaking criteria are used when necessary, mainly the “Percentage” which is based on ratio of points scored and conceded.

⁷The finals structure was changed for 2026 onwards from a final eight to final ten system. Now, only the top 6 teams are locked in and places 7-10 play a “wildcard” round for progression to the elimination finals. See https://en.wikipedia.org/wiki/AFL_finals_systems.

⁸<https://github.com/akareen/AFL-Data-Analysis>

⁹https://afltables.com/afl/afl_index.html

are included, preseason or practice matches are not.

We additionally require data about the clubs themselves. Auxiliary data files were manually created by compiling information from sources like Wikipedia¹⁰ and Austadiums¹¹. This contains, for instance, each teams’ city, state, and home-ground. However, which tenants belong to which venues seem to vary somewhat between sources. There is some inconsistency, for example, as to whether Essendon and Carlton have tenure at the MCG in addition to their primary ground at Docklands Stadium. This matter is further complicated by venues changing names due to sponsorships; venues being temporarily closed, demolished, and opened over time; teams changing home venues; and commercial deals which entail teams playing a certain number of matches at certain venues each season. This is summarised in Table 1.

Of the issues highlighted above, the most subjective is certainly the handling of these commercial deals and hence the concept of “secondary” home grounds. One of the better known contracts is with Hawthorn, who have played at York Park in Launceston, Tasmania since 2007 in a deal with the Tasmanian Government. At first, it seems only reasonable to denote these games as neutral since Hawthorn is a Melbourne-based team. A strong argument can be made, however, to treat these games with HGA in favour of Hawthorn. At time of writing, the Hawks have a win rate of over 76% at York Park, the single best record of any team across all venues.¹² In addition, the sponsorship has consistently brought Melbourne-based Hawks fans to Launceston for the matches and has increased Hawks fandom among Tasmanians. Things are complicated again, however, considering the impending introduction of the Tasmania Devils to the league (Scanlon, 2025).

Admittedly, the example above is somewhat cherry-picked. In reality, most of the dozens of venue deals seem to last only a few years and do not spark particularly strong fandoms among the local community. For instance, before the Gold Coast Suns joined the AFL as a tenant of Carrara Stadium, the venue had a complicated history of short deals. North Melbourne was supposed to play three home games per season at Carrara from 2007–2009; however, after the club turned down a \$100 million offer from the League to relocate to the Gold Coast in 2008, crowd attendance was expectedly poor.¹³ As a more perplexing example, Cazaly’s Stadium in Cairns saw both Richmond (2011–2013) and the Western Bulldogs (2014–2017) as tenants via contracts, though they only ever played against the Gold Coast Suns. To avoid debating case-by-case whether each commercial deal effects a HGA, we denote all of these sponsored games

¹⁰https://en.wikipedia.org/wiki/List_of_Australian_Football_League_grounds

¹¹<https://www.austadiums.com/sport/comp/afl/teams>

¹²See <https://afltables.com/afl/venues/overall.html>

¹³The League released the contract soon after and the 2009 home games were instead hosted by Carlton, Richmond, and St Kilda.

Table 1: Primary home venues of AFL teams over 2005–2026.

Team	State	Home Ground	Years
Adelaide	SA	Football Park * Adelaide Oval	1991-2013 2014-
Brisbane Lions	QLD	The Gabba	1997-
Carlton	VIC	Princes Park Docklands Stadium MCG	1897-2005 2005- 2005-
Collingwood	VIC	MCG	2000-
Essendon	VIC	MCG Docklands Stadium	1992- 2000-
Fremantle	WA	Subiaco Oval * Perth Stadium	1995-2017 2018-
Geelong	VIC	Kardinia Park	1941-
Gold Coast †	QLD	Carrara Stadium	2011-
GWS Giants ‡	NSW	Sydney Showground	2012-
Hawthorn	VIC	MCG	2000-
Melbourne	VIC	MCG	1889-
North Melbourne	VIC	Docklands Stadium	2000-
Port Adelaide	SA	Football Park * Adelaide Oval	1974-2013 2014-
Richmond	VIC	MCG	1965-
St Kilda	VIC	Docklands Stadium	2000-
Sydney	NSW	SCG Stadium Australia	1982- 2002-2015
West Coast	WA	Subiaco Oval * Perth Stadium	1995-2017 2018-
Western Bulldogs	VIC	Docklands Stadium	2000-

* Demolished and redeveloped

† Debuted in 2011

‡ Debuted in 2012

as neutral. This is justifiable as they only account for around 6% of matches over 2005–2024.

Data validation, cleaning, and EDA was done before converting to win matrices. The raw dataset was read into an intermediate DataFrame with the home and away teams, scores, a Boolean flag for neutral venue, and the year. Cleaning involved resolving aliases for teams and venues,¹⁴ identifying and correcting missing data,¹⁵ and other miscellaneous tasks. There are some key facts to note for the 2005–2024 matches:

- In total, 3970 games were played over this 20-season period.
- Two teams were introduced: the Gold Coast Suns in 2011 and GWS Giants in 2012.
- The premiership season was a 22-game season until becoming a 23-game season in 2023. The number of rounds per season varied from 22 to 25.
- In 2010, an additional game was played due to the Grand Final Replay between Collingwood and St Kilda.
- In 2015, the Round 14 match between Adelaide and Geelong was cancelled due to the death of Crows coach Phil Walsh just two days prior.
- In 2020, the COVID-19 pandemic saw the season suspended after Round 1 until resuming in June with only a 17-game season. This is discussed more in 4.3.
- Gather Round was the additional round introduced in 2023 onwards, where all home-and-away games are played in one city. Adelaide hosted this in 2023 and 2024.
- The special Opening Round was introduced in 2024 but did not add any more games per season overall.
- 26 unique venues were used over this period, with the MCG hosting the most games (961) followed closely by Docklands Stadium (891), as expected.
- Only 36 games (about 0.91%) were tied.

Overall, 1,153 games (about 29%) were categorised as neutral $v_{ij} = 0$. Of these, 673 matches were played by co-tenants (both teams home) and 480 matches were truly neutral (both teams away). For the former case, the MCG and Docklands Stadium were obviously the most common, where about 40% and 23% of games at these venues were between co-tenants, respectively. The latter case consists of: various commercial deals; matches outside Victoria due to COVID-19 restrictions; matches at multiple venues in Adelaide for Gather Round (2023 onwards); grand final matches almost always played at the MCG; and a handful of special exhibition matches like in China and New Zealand.

¹⁴For example, North Melbourne was called the Kangaroos Football Club until 2008.

¹⁵For example, the scraper seems to have a bug when parsing drawn final matches played into extra time, such as the 2007 semi-final for West Coast v Collingwood.

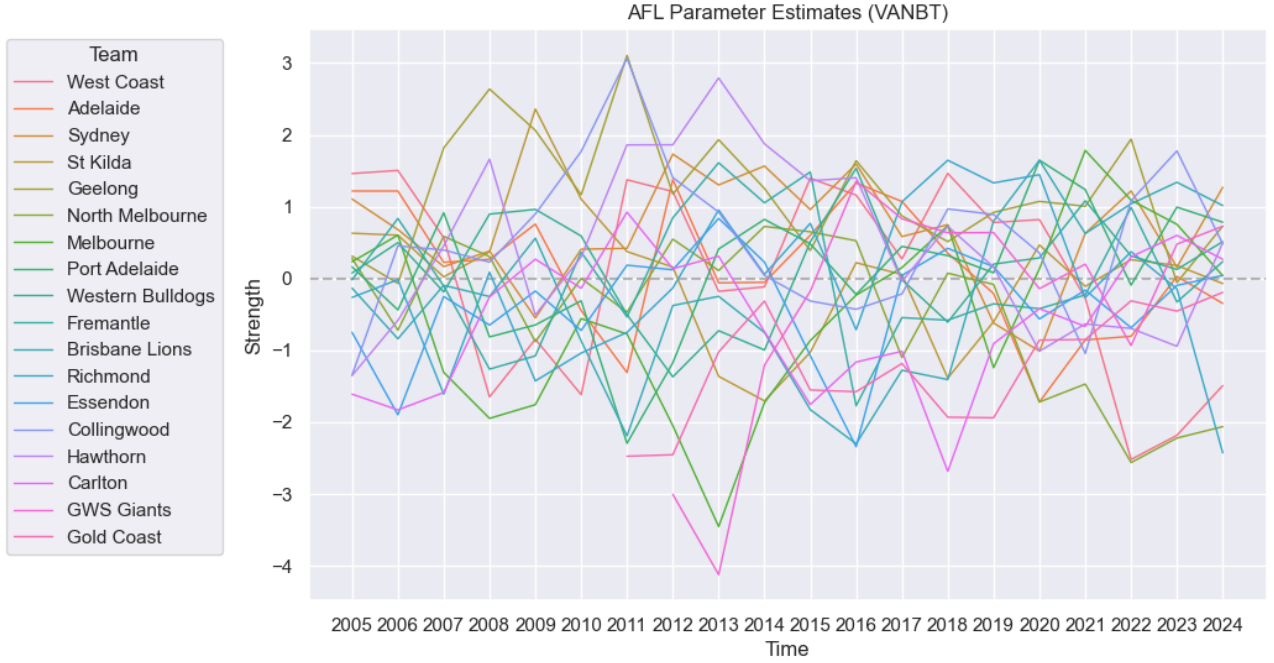


Figure 8: Parameter estimates for the AFL (2005–2024), vanilla BT model.

4.1.3 Results: Basic modelling

We first fit our ‘default’ suite of models: VAN, CHA, and TSA. These do not require us to specify a relationship matrix, so give a good place to start. Figure 8 shows the estimated team strengths over time for the vanilla model. Recall that, in our nonparametric dynamic approach, this model essentially corresponds to fitting T independent Bradley–Terry models, one on each season. There are $p = 327$ strength parameters to estimate for the $I = 18$ teams over $T = 20$ years, excluding those inactive. By default, the global reference team is Adelaide; however, to aid interpretability across time, we have applied mean-centering in all plots such that strengths are instead relative to the average team in each season. Unsurprisingly, though, the dynamic plots are still somewhat difficult to read due to the many teams and length of time being analysed. As such, plots of the average strengths are also provided, as in Figure 9.

Although the vanilla model primarily serves as a baseline for the others, we can still observe some interesting results in Figures 8 and 9. Geelong and Gold Coast are the strongest and weakest teams on average, respectively, and both by considerable margins. For the individual seasons, the highest estimated strength was Geelong in 2011, who won 19 out of 22 home-and-away games and subsequently swept the finals to claim the premiership. In very close second was Collingwood in the same season. They had a better home-and-away record of 20–2 to claim the minor premiership¹⁶ — with both losses to Geelong — and then lost to Geelong once more in the grand final. According to the vanilla model, the probability that Geelong beats Collingwood in 2011 is about 51.1%.

¹⁶the team that finishes on top of the ladder in the regular season

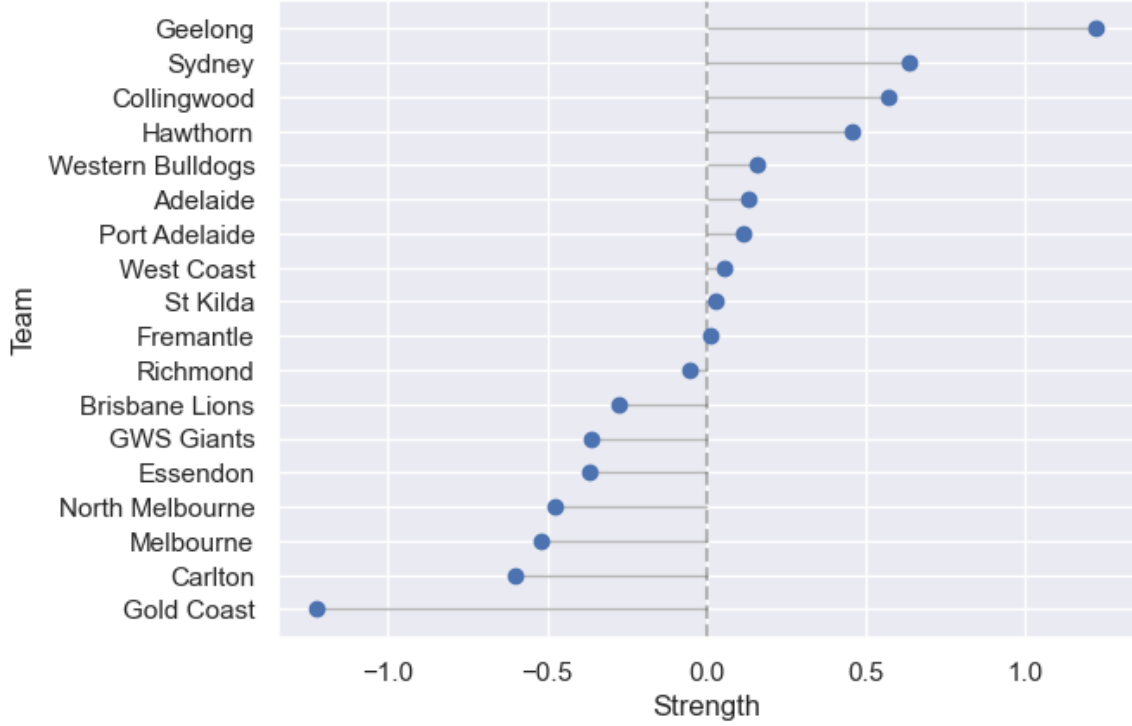


Figure 9: Average estimated team strength for the AFL (2005–2024), vanilla BT model.

The worst estimated strength was for the GWS Giants in 2013, who in their second season unfortunately claimed their second wooden spoon¹⁷ after losing all but one game. Their single victory was over Melbourne with a similarly shocking record of 2–20, making them the second worst estimated strength over this period. The largest season-wise difference in strength estimates was also in 2013, where premiers Hawthorn have a 99.9% chance to beat the GWS Giants. Importantly, Figure 8 shows how volatile the league can be; further discussion on this is left for Section 4.4.

The results for the common home-ground model are shown in Figures 10 and 11. The model adds a single home-ground parameter which is both common to all teams and assumed constant over time. As indicated on the scale bar, this was estimated to be $\hat{\alpha} = 0.533$ with SE 0.048, i.e., there is a $\exp(0.533) \approx 1.70$ times increase in the odds in favour of the home team. That is, for two evenly matched teams, the home team has a 63% chance of winning on average. This is slightly higher than the ratio of home wins to losses, which is about 58%. Compared to the vanilla model, the estimated team strengths appear largely the same after factoring for home advantage. The model suggests that there is significant evidence of a common home-ground effect in favour of none at all ($D = 126.5$ on 1 df, $p < 0.001$). This is an expected result and corroborates findings in previous work.

The TSA model supposes each team has a unique home-ground advantage, but still constant

¹⁷a fake “award” given to last place

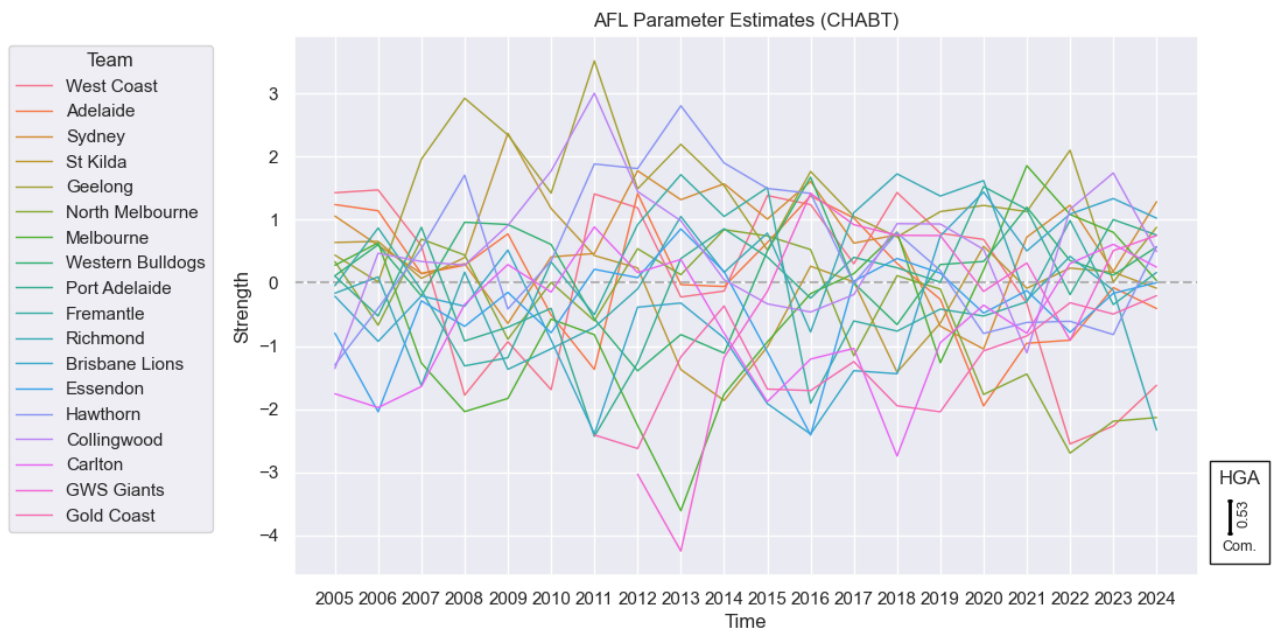


Figure 10: Parameter estimates for the AFL (2005–2024), common HGA.

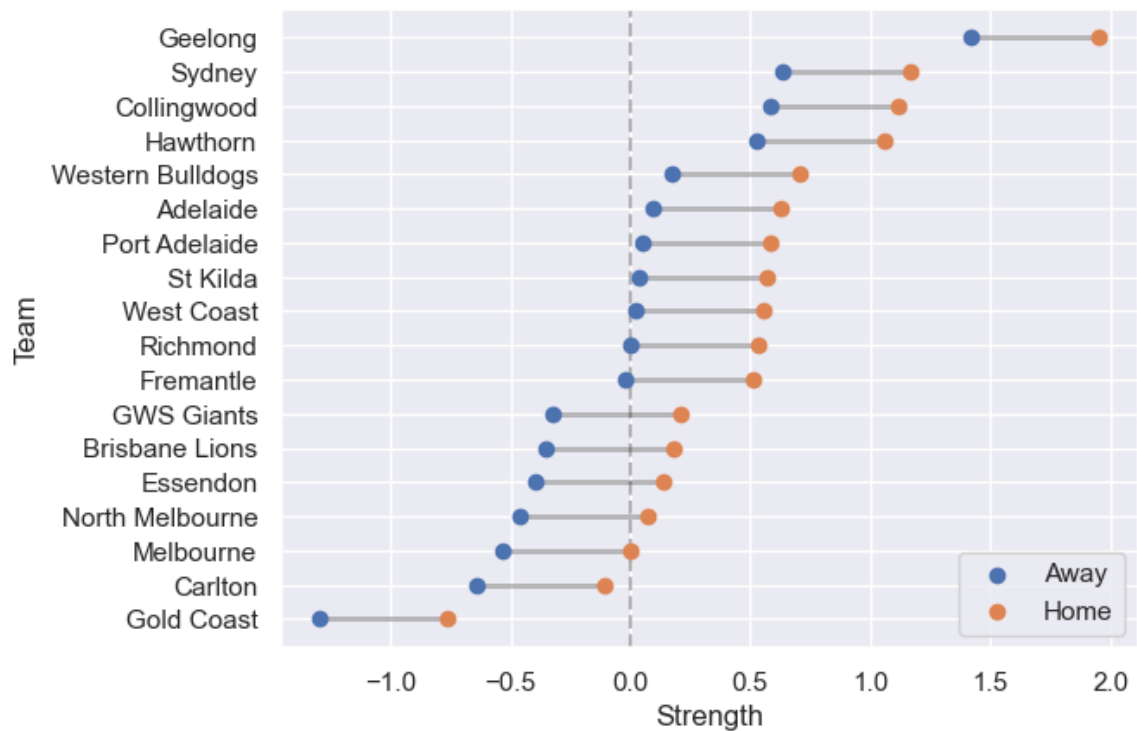


Figure 11: Average estimated team strength for the AFL (2005–2024), common HGA.

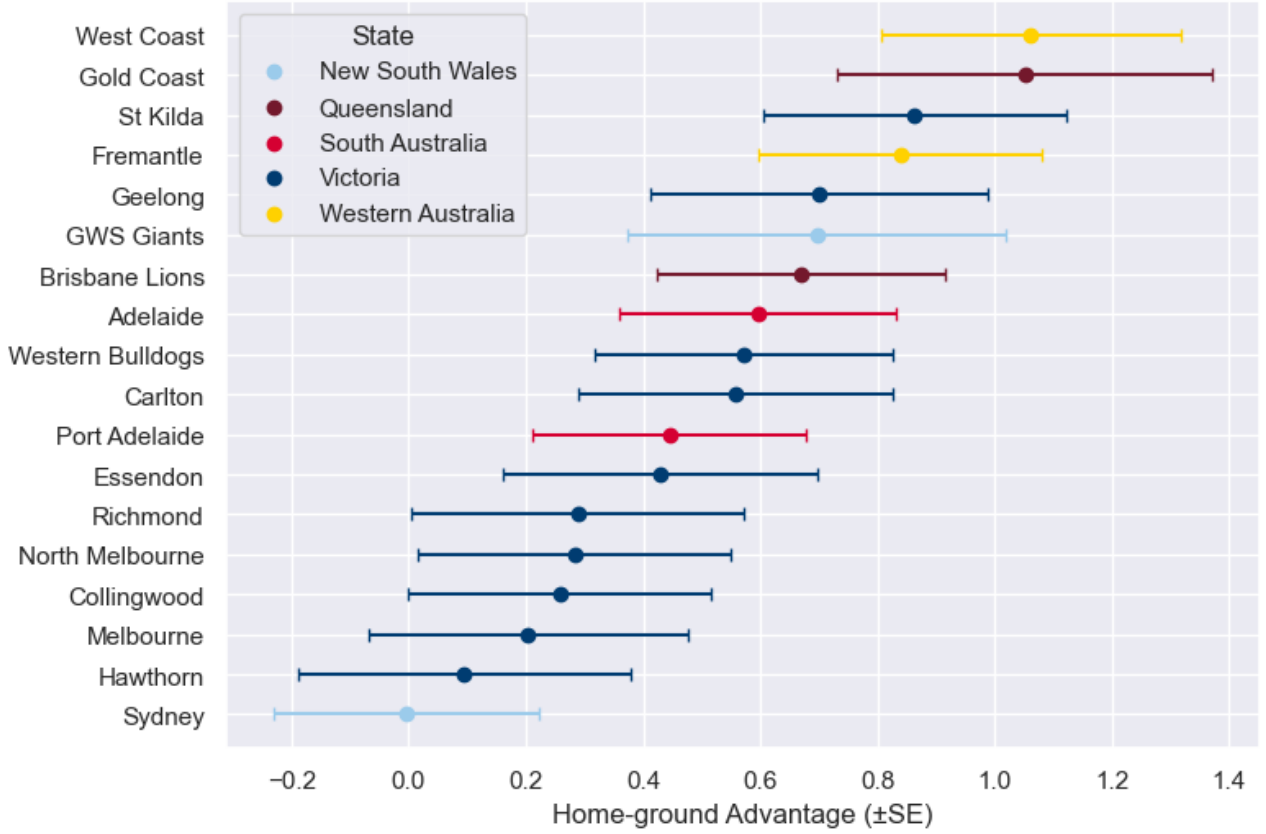


Figure 12: Team-specific HGA estimates for the AFL (2005–2024), coloured by state.

over time. The strength estimates are again similar to those in the vanilla model, so we do not include them here. Instead, the team-specific HGAs are plotted in Figure 12. We can observe quite a spread of home-ground effects. The largest advantage is a $\exp(1.062) \approx 2.89$ times increase in the odds for West Coast when playing at home. In general, the results appear to indicate that non-Victorian teams have a greater advantage at home than Victorian teams. However, there is no evidence to suggest the TSA model provides a better fit to this AFL data than the CHA model ($D = 22.1$ on 17 df, $p > 0.1$). The CHA model also has the lowest AIC and BIC of the three fitted thus far, supporting its fit adequacy.

4.1.4 Results: Nominal home-ground

Before we look for differences in HGA across states, we need to consider **nominal home effects**. The AFL fixture designates a “home” team in each match, even when we have $v_{ij} = 0$. Firstly, we must ask ourselves whether we should treat the 17% of games between co-tenants as neutral. It is not unreasonable to suggest that some order effects exist in favour of these nominal home teams, despite nullifying any edge in venue familiarity. Even for co-tenants, the “home” team usually has greater control over reserved-seat allocations which could require visiting members to relocate. Besides influencing crowd composition, the home team may benefit from preferred amenities like changerooms, kit and bench choices, and other administrative or

revenue perks like control over match-day atmosphere (music, branding, announcements). In 2023, for instance, Collingwood received much backlash from fellow MCG clubs in response to a preferential deal granting the Magpies with unique advantages in seating allocation and consistent dressing rooms (Fox Sports, 2023).

By allowing for nominal home advantages, the definition of a “neutral” game becomes quite blurry. Should we maintain that a match is still neutral when both teams are away? Recall that these minority “truly neutral” games consist of a few different scenarios: exhibition matches, COVID-19 arrangements, finals series matches, and commercial deals for out-of-state venues like those in Tasmania, Canberra, Darwin, Alice Springs, and Ballarat. For example, the argument might hold for finals series games at the MCG, where the League allocates equal club tickets and controls administrative features like advertising. Furthermore, in out-of-state venues there is often no large home-member presence let alone home club ticketing priority, hence crowd advantages are perhaps negligible. As discussed before, however, the presence of HGA at sponsored matches can be debated both ways, as in the case of Hawthorn where a home advantage in Launceston has developed over time.

Yet another issue is the existence of 131 seemingly “swapped” matches in the dataset, where the nominal home team is not a tenant of the venue but the opponent is! These games almost entirely contradict our assumption of home advantage belonging to the tenant. It is unclear whether these matches are indeed quirks of the fixture creation or simply errors in the dataset, or a combination of both. On one hand, it makes some sense that Geelong plays a couple nominal home games at the MCG per season — even against MCG tenants — due to their relative proximity to the venue and large fanbase in Melbourne. It is also understandable that MCG and Docklands tenants play nominal home games at each other’s grounds, having a total 9 tenants between them yet the two venues are only 3km apart.¹⁸ On the other hand, though, it makes little sense for out-of-state teams to be given a “home” game when playing MCG tenants at the MCG, and vice versa. As these swapped records only constitute a handful of matches, and a majority of them make at least some sense to be nominal home games, we leave them unaltered in the analysis here.

To summarise, the allocation of home games in the fixture sometimes conflicts with our definition of neutral games. To investigate this, let us disregard the advantage indicators v_{ij} and take the nominal home team as that with the home advantage. In other words, the physical venue of the match no longer dictates the direction of home advantage, but rather the nominal home team is assumed to always benefit from the effect even in situations we previously considered

¹⁸by far the closest pair of primary home venues in the AFL

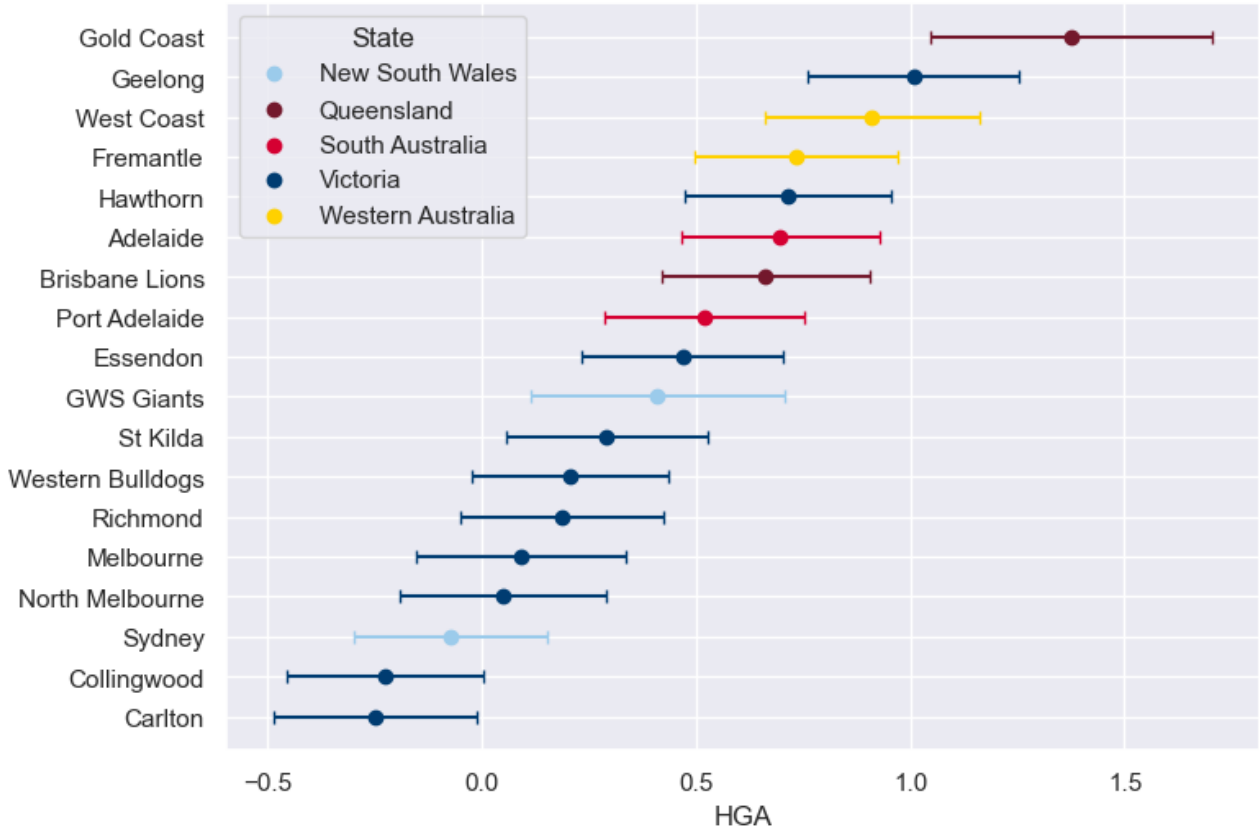


Figure 13: Team-specific HGA estimates for the AFL (2005–2024), coloured by state, under the nominal home team assumption.

neutral. We therefore re-fit our HGA models to look for any differences in home-ground effects with this alteration.¹⁹

The CHA model produces an estimated home-ground advantage of $\hat{\alpha} = 0.400$ with SE 0.039 for the nominal home team. This is still a significant effect ($D = 106.4$ on 1 df, $p < 0.001$) but weaker than the HGA found earlier when neutral games were factored, as expected. The estimated HGAs for the TSA model are shown in Figure 13. The general trend appears the same as before, as the plot suggests non-Victorian teams have higher individual advantages. Most of the Melburnian teams' HGA estimates have dropped considerably compared to Figure 12, which agrees with the idea that these co-tenant games do not particularly benefit the nominal home team. In contrast, Hawthorn is the only Melbourne team to instead see an increase in estimated HGA, likely due to the inclusion of their long and successful record at York Park as home games. Interestingly, the TSA model now offers a better fit than the CHA model ($D = 49.8$ on 17 df, $p < 0.001$) under this nominal home team approach. Of the five models fit thus far, the CHA model with neutral matches (Figure 10) still has the lowest AIC and BIC.

¹⁹The VAN model will be the same as it does not estimate HGA parameters

Table 2: Estimated state-specific HGAs for the AFL (2005–2024).

State	Allow Neutral Games		Nominal Home Team	
	HGA	[SE]	HGA	[SE]
New South Wales	0.356	0.183	0.422	0.182
Queensland	0.687	0.185	0.835	0.170
South Australia	0.515	0.174	0.610	0.158
Victoria	0.425	0.089	0.183	0.064
Western Australia	0.939	0.184	0.812	0.166

4.1.5 Results: Groupings

From our basic analysis, we observed that the AFL data is best explained by a common HGA when accounting for neutral games. Instead assuming the nominal home team always has the HGA, the team-specific model explained the data best and indicated that Victorian teams have weaker home effects. We can further investigate this idea with our hierarchical BT models by defining relationship matrices based on the home state of each team. Though these divisions are not fundamentally built into the AFL season structure like in the NBA, differences in team locations and the disproportionate number of Victorian teams certainly still influence the fixture. We will first examine differences across states and then specifically compare Melburnian and non-Melburnian teams.

To compare HGAs across the five states in the AFL, we simply need to specify an asymmetric relationship matrix where teams in the same state use the same HGA parameter. We are still interested in comparing the assumptions for allowing neutral matches against always taking the nominal home team, so have fit two sets of models. The estimated HGAs are presented in Table 2. The results suggest that WA and QLD teams have the strongest home-ground effects, and Victoria the weakest, which is something we observed in the team-specific models prior. In particular, this seems to support the idea of a travel burden, as WA and QLD venues are geographically furthest from all others. Furthermore, the Victorian HGA is much weaker when including nominal home games, while the other states stay roughly similar across the two assumptions. The state-specific model provides no better fit to the data than the CHA model when allowing neutral matches ($D = 7.15$ on 4 df, $p > 0.1$); though it does improve over CHA under the nominal home assumption ($D = 21.8$ on 4 df, $p < 0.001$), and the nominal TSA model still also remains an improvement ($D = 28.0$ on 13 df, $p < 0.01$).

Lastly, we group the teams simply as either Melbourne or non-Melbourne. To examine these differences, we use a within-vs-across model. Recall that half the teams in the AFL are from Melbourne; though Geelong is also in Victoria, the city is over 60km away from Melbourne so we assume the travel burden is still substantially different. It does not make much sense

Home	Adelaide	0	1	2	2	2	1	1	1	1	2	2	2	1	2	2	1	1	2
	Brisbane Lions	1	0	2	2	2	1	1	1	1	2	2	2	1	2	2	1	1	2
	Carlton	4	4	0	3	3	4	4	4	4	3	3	3	4	3	3	4	4	3
	Collingwood	4	4	3	0	3	4	4	4	4	3	3	3	4	3	3	4	4	3
	Essendon	4	4	3	3	0	4	4	4	4	3	3	3	4	3	3	4	4	3
	Fremantle	1	1	2	2	2	0	1	1	1	2	2	2	1	2	2	1	1	2
	Geelong	1	1	2	2	2	1	0	1	1	2	2	2	1	2	2	1	1	2
	Gold Coast	1	1	2	2	2	1	1	0	1	2	2	2	1	2	2	1	1	2
	GWS Giants	1	1	2	2	2	1	1	1	0	2	2	2	1	2	2	1	1	2
	Hawthorn	4	4	3	3	3	4	4	4	4	0	3	3	4	3	3	4	4	3
	Melbourne	4	4	3	3	3	4	4	4	4	3	0	3	4	3	3	4	4	3
	North Melbourne	4	4	3	3	3	4	4	4	4	3	3	0	4	3	3	4	4	3
	Port Adelaide	1	1	2	2	2	1	1	1	1	2	2	2	0	2	2	1	1	2
	Richmond	4	4	3	3	3	4	4	4	4	3	3	3	4	0	3	4	4	3
	St Kilda	4	4	3	3	3	4	4	4	4	3	3	3	4	3	0	4	4	3
	Sydney	1	1	2	2	2	1	1	1	1	2	2	2	1	2	2	0	1	2
	West Coast	1	1	2	2	2	1	1	1	1	2	2	2	1	2	2	1	0	2
	Western Bulldogs	4	4	3	3	3	4	4	4	4	3	3	3	4	3	3	4	4	0
		Adelaide	Brisbane Lions	Carlton	Collingwood	Essendon	Fremantle	Geelong	Gold Coast	GWS Giants	Hawthorn	Melbourne	North Melbourne	Port Adelaide	Richmond	St Kilda	Sydney	West Coast	Western Bulldogs
		Away																	

Figure 14: Relationship matrix for the within-vs-across model where teams are grouped as either from Melbourne or not. Any symmetry constraints are relaxed, i.e., such that $\alpha_1 \neq \alpha_3$ and $\alpha_2 \neq \alpha_4$. For instance, the HGA used when a Melbourne team is home against a non-Melb team (α_4) is not necessarily the same as that used when the situation is swapped (α_2).

Table 3: Estimated within-vs-across HGAs for teams grouped as either Melbourne (Melb) or non-Melbourne (non-Melb) for the AFL (2005–2024).

Home	Away	HGA	[SE]
Melb	Melb	0.054	0.070
Melb	non-Melb	0.305	0.131
non-Melb	Melb	0.746	0.129
non-Melb	non-Melb	0.516	0.075

to constrain the HGA for Melb–Melb games to be the same as non-Melb–non-Melb, hence we are essentially using different parameters depending on who travels. As such, we only use the nominal home team assumption here. Otherwise, since most Melb-vs-Melb team matches are classified as neutral, there would be very little data to estimate the HGA parameter for this case. To help visualise this model, its relationship matrix is shown in Figure 14.

The modelling results are given in Table 3 and show some differences between the four rela-

Table 4: Some HBT model fit results for the AFL (2005–2024).

Model	p	Type	-LLH	AIC	BIC
Vanilla	327	-	2068.7	4791.4	6847.1
Common HGA	328	True home	2005.4	4666.9	6728.9
State-specific HGA	332	True home	2001.9	4667.8	6754.9
Team-specific HGA	345	True home	1994.4	4678.8	6847.7
Common HGA	328	Nominal home	2015.5	4686.9	6748.9
State-specific HGA	332	Nominal home	2004.6	4673.2	6760.3
Team-specific HGA	345	Nominal home	1990.6	4671.2	6840.0
Melbourne-based within-vs-across	331	Nominal home	2001.8	4665.7	6746.5

tionship levels. Notably, the HGA in a match between two Melbourne teams is essentially zero, again supporting the claim that home effects are largely negligible for such games. Conversely, the strongest HGA is seen when non-Melbourne teams host Melbourne teams. In the other two cases — when a non-Melbourne team is away — milder HGAs are estimated. This may suggest that Melbourne-based teams are the most negatively effected by travel burdens, and perhaps that the effects of crowd cheer are stronger for non-Melbourne teams playing at home. Compared to the CHA model with nominal home assumption, this does again offer an improvement ($D = 27.3$ on 3 df, $p < 0.001$). Rather importantly, this model has the lowest AIC of all fit on the AFL data thus far.

4.1.6 Discussion

A variety of BT models have been fit to the AFL data to examine home-ground advantage. Based on our analyses of match results from 2005–2024, there is significant evidence to support the existence of HGA. Furthermore, we have seen that the magnitude of home effects can vary greatly across states and even teams, most prominently when comparing Melburnian teams to others. We have also considered differences between “true” home games — where we distinguish neutral matchups — and nominal home games — where the designated home team is presumptively given an advantage irrespective of the venue and opponent. Under the nominal HGA assumption, we observed that the advantage for Victorian teams becomes noticeably weaker than when estimating using only “true” home games, likely weighed down by the relatively negligible order effects in games between co-tenants. Overall, the data seems well explained by a common HGA under the “true” home game assumption; however, the data with nominal home assumption is fit adequately by even a team-specific HGA model. Fit statistics for the models are collated in Table 4.

These results corroborate with previous work, the wider literature, and general knowledge for most sports fans. Using the nominal home approach, Jones (2021) found a common HGA was

supported by a BT model fit to the 2010–2019 AFL results, assuming that team abilities were constant over time. Linear score margin models by Clarke (2005) also suggested differences in HGA for individual clubs, based on 1980–1998 AFL data. In particular, Clarke finds “a strong isolation effect” whereby non-Victorian teams have significantly larger advantages than Victorian clubs, concluding that “crowd effects and ground familiarity are a major determinant of home advantage.” This especially aligns with our analysis when grouping teams into Melbourne and not, though we have parameterised further based on the group of the opponent too. Soffner and Leonelli (2024) seem to delineate between AFL matches played as the nominal home team and matches actually played at home, as we have. They also find significant benefits for being the home team, albeit only assuming a common HGA, with a rolling-window BT model approach on 2015–2023 data.

4.2 National Rugby League (NRL)

4.2.1 Background

The National Rugby League (NRL) is a professional men’s rugby league competition in Australia and New Zealand. There are currently 17 teams from New South Wales, Queensland, Victoria, ACT, and New Zealand.²⁰ The League has seen 25 teams since its inaugural 1998 season, though only 11 have played in every year to date. It is another prominent spectator sport — with fans growing across Oceania — and the second richest league in Australia behind the AFL. The NRL and AFL are frequently subjects of the long-standing debate on which sport is most popular in Australia (Harcourt, 2025).

The competition traces back to 1908 as the New South Wales Rugby League (NSWRL), originally established as a professional breakaway from rugby union in Sydney. After decades of steady growth, the competition underwent a period of significant geographical expansion during the 1980s, culminating in the Australian Rugby League (ARL) taking control in 1995. However, commercial tensions bubbled in the mid-1990s in the wake of the tumultuous “Super League war.” Specifically, conflict over broadcasting rights saw Rupert Murdoch’s News Corporation form a rival “Super League” to challenge the newly established ARL, ultimately becoming a struggle for control of the sport during its rapid nationalisation. The schism resulted in two separate, concurrent premiership competitions being played in 1997. This was (unsurprisingly) a very nuclear decision, stretching spectators and sponsors thin and subsequently pushing many clubs into financial trouble. The rivalry quickly reached a peace agreement before the 1998

²⁰The Perth Bears and PNG Chiefs are expected to debut in 2027 and 2028, respectively.

season, merging the competitions to form the National Rugby League (NRL), which has since remained the preeminent rugby league competition in the southern hemisphere.

A standard NRL match consists of two teams playing over 40-minute halves. Teams compete to score points in a few ways: a try (4 points), a conversion (2 points) following a try, a penalty goal (2 points), or a field goal (1 point). Games tied after full time proceed to the “golden point” extra time, a sudden death format over 5-minute halves where the first team to score any points wins. Should neither team score in this period, then the match is recorded as a draw.²¹ It has been used in the NRL since 2003.²²

The NRL currently runs a 24-match regular season spanning 27 rounds, from March to September. There are some themed rounds to note like ANZAC Round and Magic Round, as well as test matches played separately from the premiership season like against Pacific Island nations and the famous State of Origin series. Teams are ranked on a ladder based on match results: two points for a win or bye, 1 point for a draw, and none for a loss.²³ The top 8 teams at the end of the regular season progress to the finals series, with the grand final deciding the season’s premiers. As in the AFL, the season fixture is inherently imbalanced. The League must schedule matches to balance matchup requirements, player rest time, special rounds, logistical disruptions like State of Origin, and viewership maximisation. Just as the AFL’s history has led to half the teams calling Melbourne home, currently 9 of the 17 NRL teams are based in the Greater Sydney area. The travel burden hence disproportionately affects teams like the North Queensland Cowboys and New Zealand Warriors.

4.2.2 Dataset and EDA

We wish to perform similar analyses on the NRL match results over the past 20 years. Unfortunately, this data seems less conveniently available than the AFL data. We have chosen to use the Kaggle dataset by Mitchell Lohmann²⁴ which contains match data from 1990-2025. The dataset itself is scraped verbatim from the NRL draw website.²⁵ While close to meeting our needs, it is however missing venue information for pre-1998 data. There are also some issues with venue names changing over time due to sponsorships and aliases, making it difficult to track home grounds. For the sake of scope, we will use this data with the nominal home assumption. In future, we would ideally scrape more consistent data sources like AFLTables²⁶

²¹From 2015, the golden point is only activated in final series games if scores remain tied after an additional 10 minutes.

²²Golden point was added for drawn AFL grand finals in 2016–2019, but never used.

²³Again, ties are broken by metrics like match points differential

²⁴<https://www.kaggle.com/datasets/mitchelllohmnn/nrl-fixtures-1990-2025>

²⁵<https://www.nrl.com/draw>

²⁶https://afltables.com/rl/rl_index.html

or the Rugby League Project²⁷ as both appear very well maintained. The current dataset is nonetheless sufficient to explore home-ground effects in the NRL, something sorely missing from the current literature.

Additional club information was manually compiled again from Wikipedia²⁸ and Austadiums²⁹. As before, there are inconsistencies with the tenants across sources. Furthermore, some NRL teams also participate in commercial deals to host games outside their primary home-ground(s). Note that we do not consider preseason games nor special cups like State of Origin. A summary of the teams and their home grounds can be found in Table 5.

After data cleaning, some EDA was performed on the NRL results. We make a note of the following facts about the data:

- In total, 3988 games were played over the 20-season period. Interestingly, this is only 18 more than the AFL dataset.
- Two teams were introduced: the Gold Coast Titans in 2007 and the Dolphins in 2023.
- The NRL has consistently had a 24-game regular season over this period. The only exception was of course the 2020 season, where the COVID-19 pandemic postponed the season for two months until continuing in May as a 20-game campaign.
- The number of rounds per season has varied from 25-27, though only by changing the number of byes rather than having additional matches.
- At least 50 unique venues have been used since 2005, about double that of the AFL.³⁰
- Suncorp Stadium (Lang Park, Brisbane) has held the most games (345), followed by Accor Stadium (Stadium Australia, Sydney) in second (262). The latter venue has hosted all NRL grand finals since 1999, except in 2021 when it was hosted at Lang Park due to a COVID-19 lockdown in NSW.
- Only 16 games (about 0.40%) were tied (after golden point). Although the dataset does not state which games went into golden point, it was found that at least 175 matches (about 4%) have used it since its introduction in 2003.

4.2.3 Results: Nominal home-ground

The vanilla model was fit to the NRL data from 2005–2024 to give us a baseline for investigating home-ground. Figure 15 shows the estimated team strengths in each season and Figure 16 shows the average strengths over the 20 seasons. The basic suite of models are fitted first to investigate

²⁷<https://www.rugbyleagueproject.org/competitions/nrl/data.html>

²⁸https://en.wikipedia.org/wiki/National_Rugby_League#Teams

²⁹<https://www.austadiums.com/sport/comp/nrl/teams>

³⁰The current dataset makes it hard to pin the true number due to its use of multiple aliases.

Table 5: Primary home venues of NRL teams over 2005–2026.

Team	State	Home Ground	Years
Brisbane Broncos	QLD	Lang Park	2003–
Canberra Raiders	ACT	Canberra Stadium	1990–
Canterbury-Bankstown Bulldogs	NSW	Stadium Australia Belmore Sports Ground	2005– 2015–
Cronulla-Sutherland Sharks	NSW	Endeavour Field * Jubilee Stadium	1968– 2020–2021
Dolphins ‡	QLD	Dolphin Stadium Lang Park	2023– 2023–
Gold Coast Titans †	QLD	Carrara Stadium Robina Stadium	2007 2008–
Manly Warringah Sea Eagles	NSW	Brookvale Oval	2003–
Melbourne Storm	VIC	Olympic Park Stadium * Melbourne Rect. Stadium	1998–2009 2010–
New Zealand Warriors	NZ	Mount Smart Stadium	1995–
Newcastle Knights	NSW	Newcastle Intl. Sports Centre	1988–
North Queensland Cowboys	QLD	Willows Sports Complex * North Queensland Stadium	1995–2019 2020–
Parramatta Eels	NSW	Parramatta Stadium * Stadium Australia Western Sydney Stadium	1986–2016 2017–2018 2019–
Penrith Panthers	NSW	Penrith Stadium * Western Sydney Stadium	1967–2024 2025–
South Sydney Rabbitohs	NSW	Sydney Football Stadium * Stadium Australia	2002–2005 2006–
St. George Illawarra Dragons	NSW	Wollongong Showground Jubilee Stadium Stadium Australia	1999– 2003– 2014–2017
Sydney Roosters	NSW	Sydney Football Stadium * SCG Sydney Football Stadium	1988–2018 2019–2022 2022–
Wests Tigers	NSW	Campbelltown Sports Stadium Leichhardt Oval Stadium Australia Sydney Football Stadium * Stadium Australia Western Sydney Stadium	2000– 2000– 2005–2008 2009–2013 2014–2018 2019–

* Demolished and/or redeveloped

† Debuted in 2007

‡ Debuted in 2023

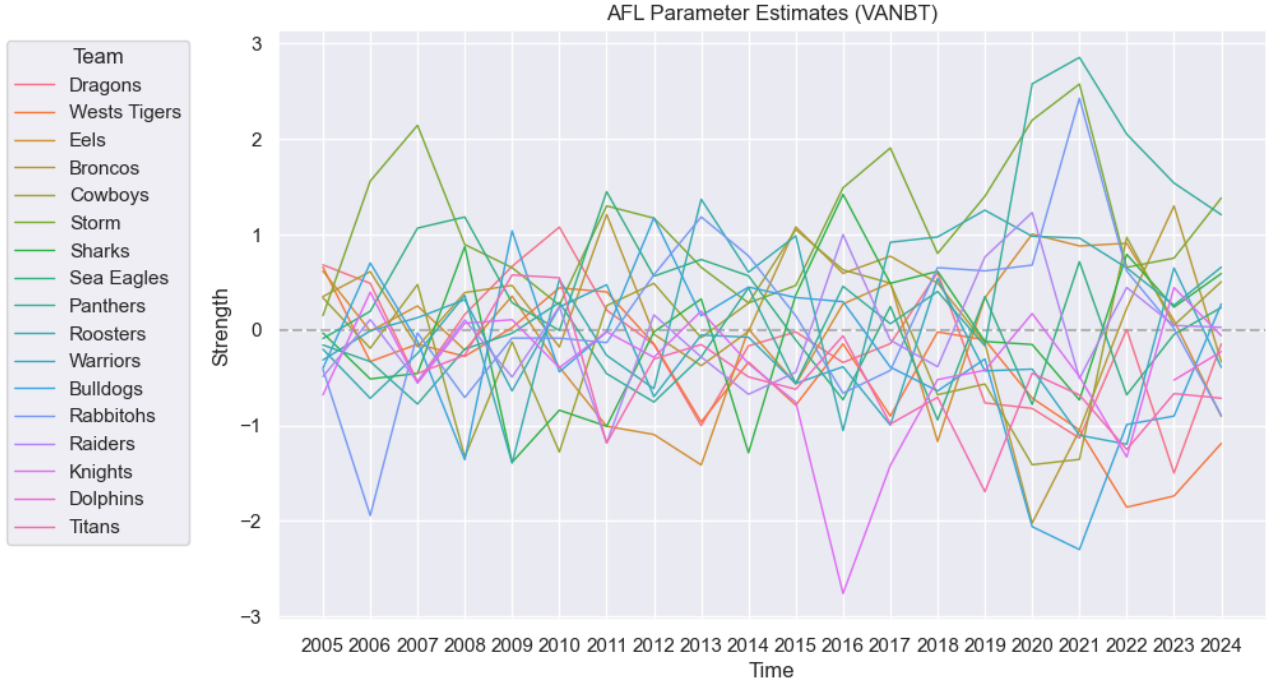


Figure 15: Parameter estimates for the NRL (2005–2024), vanilla BT model.

if the NRL exhibits a nominal home team advantage. There are $p = 300$ strength parameters to estimate for the $I = 17$ teams over $T = 20$ seasons, excluding inactive parameters. The Broncos are the default global reference team in this case, though mean-centring of the strengths is applied to all figures again. Before we quantify any nominal HGA, there are some results worth mentioning for the vanilla model.

Firstly, Figure 16 indicates that the strongest team on average is the Melbourne Storm, by quite a significant margin. The Storm were undoubtedly dominant, achieving 5 premierships, 6 runners-up, and 9 minor premierships since 2005.³¹ The remaining teams are relatively closer in average strength, with the Penrith Panthers in second and Gold Coast Titans at the bottom.³² Looking at the individual season strength estimates, the Panthers take the top two spots in 2021 and 2020, respectively. The club had a commanding record in these years affected by COVID-19, making the finals in both seasons and winning in 2021. The weakest estimated ability is the Knights in 2016, winning only 1 game all season.³³

Parameter estimates for the common HGA model are plotted in Figure 17 and averages in Figure 18. The CHA model produced an estimated home effect of $\hat{\alpha} = 0.348$ with SE 0.036. In

³¹Some context is relevant here. In 2010, it was revealed that the Storm had breached NRL salary caps from 2006–2010. The club received heavy punishments as a result of the scandal, including being stripped of their premiership titles over this period and losing all competition points in the 2010 season. The model is blind to these facts, however, and estimates the Storm’s ability purely based on their match results. See for instance https://en.wikipedia.org/wiki/Melbourne_Storm_salary_cap_breach.

³²Unfortunately, this means the Gold Coast AFL and NRL teams are both estimated to be the worst clubs in their respective leagues over the past 20 years.

³³This would be the second of three consecutive wooden spoons for the Knights.

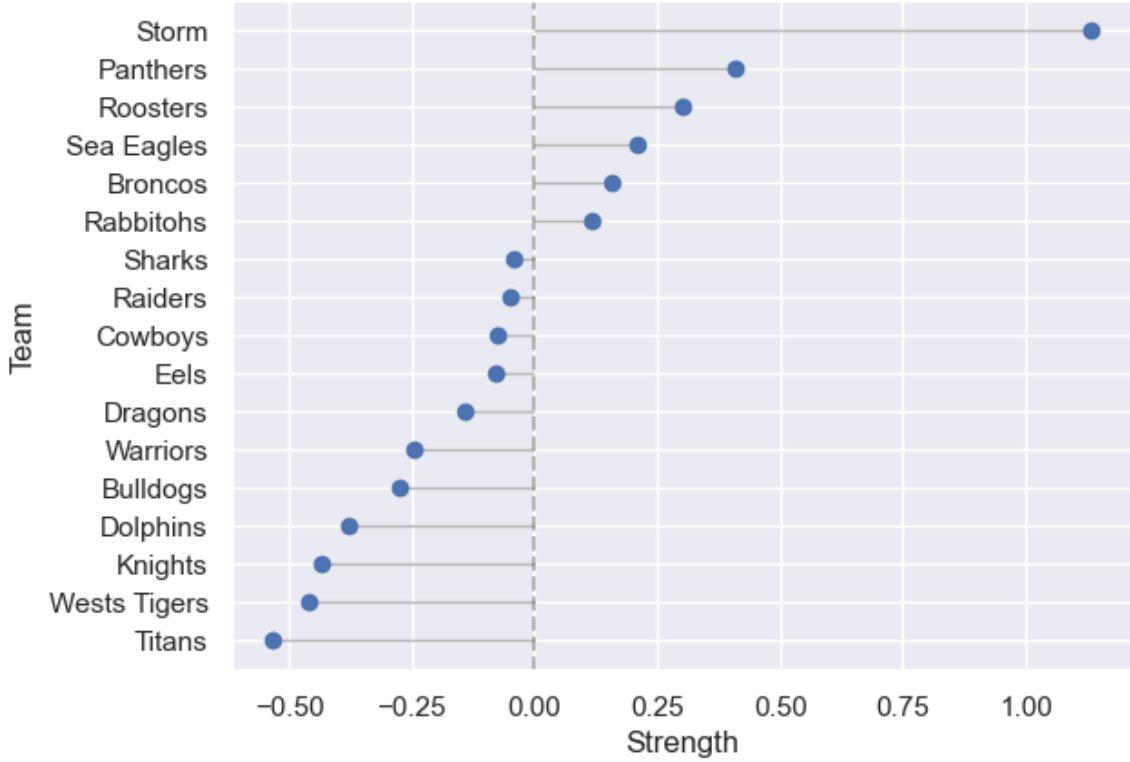


Figure 16: Average estimated team strength for the NRL (2005–2024), vanilla BT model.

other words, the home team enjoys a $\exp(0.348) \approx 1.42$ times increase in their odds of winning and hence has a 59% chance of beating an evenly matched opponent when at home. This is only slightly higher than the ratio of home wins based on the data, which is about 57%. As before, the strength estimates do not appear to change greatly from those estimated by the vanilla model. The addition of a common HGA is significant, with very strong evidence that the CHA model provides a better fit to the data ($D = 93.5$ on 1 df, $p < 0.001$).

The results of the TSA model are given in Figure 19. It seems that at least a few teams have certainly benefited from playing at their own grounds in the past 20 seasons. The St George Illawarra Dragons appear to have the strongest home advantage, seeing a $\exp(0.83) \approx 2.30$ times increase to their odds of beating an away opponent. This is interesting as the Dragons appear quite ordinary in terms of average estimated team ability; the same could be said of the Cowboys and Eels, judging by Figure 18. We observed a similar phenomenon in the AFL, where the Gold Coast Suns had the strongest (nominal) home advantage despite being the worst team in the league. While the Suns are one of two Queensland teams in the AFL — and hence arguably suffer from relatively higher travel burden — the Dragons are one of 10 New South Welsh teams in the NRL. The Dragons also have two primary home grounds, owing to the joint nature of the club.³⁴ Furthermore, these two venues are over 50km apart, suggesting that the Dragons fanbase may be surprisingly intense at both stadiums.

³⁴The St George Dragons (Sydney) and Illawarra Steelers (Wollongong) merged in 1999.

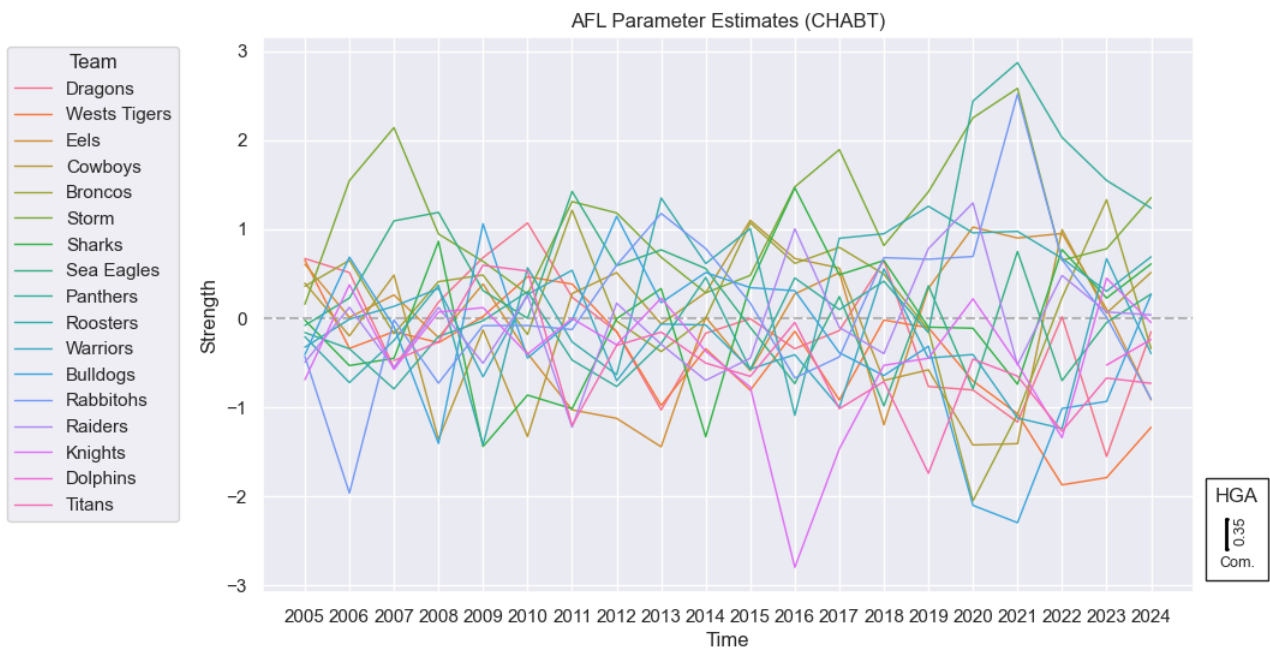


Figure 17: Parameter estimates for the NRL (2005–2024), common HGA model.

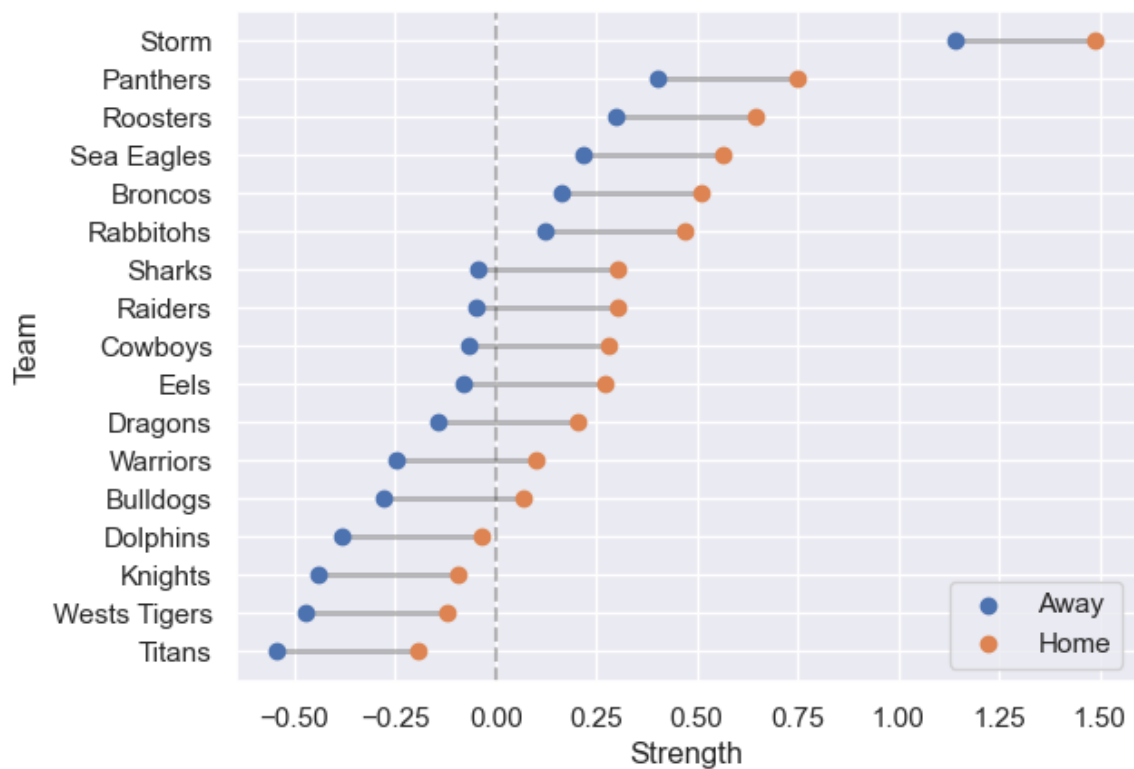


Figure 18: Average estimated team strength for the NRL (2005–2024), common HGA model.

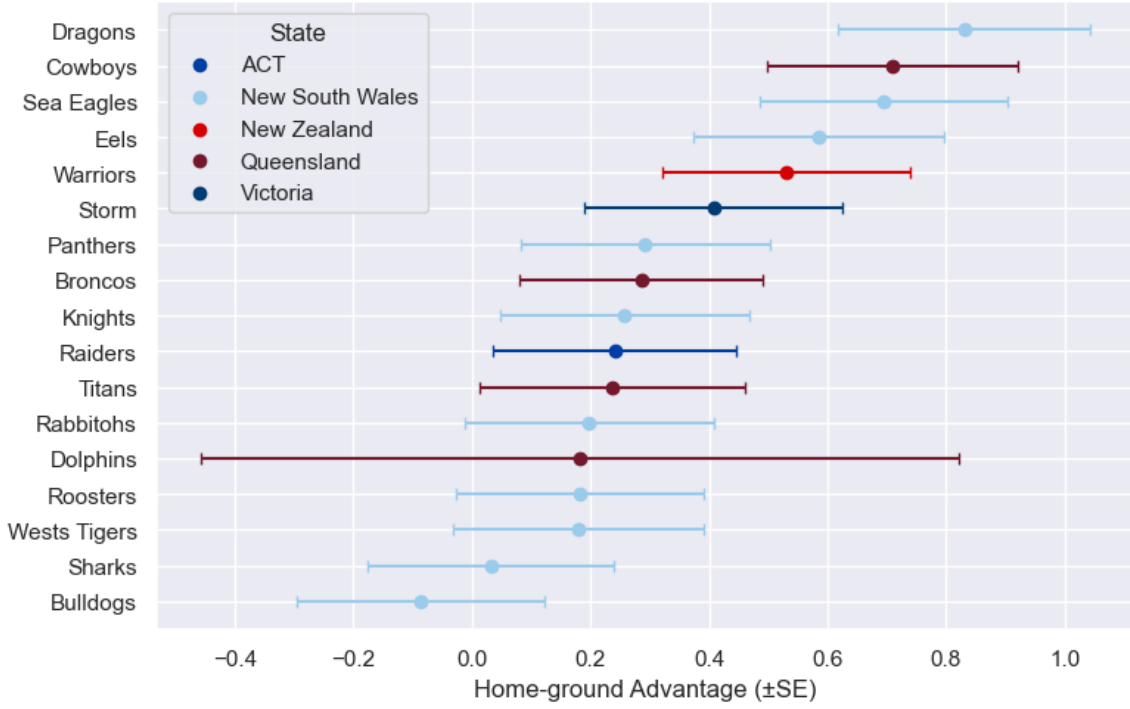


Figure 19: Team-specific HGA estimates for the NRL (2005–2024), coloured by state.

More research could be done on characteristics of these stadiums, as we see that these more regional venues often allow for greater advantages than “major” grounds like the MCG and Stadium Australia. When considering the home states of these teams, there does not appear to be as clear a trend like that seen for the AFL. Specifically, the New South Welsh teams display a variety of HGAs in Figure 19. The non-NSW teams are similarly diverse; we pry further in the next section. Overall, there is no evidence to suggest that the nominal home advantage in the NRL is team-specific rather than a common effect to all teams ($D = 20.8$ on 16 df, $p > 0.1$). Note that the standard error for the Dolphins’ HGA is of course much larger than the others as they only have two years of data in the model.

4.2.4 Results: Groupings

From the TSA model, we saw there was no obvious disparity between NSW and non-NSW teams for the NRL. By proposing some candidate relationship matrices, we can use our hierarchical BT model to investigate this further. As with the AFL, the NRL has no explicit divisions that dictate the season structure; however, there is a disproportionate number of New South Welsh teams which likely impacts fixture scheduling and hence how home-ground advantage is experienced across the different states. We take the same general approach as before, first fitting a state-specific HGA model and then looking for differences between teams within and outside Greater Sydney. As a reminder, with our current NRL dataset, we are only looking at order effects for the nominal home team in each match, rather than explicitly distinguishing

Table 6: Estimated state-specific HGAs (nominal) for the NRL (2005–2024).

State	HGA	[SE]
ACT	0.228	0.205
New South Wales	0.319	0.054
New Zealand	0.524	0.208
Queensland	0.408	0.114
Victoria	0.405	0.217

“true” home games from neutral games based on the venue.

In the AFL, there are 10 Victorian teams and an even 2 teams in the other participating states. The NRL is even less balanced, with a majority 10 teams in NSW, then 4 Queensland teams, and only one team from each of Victoria, the ACT, and New Zealand. Therefore, when we fit a state-specific model, the estimated HGAs for the latter three states will be just the HGAs of the respective teams from the TSA model before. We thus expect that this model may not be particularly useful compared to the TSA model. The results assuming a state-specific HGA are given in Table 6. The table suggests that New Zealand (i.e., the Warriors) has the biggest home advantage, perhaps as it is the furthest location geographically and hence condemns visiting teams to greater travel fatigue. Of course, the Warriors are currently the only NRL team that is not Australian, promoting a cross-country rivalry which hence may also contribute to supportive NZ crowds.

Queensland and Victorian teams (i.e., the Storm for the latter) share a similar estimated home boost, both above that of NSW teams. The ACT (i.e., the Raiders) is the only state to have a weaker home effect than NSW; as the ACT is an enclave of NSW, these results indeed point toward a correlation with travel distance. However, model comparison finds no evidence to suggest a state-specific HGA explains the data better than a common HGA ($D = 1.5$ on 4 df, $p > 0.1$). Furthermore, a team-specific home ground does provide a somewhat better fit than the state-specific model ($D = 19.3$ on 12 df, $p < 0.1$). This was expected, as this model essentially only constrains the NSW and QLD home-ground parameters, and the teams seemed to have fairly diverse individual HGAs in Figure 19. As such, we can attempt to fit a model with more constraints on the teams’ locations.

We propose now a model with only two HGA parameters: one for Sydney-based teams and one for all other teams. Note that we are grouping together teams within the Greater Sydney area and those outside. Much like Geelong in the AFL, there is one New South Welsh team not close to Sydney: the Newcastle Knights’ home ground is about 120km away from Stadium Australia. The model estimates a Sydneysider home-ground of $\hat{\alpha}_{\text{Syd}} = 0.325$ with SE 0.058 and

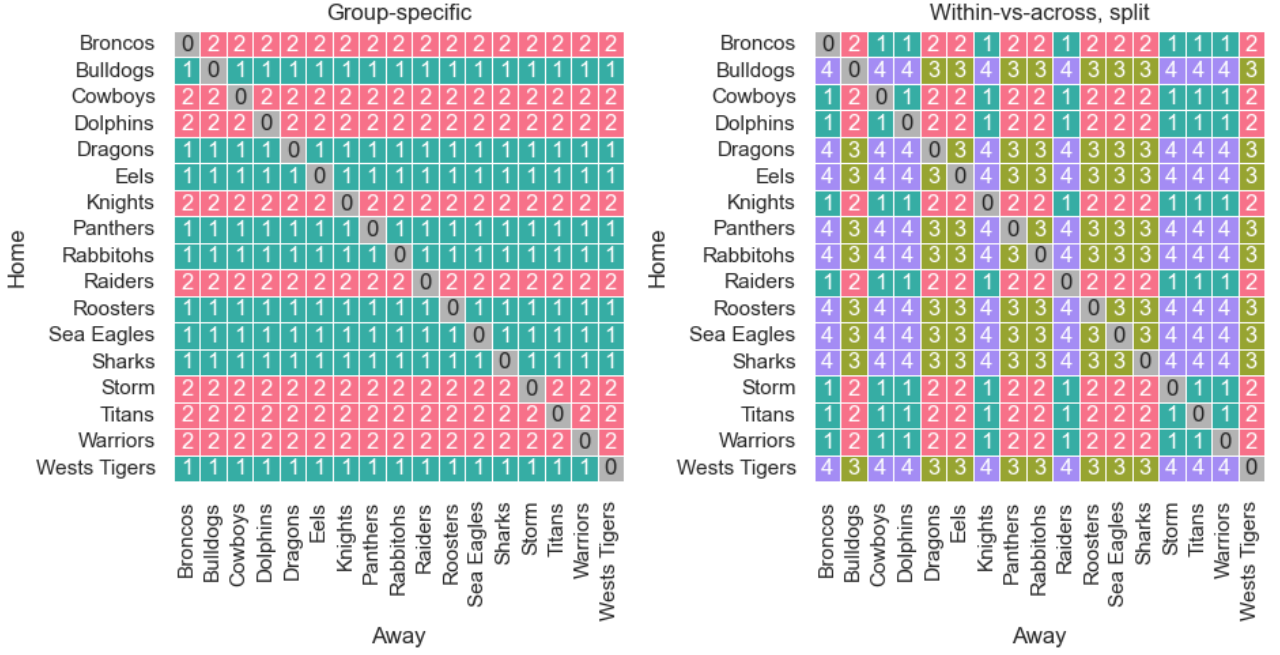


Figure 20: Relationship matrices for the group-specific model (left) and within-vs-across model (right), where teams have been grouped as either Sydney or not. The latter is the same as we did for the AFL but was for Melbourne-based clubs instead.

Table 7: Estimated within-vs-across HGAs for teams grouped as either Sydney or non-Sydney for the NRL (2005–2024).

Home	Away	HGA	[SE]
Sydney	Sydney	0.489	0.079
Sydney	non-Sydney	0.433	0.130
non-Sydney	Sydney	0.392	0.060
non-Sydney	non-Sydney	0.104	0.131

non-Sydney HGA of $\hat{\alpha}_{\text{non-Syd}} = 0.377$ with SE 0.069. Clearly, there is no significant difference in HGA for Sydney teams versus others in favour of the CHA model ($D = 0.3$ on 1 df, $p > 0.1$).

The last HIE model we will fit is the within-vs-across model, again grouping teams as either having a Sydney home ground or not. To be clear, the difference with the within-vs-across model and the “Sydney-sider-or-not” model we just used is that the former assumes the HGA depends on the matchup, not only the home team’s group. We show the two relationship matrices for these models in Figure 20 to further illustrate this difference. The results of the within-vs-across model are given in Table 7.

Yet again, the differences are not as dramatic as we saw with AFL clubs. In fact, we see some home effects are quite the opposite, notably because the Sydney-vs-Sydney matchup has the highest estimated HGA. This is certainly counterintuitive, as one might expect that presumed factors like disproportionate crowd composition and travel fatigue be neutralised in games between two Sydney-based teams. Perhaps the greater diversity of venues among

Table 8: Some HBT model fit results for the NRL (2005–2024).

Model	p	-LLH	AIC	BIC
Vanilla	300	2312.3	5224.7	7112.0
Common HGA	301	2265.6	5133.1	7026.7
Sydney-vs-non-Sydney	302	2265.4	5134.9	7034.8
State-specific HGA	305	2264.8	5139.6	7058.4
Team-specific HGA	317	2255.2	5144.3	7138.6
Sydney-based within-vs-across	304	2262.7	5133.5	7046.0

Sydney teams (as seen in Table 5) contributes to a substantial difference in venue familiarity and bolstered rivalries between these teams, at least when compared to the AFL. Additionally, the HGA for matches between two non-Sydney teams is estimated to be the weakest, where again we would instead assume the effects of travel burden in this scenario improve home performances the most. Nonetheless, this HGA parameterisation does not improve upon the CHA model ($D = 5.7$ on 3 df, $p > 0.1$).

4.2.5 Discussion

We have fit several BT models on the NRL data to analyse whether the nominal home team experiences an advantage. For match outcomes from 2005–2024, we have found strong evidence to support the existence of a mild advantage in favour of the home team. The models are summarised in Table 8. From those fit in our analyses, a common home-ground advantage was found to be most adequate for the data. The NRL has not been analysed in previous work, and it was mentioned in Section 1.4 that paired comparison models — to the best of our knowledge — have not been applied to the NRL in the literature either. The general notion of home-ground advantage in professional rugby league has received some attention, however, which we can corroborate our findings with.

Most of these analyses on HGA are either qualitative (e.g., psychological studies on player perceptions) and/or use relatively simple observational statistics (e.g., the percentage of home wins over some time period) (McGuckin et al., 2015). We have already mentioned that our findings agree with these summary statistics. Longden and Kannard (2014) review home advantage in the context of salary cap violations during the 2001–2012 seasons. They find that, in 2012, home-win probabilities varied by stadium type and opponent, where traditional Sydney grounds conferred more advantage than shared and neutral venues. Furthermore, they note that Sydney teams had the highest chance of a home win when playing against a non-Sydney opponents, controlling for team quality. This seems to agree with our within-vs-across model; we found that Sydney home games, either against fellow Sydneysiders or otherwise, have the

strongest estimated HGAs.

Some of the hypothesised HGA influences such as crowd cheer and ground familiarity are also supported in the literature. For example, Woods and Robertson (2021) discuss such with respect to the AFL, NRL, and Super Rugby. Research furthermore agrees with what we observed in the team-specific HGAs, where there is indeed often little correlation between the strength of a team and the magnitude of its home advantage (Clarke, 2005). Compared to our AFL analysis, we found a slightly weaker nominal HGA for the NRL; hypothesis testing and AIC suggest that both competitions are well-modelled by a common home-ground advantage. Despite their similarities, we found some substantial differences in HGAs when testing common relationship structures. While the AFL suggested non-Victorian teams have higher home advantages, the NRL models found that differences across states were much smaller if not in favour of Sydney-based teams. Overall, our home-ground analyses for both sports coincide with historical win rates and aforementioned literature.

4.3 Case Study: COVID-19 Pandemic

4.3.1 Motivation

The COVID-19 pandemic had a profound effect on the professional sports industry. Competitions across the world were faced with prolonged suspensions and unprecedented conditions that greatly altered their operation. The World Health Organisation declared a global pandemic on March 11th, 2020, at which time the AFL and NRL seasons were in only just commencing for the year. These sports were faced with unusual disruptions in the form of recurrent lockdowns, border closures, and the implementation of “ghost games” played behind closed doors.³⁵ This offers a unique case study for us: how did these radical adjustments to fixtures, stadium usage, and the removal of crowds affect the home-ground advantage in these leagues? We first recap the major changes to each league, then investigate and discuss results using our HBT models.

The pandemic subjected the AFL to a very erratic fixture. After playing only the first round of the 2020 season, matches were suspended until recommencing in June under heavy restrictions. The affect on spectators was dire, as total attendance for the season reduced to only 13.7% of the previous year (Atkinson & Lawson, 2021). Victoria endured notably stringent quarantine measures and hence teams fled elsewhere to continue the season in (relatively) less strict locations like Queensland. Of course, the mass relocation stripped many teams of true home games.

³⁵This was the first time in the history of both codes that premiership matches were played in empty stadiums; broadcasters even experimented with adding artificial crowd cheer and atmosphere to their productions.

This was most evident for the grand finals: the 2020 final was moved to the Gabba in Brisbane and then to Perth Stadium in 2021.³⁶ Undoubtedly, then, the league’s home advantage not only diminished from lack of crowds but also frequent transfers to neutral interstate venues.

The NRL faced similar adversity, though also had some unique differences to the AFL. In particular, the NRL was one of the first professional sports globally to resume play, restarting matches after only a couple months in May. The logistical challenges were nonetheless burdensome for many teams. Quarantine restrictions in New Zealand, for example, forced the Warriors to play the entire season out of Kingscliff, NSW. Similarly, the Storm fled to the Sunshine Coast due to the prolonged lockdowns in Victoria.³⁷ Another key difference to the AFL is the distribution of teams, as Victorian restrictions were much tighter than NSW. As such, the 2021 NRL season was relatively less impacted than the AFL; however, the finals were still eventually forced to Queensland by a later NSW outbreak. To summarise, the fixtures and hence home games were fundamentally altered in both leagues over the first two years of the pandemic.

4.3.2 Analysis

We wish to model home-ground effect around the pandemic-affected seasons to see if it diminished during restrictions and how it changed in the seasons after. To do this, we will fit over three time periods: a three-year window prior to the pandemic (2017–2019), the major pandemic years (2020–2021), and a three-year window after restrictions eased (2022–2024). Let us refer to these windows simply as “before,” “during,” and “after,” respectively. The choice of the 2022 season as “post-pandemic” for our purposes is justified, for example, based on statements in the AFL’s annual report (Australia Football League, 2022). In this season, “crowds enthusiastically returned to venues” and the grand final was held at the MCG again. For comparison, we will also fit a model to the entire period (2017–2024) under the constant HGA assumption. Our null hypothesis is thus that the home advantage did not significantly change before and after the pandemic. This process is done for both leagues under the nominal home assumption.³⁸

Let us first consider the AFL. Using the three-season window scheme outlined above, we start with the common HGA model. Fitting these models, we get the following estimated HGAs (and SEs): $\hat{\alpha}_{\text{before}} = 0.352$ (0.095) pre-pandemic, $\hat{\alpha}_{\text{during}} = 0.192$ (0.123) during, and $\hat{\alpha}_{\text{after}} =$

³⁶This was the first (and second) time in VFL/AFL history that the grand final was played outside Melbourne.

³⁷The Storm would still win the 2020 season, reiterating an independence between team strengths and unique home advantage we mentioned previously.

³⁸The venue-specific win matrices for pandemic-affected years are certainly not strongly connected under the true home assumption.

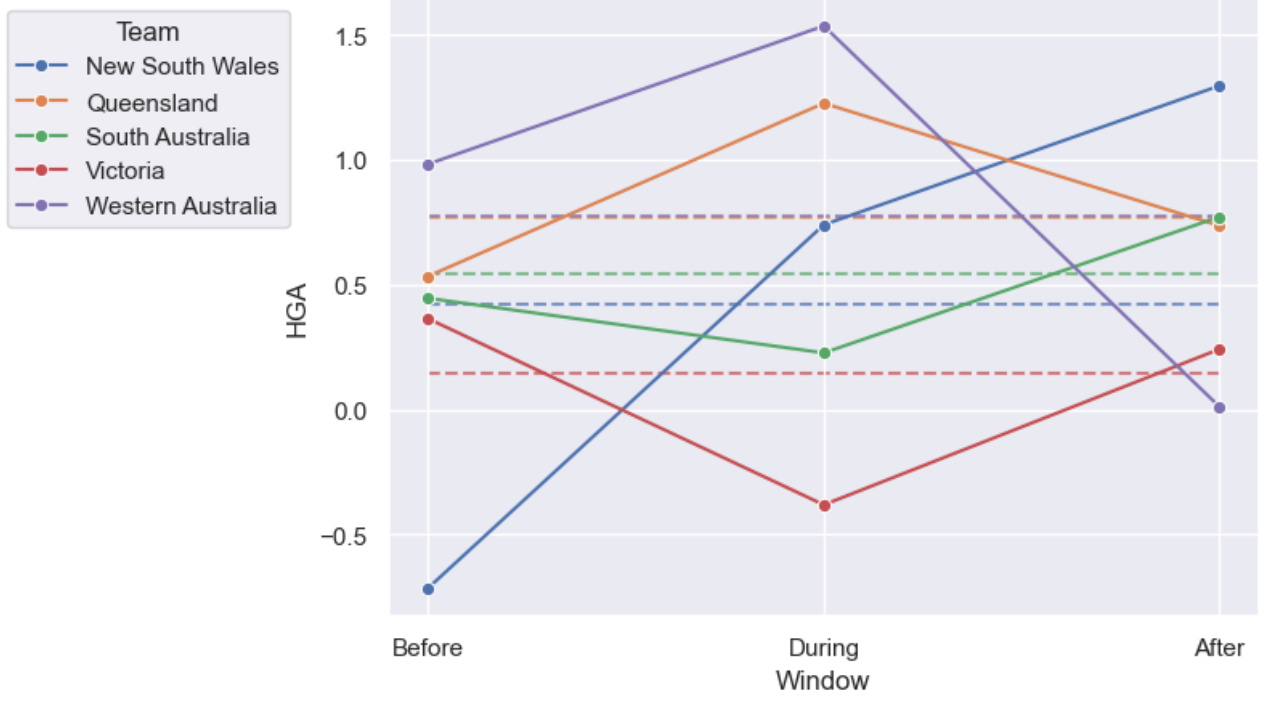


Figure 21: Estimated HGAs in AFL for the state-specific model fit before, during, and after the pandemic. Dashed lined represent the results for the model fit to the entire period (2017–2024).

0.459 (0.097) post-pandemic. Furthermore, the constant HGA is estimated to be $\hat{\alpha}_{\text{entire}} = 0.356$ (0.059) across the whole time period. As expected, the magnitude of HGA dropped during the pandemic, and appears to have returned even stronger in the years following. We can perform a comparison of deviance by treating the three window models as one alternative to the constant HGA model. The test suggests that the home effect has not changed over this period ($D = 2.87$ on 2 df, $p > 0.1$).

Recall from our investigation of nominal HGA earlier that the state-specific and team-specific HGA models provided better fits to the data than the CHA model. It is reasonable to assume that the HGAs were impacted in these more complex models because different states experienced variable strictness of lockdown conditions and hence not all teams were forced to relocate. We therefore expect that HGA does significantly change across the windows in these parameterisations. The estimated HGAs for the state-specific model are plotted in Figure 21; the results for the TSA model lack visual clarity and instead can be found in Appendix C. We see that the Victorian HGA is the only one to drop below zero (a home *disadvantage*) during the pandemic, as expected, and the refuge states of QLD and WA saw increased advantages. Compared to their constant HGA analogues, there is indeed strong evidence of a difference across the windows on a state-specific level ($D = 25.0$ on 10 df, $p < 0.01$) and weak evidence on a team-specific level ($D = 50.1$ on 36 df, $p < 0.1$).

We can also test for more structural, “long-term” HGA change — as opposed to verifying the

“shock” during the pandemic — by ignoring the middle window. This is a slightly different research question, instead looking for some “hangover” effect on HGA in the league following the resumption of a relatively normal fixture post-pandemic. Again, there is no evidence of a difference in common HGA before and after the pandemic ($D = 0.62$ on 1 df, $p > 0.1$). Conversely, the HGA does appear to have changed after the pandemic on a state-specific level ($D = 14.1$ on 5 df, $p < 0.05$) and team-specific level ($D = 28.9$ on 18 df, $p < 0.05$).

We repeat these tests for the NRL, beginning with a common HGA. The individual-window models produced estimates of $\hat{\alpha}_{\text{before}} = 0.273$ (0.092) pre-pandemic, $\hat{\alpha}_{\text{during}} = 0.204$ (0.137) during, and $\hat{\alpha}_{\text{after}} = 0.388$ (0.094) post-pandemic. One notes the same trend of HGA dropping in the pandemic years and then rising higher than pre-pandemic levels; however, the differences are notably smaller than seen in the AFL. Assuming a constant HGA over this period, the model estimates $\hat{\alpha}_{\text{entire}} = 0.306$ (0.059) and we find no evidence of a change in common HGA over the pandemic ($D = 1.4$ on 2 df, $p > 0.1$).

From our earlier NRL analysis, we found that the nominal home advantage is no better explained by a state-specific or team-specific model. Fitting and testing these models regardless, the results also support a constant HGA on both the state level ($D = 11.5$ on 10 df, $p > 0.1$) and team level ($D = 33.3$ on 34 df, $p > 0.1$). The state-specific HGAs are plotted in Figure 22. Some interesting things can be observed, such as the Warriors (NZ) and Storm (VIC) having higher home advantages during the pandemic despite moving from their primary homes! The results are much the same when only testing for long-term HGA change. There is no evidence to suggest that current home effects in the NRL are different to those pre-pandemic for the common ($D = 0.8$ on 1 df, $p > 0.1$), state-specific ($D = 2.5$ on 5 df, $p > 0.1$), nor team-specific models ($D = 12.6$ on 17 df, $p > 0.1$).

4.3.3 Discussion

The effects of the COVID-19 pandemic on (nominal) home-ground advantage in the two leagues are complicated. For the AFL, we found differences across the pandemic period were significant on state- and team-specific levels. In the NRL, however, testing indicated that the HGA had not substantially changed compared to a constant assumption for the common, state-specific, and team-specific models. Ignoring the lockdown-affected window, our models suggest that there is perhaps a long-term effect on team-specific HGA in the AFL but not the NRL.

These results are mostly as expected based on our understanding of the severity of lockdown measures across Australia and New Zealand. It is well-known that COVID-19 annulled the home advantage of many teams during the pandemic, with the adoption of ghost games and

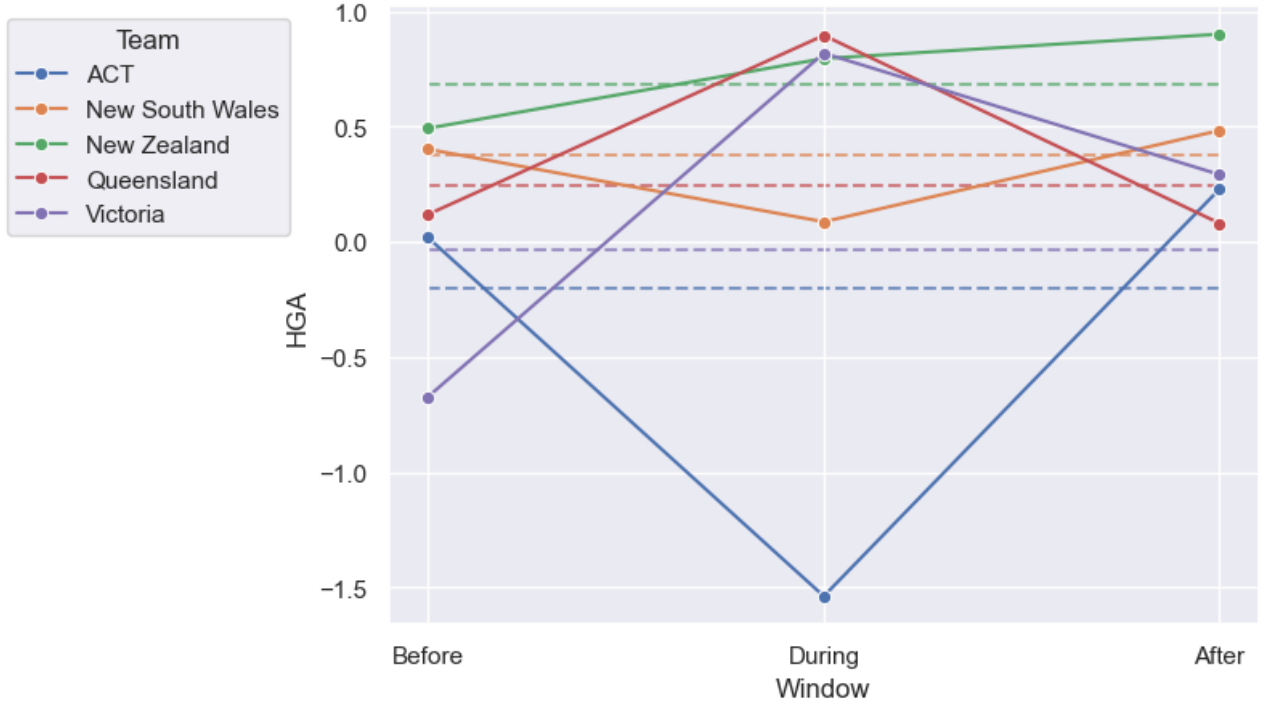


Figure 22: Estimated HGAs in NRL for the state-specific model fit before, during, and after the pandemic. Dashed lined represent the results for the model fit to the entire period (2017–2024).

season-long relocation to interstate hubs. As mentioned in Section 1.4, Liu et al. (2025) confirmed reduced team-specific HGAs in EPL which “gradually recovered as fans returned” and restrictions relaxed. This certainly corroborates with crowd effects being a key driver of home advantage (Ponzo & Scoppa, 2018). The NRL remains a unique case as one of the least-impacted competitions; the season was among the first to resume globally and experienced milder quarantine restrictions than the Victoria-centered AFL. This is reflected by the adequacy of the constant HGA model on the NRL matches over the windows.

Furthermore, we noted that home advantage tended to rise on average in the post-pandemic season. Compared to pre-pandemic estimates, HGA seemed to increase by roughly 0.1 on the log-odds in both leagues in the following years.³⁹ This likely points to impassioned crowds returning as social distancing regulations alleviated in 2022 onward; indeed, most clubs have enjoyed record-breaking membership uptake after declines in 2020 (AAP, 2024). Further models could certainly be tested, such as within-vs-across group relationship structures fitted previously. For brevity sake we curtail our investigation here.

³⁹i.e., odds increased by a factor of approx. 1.1 in favour of the home team compared to pre-pandemic conditions

4.4 On Stationarity

4.4.1 Motivation

The last consideration for this chapter regards the **stationarity** of the AFL and NRL. Specifically, this section briefly explores the extent to which our Bradley–Terry model parameters can be assumed static for each competition. As such, we hope to justify our initial claim that home-ground advantage is constant over time. To this end, recall that our implementation can easily recover the fully-dynamic approach as discussed in Section 2.6.

For context, previous work by Jones (2021) analysed 10 seasons of AFL data (2010–2019) using static HBT models. It was posited that “a three season window may be best for modelling the AFL” based on some simple analyses; the same approach was taken by Soffner and Leonelli (2024). These decisions are mainly spurred by the AFL draft system, whereby clubs’ placements on the previous season’s ladder determines their pick priority for unsigned players. Intuitively, then, this system enforces a natural mechanism for equalising team abilities and keeps the competition healthy from one season to the next. Conversely, the NRL does not currently use a draft system, yet both competitions remain dynamic due to other factors like roster adjustments, captaincy and coaching changes, and player injuries.⁴⁰ Altogether, the volatility of these competitions begs the question of whether parameters should be assumed constant for any duration.

We can get a feel for this from the variance of strength estimates in the models fit beforehand, details of which are provided in the Appendix. In the AFL, for example, the GWS Giants have the most erratic performance, starting with multiple wooden spoons until settling as a decent team after a few seasons. Teams such as West Coast and Melbourne have experienced some larger dips in performance, whereas Sydney and Essendon have remained very consistent every year (for better or for worse). In the NRL, the Panthers have the most volatile estimated strengths due to their dominance only recently with three consecutive premierships. The least variable NRL performances belong to the Warriors and Titan (excluding the Dolphins). Qualitatively speaking, team strengths in the AFL seem to fluctuate much more than the NRL, which may be due to the draft system in the former.

⁴⁰To illustrate, 15 of the 18 AFL teams and 15 of the 17 NRL teams have made the grand finals at least once since 2000.

Table 9: Results of testing static vs. dynamic team strengths with the CHA model on 3-season windows for the AFL (2010–2019).

Window #	Start Year	End Year	D	df	p	(sig.)
1	2010	2012	89.5	31	<0.001	***
2	2011	2013	91.7	33	<0.001	***
3	2012	2014	60.9	34	0.003	**
4	2013	2015	72.5	34	<0.001	***
5	2014	2016	101.9	34	<0.001	***
6	2015	2017	76.5	34	<0.001	***
7	2016	2018	77.5	34	<0.001	***
8	2017	2019	59.3	34	0.005	**

Table 10: Results of testing static vs. dynamic team strengths with the CHA model on 3-season windows for the NRL (2010–2019).

Window #	Start Year	End Year	D	df	p	(sig.)
1	2010	2012	70.6	30	<0.001	***
2	2011	2013	59.2	30	0.001	**
3	2012	2014	45.0	30	0.039	*
4	2013	2015	43.3	30	0.055	.
5	2014	2016	75.0	30	<0.001	***
6	2015	2017	52.2	30	0.007	**
7	2016	2018	71.5	30	<0.001	***
8	2017	2019	55.0	30	0.004	**

4.4.2 Analysis and discussion

We apply our new dynamic framework to do some exploratory analysis of stationarity of team strengths and home advantage in both leagues. To keep our scope reasonable and exclude the pandemic, we limit ourselves to a time period of 2010–2019 for more direct comparison to (Jones, 2021). We first test whether a 3-season window is sufficient for the constant team strength assumption. For each window, we compare the static model assuming no time-dependent parameters to our dynamic model with independent team strengths, then record the results of the corresponding hypothesis test. Test outputs are provided in Tables 9 and 10.

These results show that dynamic team strengths provide better fits in all 3-season windows tested. The evidence is markedly strong for the AFL. The NRL shows relatively weaker evidence of dynamic abilities for some windows, but this was expected due to its lack of the same draft system. Regardless, there is still at least some level of improvement when using dynamic team strengths in each season. Overall, this quick analysis suggests that even a period as short as 3-seasons is inadequate for capturing the variability of team strengths in the AFL and NRL.

We have demonstrated that our incorporation of dynamic team strengths into the HBT frame-

Table 11: Results of testing static vs. dynamic HGA with the CHA and TSA models on 2010–2019 AFL and NRL data.

League	Model	D	df	p	(sig.)
AFL	CHA	4.5	9	0.875	
AFL	TSA	173.4	159	0.206	
NRL	CHA	6.2	9	0.718	
NRL	TSA	167.8	144	0.085	.

work is necessary. To conclude, we now perform similar testing on the stationarity of home-ground advantage. This is done by comparing our proposed model (dynamic strengths, constant HGA) and the fully-dynamic model fit to the 2010–2019 match data for both leagues. We keep this modelling brief by only fitting the CHA and TSA models, leaving any time dependence of hierarchical home effects for future study. Results are given in Table 11.

The estimated home effects for the fully-dynamic common HGA models are plotted in Figures 23 and 24 for reference. For the AFL, testing suggests that assuming a constant home effect is sufficient for both the common and team-specific HGA models. Of course, there is some fluctuation from season to season; however, the results show that the fully-dynamic model provides no meaningful improvement over our semi-dynamic approach. The results are similar for the NRL, with no evidence of a better fit using a dynamic common HGA model and perhaps weak evidence in favour of dynamic team-specific home effects. Therefore, this testing supports our original claim that home advantage may be treated as constant, and thus our proposed dynamic extension for HBT models is justifiable.

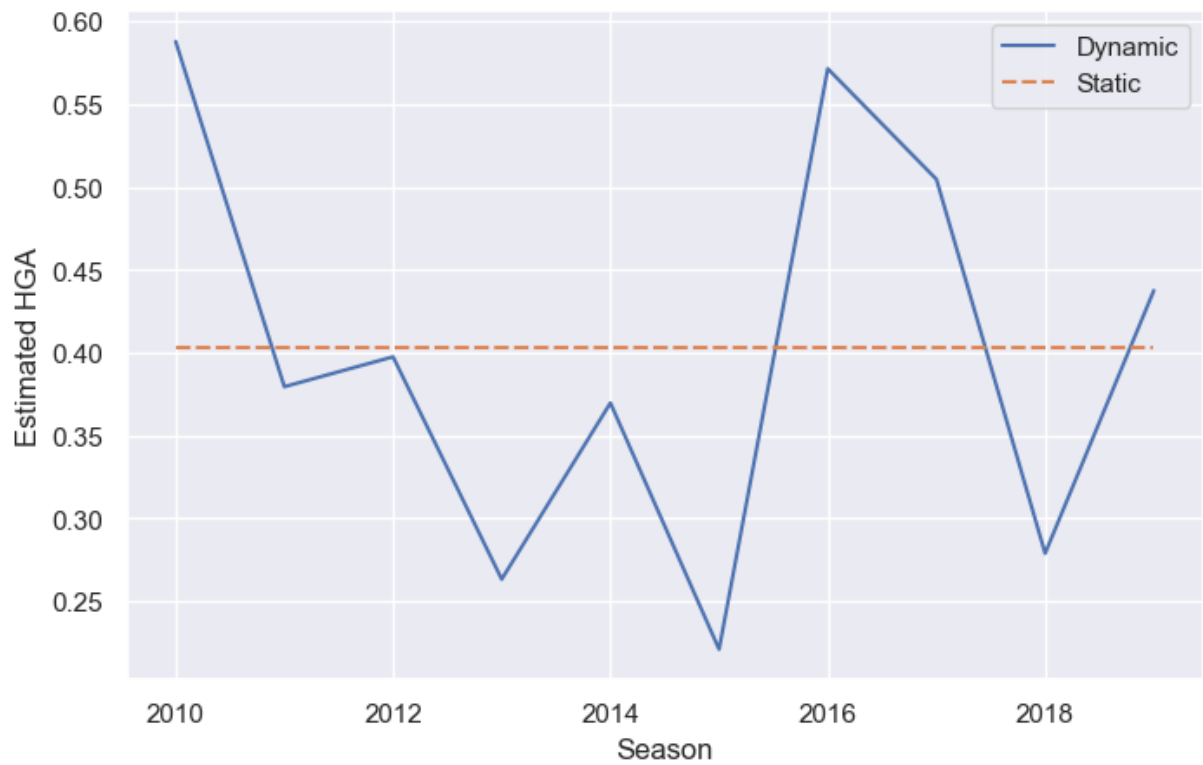


Figure 23: Estimated HGAs for the fully-dynamic (blue) and constant HGA (orange) CHA models, fit to the AFL (2010–2019).

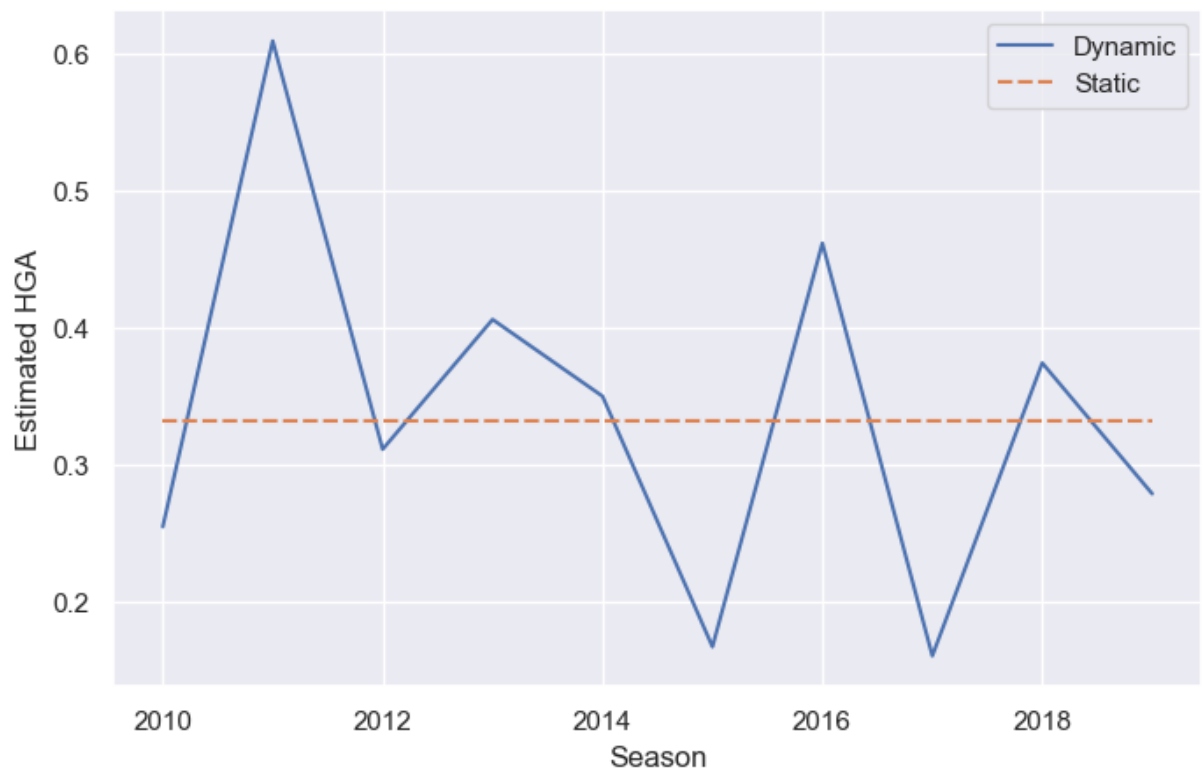


Figure 24: Estimated HGAs for the fully-dynamic (blue) and constant HGA (orange) CHA models, fit to the NRL (2010–2019).

Chapter 5: Conclusion

5.1 Summary

In this thesis, we have explored paired comparison models and their application to significant Australian sporting leagues. The history and current state of the literature has been thoroughly surveyed, including popular packages and prior relevant applications. Previous contributions to Hierarchical Bradley–Terry models by Jones (2021) and Zhang (2024) have been critiqued, primarily noting the need for a dynamic extension and revised implementation. As such, we proposed a modelling framework where team strengths are independent between time blocks, while order effects — namely home-ground advantages — are presumed constant. Key modelling results have been provided for this dynamic extension, and we have furthermore unified all treatments of home-ground advantage under a fully-generalised Hierarchical BT model. Using arbitrary relationship matrices, the modeller can easily specify any desired relationship structure between competitors up to a pair-specific level. Hypothesis testing on fit adequacy is then straightforward, with a simple algorithm outlined to check for nestedness.

The new implementation was discussed, highlighting the benefits of an object-oriented approach and various improvements upon the previous codebase developed by Zhang (2024). We divulged the critical practical details for data structures, model initialisation, parameter estimation, and additional support tools. The models have been applied to examine HGA in two professional sports leagues — the AFL and NRL — featuring both general analyses of match outcomes from the past 20 seasons and select case studies. These applications included fitting a basic suite of BT models, investigating a selection of intuitive relationship structures, and differences between a nominal and “true” home-ground advantage. Home effects were found to be significant in both leagues, in agreement with findings in the literature, and some results supported fits beyond a common HGA. Most notably, perhaps, were the seemingly contrary home advantages suggested by our matchup models based on teams’ residency in the “hub” states of each league.

Furthermore, our analysis of the COVID-19 pandemic period confirmed reduced home advantages in pandemic years and subsequent resurgence in the seasons after. The effects of different lockdown severity on the two leagues were considered, finding no significant changes to home effects in the NRL. In contrast, the AFL did indeed suffer harsher restrictions, with evidence

to suggest HGA changed on a state-specific and team-specific level as the majority Victorian teams were forced to move interstate. Lastly, a brief examination of the volatility of both leagues verified that team strengths do significantly fluctuate over seasons, and yet home advantage remains sufficiently static. These results justified our current modelling assumption in favour of fully-dynamic HBT models. In summary, we claim to have successfully achieved the project aims listed in Section 1.6.

5.2 Future Considerations

There are several aspects of this project that could be developed further. We have already discussed some directions within the scope of our current modelling framework, such as additional HBT model technicalities in Section 2.7 and further codebase improvements in 3.4. Regarding our applications, there are always more possibilities to investigate. We have already provided rather in-depth analyses of AFL and NRL results for the past 20 years. A first priority for further analysis would likely be comparing “true” and nominal home effects for the NRL, as a restrictive dataset limited us to only consider the latter. Results could then be compared to the AFL for both the basic suite and more complex relationship structures, such as a “Sydney-vs-other” parameterisation. Obviously, this would require extra work for the data collection, validation, and formatting, noting the inconvenience of venue naming and so-on. Another idea would be distinguishing between matches played during the premiership season and finals series.

We demonstrated the flexibility of the models with the case studies on COVID-19 and stationarity. Specifically, these showed how one could construct certain windows or recover fully-dynamic analogues to reflect their hypotheses, where the richness of these nested models becomes apparent. Of course, more rigorous analyses with increasingly complicated relationship models could be done. There are also several other interesting cases to consider, such as changes in home advantage during the rapid expansion of both leagues, i.e., the transition from VFL to AFL or ARL to NRL. The effects of the Super League war on the latter would be especially noteworthy.

Going further beyond our scope, these models could obviously be applied to numerous other professional sports where home-ground effects are of interest. Owing to previous work, this would primarily be the NBA. As shown in various examples throughout Chapter 2, the explicit categorisation of teams into divisions and conferences directly effects the NBA fixture and thus provides some interesting relationship structures to explore. More local to Australia, other popular professional head-to-head competitions are clear candidates for analysis, such as the Big Bash League (BBL) in cricket, A-Leagues in soccer, and the National Basketball League

(NBL).¹ These sports highlight potential modelling concerns, such as the handling of ties, which would require extending the theory as necessary. Furthermore, it may be interesting to contrast our results with HGA in women’s professional sport, such as the AFLW and NRLW, which experience limited media coverage and gender-based consumer bias compared to their historically male-oriented counterparts (Fujak et al., 2025; Glazbrook et al., 2025).

Besides the myriad of possible applications, there are also some future directions we would like to take our framework itself. For example, our models currently assume independence between team strengths over time, but it would be judicious to enforce some proper dependence structure when examining historical team performance. We surveyed several non-stochastic and stochastic extensions in Section 1.2.4. Of these, the AR-1 model (1.13) seems like an appropriate extension to consider first, as such stochastic models may suitably explain league volatility, for example, due to draft systems. More robust models, such as GP priors (Bayesian) and Glicko2 (rating systems), become reasonable options when our datasets grow larger. Based on the factors commonly attributed to HGA, we may also wish to consider covariates for attendance, club membership, or other venue-specific features. Nonetheless, the framework and implementation developed in this project provide a strong foundation for historical analysis of user-specified home advantage structures in head-to-head competitions.

¹The NBL does not have divisions like the NBA

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Chapter A: Additional Model Details

A.1 On Probability Identities

A.1.1 The logistic function

In this section, we highlight some key functions and related identities that were used in the preceding derivations. The Bradley–Terry model is fundamentally defined by the *logistic* function $f : \mathbb{R} \rightarrow \mathbb{R}$, also called the *expit*,

$$f(x) := \frac{1}{1 + e^{-x}} = \frac{e^x}{1 + e^x} \quad (\text{A.1})$$

It is easy to understand *why* upon recognising that it is the CDF of a standard logistic distribution, and hence brings all the useful notions for defining probabilities. For example, complementary probabilities are clear from the symmetry $f(-x) = 1 - f(x)$. A key result for us are the following derivatives

$$\frac{d}{dx} f(x) = f(x)(1 - f(x)) \quad (\text{A.2})$$

$$\implies \frac{d}{dx} \log f(x) = 1 - f(x) \quad (\text{A.3})$$

which we will make use of shortly. Finally, the *logit* function is the inverse of the *expit*,

$$\text{logit}(x) = f^{-1}(x) = \log \frac{x}{1 - x} \quad (\text{A.4})$$

These functions have rich mathematical properties and vast applications.¹

A.1.2 Connecting probability and log-odds

The properties of the *expit* and *logit* are useful for simplifying the likelihood equations. The primary reason is that likelihoods are calculated with probabilities, but our parameters are defined on the more intuitive log-scale. Conversion is then made easy. As a simple example, consider the original Bradley–Terry model, which defines the probability

$$\pi_{ij} := \mathbb{P}(i \succ j) = \frac{e^{\theta_i}}{e^{\theta_i} + e^{\theta_j}} = \frac{1}{1 + e^{-(\theta_i - \theta_j)}} = f(\theta_i - \theta_j) \quad (\text{A.5})$$

¹see https://en.wikipedia.org/wiki/Wilks'_theorem

and so the log-odds yields

$$\log \left(\frac{\pi_{ij}}{\pi_{ji}} \right) = \text{logit}(\pi_{ij}) = \theta_i - \theta_j \quad (\text{A.6})$$

for unique $i, j = 1, \dots, I$. In general, the log-odds of note are

$$\text{logit}(\pi_{ijkt}) = \theta_{it} - \theta_{jt} \quad (\text{A.7})$$

$$\text{logit}(\pi_{ijkt}^{(h)}) = \theta_{it} - \theta_{jt} + \alpha_k \quad (\text{A.8})$$

$$\text{logit}(\pi_{ijkt}^{(a)}) = \theta_{it} - \theta_{jt} - \alpha_k \quad (\text{A.9})$$

for unique $i, j = 1, \dots, I$, any $k = 1, \dots, K$, and any $t = 1, \dots, T$. Finally, we want to note the complementary probabilities

$$1 - \pi_{ijkt} = f(-\theta_{it} + \theta_{jt}) = \pi_{jikt} \quad (\text{A.10})$$

$$1 - \pi_{ijkt}^{(h)} = f(-\theta_{it} + \theta_{jt} - \alpha_k) = \pi_{jikt}^{(a)} \quad (\text{A.11})$$

A.1.3 Differentiating probabilities

Both probabilities and log-probabilities appear in ℓ and its gradients for each model, hence the need for certain first- and second-order partial derivatives w.r.t. the various parameters. We give generalised results below, utilising (A.2) and (A.3).

First-order derivatives w.r.t. team strengths θ_{it} :

$$\frac{\partial}{\partial \theta_{it}} \pi_{ijkt} = -\frac{\partial}{\partial \theta_{jt}} \pi_{ijkt} = \pi_{ijkt}(1 - \pi_{ijkt}) \quad (\text{A.12})$$

$$\frac{\partial}{\partial \theta_{it}} \log \pi_{ijkt} = -\frac{\partial}{\partial \theta_{jt}} \log \pi_{ijkt} = 1 - \pi_{ijkt} \quad (\text{A.13})$$

and second-order derivatives:

$$\frac{\partial^2}{\partial \theta_{it}^2} \log \pi_{ijkt} = \frac{\partial^2}{\partial \theta_{jt}^2} \log \pi_{ijkt} = -\pi_{ijkt}(1 - \pi_{ijkt}) \quad (\text{A.14})$$

$$\frac{\partial^2}{\partial \theta_{it} \partial \theta_{jt}} \log \pi_{ijkt} = \pi_{ijkt}(1 - \pi_{ijkt}) \quad (\text{A.15})$$

for unique $i, j = 1, \dots, I$. All other derivatives are zero, that is with respect to any other θ_{ls} that is not θ_{it} or θ_{jt} . The generic probabilities π_{ijkt} here can be substituted with their venue-specific forms, $\pi_{ijkt}^{(h)}$ or $\pi_{ijkt}^{(a)}$, and the results hold exactly.

First-order derivatives w.r.t. α_k :

$$\frac{\partial}{\partial \alpha_k} \pi_{ijkt}^{(h)} = \pi_{ijkt}^{(h)} (1 - \pi_{ijkt}^{(h)}) \quad (\text{A.16})$$

$$\frac{\partial}{\partial \alpha_k} \pi_{ijkt}^{(a)} = -\pi_{ijkt}^{(a)} (1 - \pi_{ijkt}^{(a)}) \quad (\text{A.17})$$

$$\frac{\partial}{\partial \alpha_k} \log \pi_{ijkt}^{(h)} = 1 - \pi_{ijkt}^{(h)} \quad (\text{A.18})$$

$$\frac{\partial}{\partial \alpha_k} \log \pi_{ijkt}^{(a)} = - (1 - \pi_{ijkt}^{(a)}) \quad (\text{A.19})$$

and second-order derivatives with cross terms:

$$\frac{\partial^2}{\partial \alpha_k^2} \log \pi_{ijkt}^{(h)} = -\pi_{ijkt}^{(h)} (1 - \pi_{ijkt}^{(h)}) \quad (\text{A.20})$$

$$\frac{\partial^2}{\partial \alpha_k^2} \log \pi_{ijkt}^{(a)} = -\pi_{ijkt}^{(a)} (1 - \pi_{ijkt}^{(a)}) \quad (\text{A.21})$$

$$\frac{\partial^2}{\partial \theta_{it} \partial \alpha_k} \log \pi_{ijkt}^{(h)} = -\frac{\partial^2}{\partial \theta_{jt} \partial \alpha_k} \log \pi_{ijkt}^{(h)} = -\pi_{ijkt}^{(h)} (1 - \pi_{ijkt}^{(h)}) \quad (\text{A.22})$$

$$\frac{\partial^2}{\partial \theta_{it} \partial \alpha_k} \log \pi_{ijkt}^{(a)} = -\frac{\partial^2}{\partial \theta_{jt} \partial \alpha_k} \log \pi_{ijkt}^{(a)} = \pi_{ijkt}^{(a)} (1 - \pi_{ijkt}^{(a)}) \quad (\text{A.23})$$

for $k = 1, \dots, K$. Note that these results are given irrespective of the relationship r_{ij} . Clearly, if $r_{ij} \neq k$ then the derivatives above are zero as necessary. Of course, all derivatives of the neutral venue probabilities π_{ijkt} are also zero as home-ground effects α_k do not appear in their definitions! One can simply use the nested relationships in 2.6 to simplify these results as necessary for a specific model.

A.1.4 A method for calculating model derivatives

Calculating the score and hessian elements is quite difficult due to the nested sums, so here we demonstrate an intuitive method for finding such. The key is representing the sums with matrices and applying the respective derivatives component-wise.² Consider the common home-ground advantage (CHA) model (2.3), for which we can express the log-likelihood (2.11) as

$$\begin{aligned} \ell(\boldsymbol{\theta} \mid \mathcal{D}) &= \sum_{ijt} h_{ijt} \log \pi_{ijt}^{(h)} + a_{ijt} \log \pi_{ijt}^{(a)} + u_{ijt} \log \pi_{ijt} \\ &= \sum_{ijtx} x_{ijt} \log \pi_{ijt}^{(x)} \\ &= \sum_{tx} \mathbf{1}^T [X_t \odot P_t^{(x)}] \mathbf{1} \end{aligned}$$

Firstly, with an abuse of notation, we introduce a convenient new sum index x to denote summing over the venue types (h, a, u) . Secondly, we have expressed the double sum over team

²Inspired by Vandermonde matrices.

pairs (i, j) with a matrix product³ where $\mathbf{1} = (1, \dots, 1)^T \in \mathbb{R}^I$ and

$$P_t^{(v)} = \begin{bmatrix} 0 & \log \pi_{1,2,t}^{(x)} & \cdots & \log \pi_{1,i,t}^{(x)} & \cdots & \log \pi_{1,I,t}^{(x)} \\ \log \pi_{2,1,t}^{(x)} & 0 & \cdots & \log \pi_{2,i,t}^{(x)} & \cdots & \log \pi_{2,I,t}^{(x)} \\ \vdots & \vdots & \ddots & \vdots & & \vdots \\ \log \pi_{i,1,t}^{(x)} & \log \pi_{i,2,t}^{(x)} & \cdots & 0 & \cdots & \log \pi_{i,I,t}^{(x)} \\ \vdots & \vdots & & \vdots & \ddots & \vdots \\ \log \pi_{I,1,t}^{(x)} & \log \pi_{I,2,t}^{(x)} & \cdots & \log \pi_{I,i,t}^{(x)} & \cdots & 0 \end{bmatrix}$$

Notice that only the i -th column and row of each $P_t^{(x)}$ contain θ_{it} because each $\pi_{ijt}^{(x)}$ is a function of $\theta_{it} - \theta_{jt}$.

To derive the score, let's first find $\frac{\partial l}{\partial \theta_{it}}$. The main idea here is finding the element-wise derivatives to see which terms vanish. We know that $\frac{\partial}{\partial \theta_{it}} P_s^{(x)} = 0$ when $s \neq t$. The only nonzero components are the i -th row and column of $P_t^{(x)}$.

$$\frac{\partial}{\partial \theta_{it}} P_t^{(x)} = \begin{bmatrix} 0 & 0 & \cdots & -(1 - \pi_{1,i,t}^{(x)}) & \cdots & 0 \\ 0 & 0 & \cdots & -(1 - \pi_{2,i,t}^{(x)}) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & & \vdots \\ 1 - \pi_{i,1,t}^{(x)} & 1 - \pi_{i,2,t}^{(x)} & \cdots & 0 & \cdots & 1 - \pi_{i,I,t}^{(x)} \\ \vdots & \vdots & & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -(1 - \pi_{I,i,t}^{(x)}) & \cdots & 0 \end{bmatrix}$$

So we will have

$$\begin{aligned} \frac{\partial l}{\partial \theta_{it}} &= \sum_x \mathbf{1}^T \left[X_t \odot \frac{\partial P_t^{(x)}}{\partial \theta_{it}} \right] \mathbf{1} \\ &= \sum_x \mathbf{1}^T \left[X_{i \bullet t} \odot \frac{\partial P_{i \bullet t}^{(x)}}{\partial \theta_{it}} + X_{\bullet it} \odot \frac{\partial P_{\bullet it}^{(x)}}{\partial \theta_{it}} \right] \\ &= \sum_{jx} x_{ijt} (1 - \pi_{ijt}^{(x)}) - x_{jit} (1 - \pi_{jit}^{(x)}) \\ &= \sum_j h_{ijt} (1 - \pi_{ijt}^{(h)}) - h_{jit} \pi_{ijt}^{(a)} + a_{ijt} (1 - \pi_{ijt}^{(a)}) - a_{jit} \pi_{ijt}^{(h)} + u_{ijt} (1 - \pi_{ijt}^{(u)}) - u_{jit} \pi_{ijt}^{(u)} \\ &= \sum_j \underbrace{h_{ijt} + a_{ijt} + u_{ijt}}_{\# \text{ games } i \text{ beats } j} - \underbrace{(h_{ijt} + a_{jit})}_{\# \text{ games } i \text{ home vs } j} \pi_{ijt}^{(h)} - \underbrace{(a_{ijt} + h_{jit})}_{\# \text{ games } i \text{ away vs } j} \pi_{ijt}^{(a)} - \underbrace{(u_{ijt} + u_{jit})}_{\# \text{ games } i \text{ neut. vs } j} \pi_{ijt}^{(u)} \end{aligned}$$

Similarly, now consider deriving $\frac{\partial l}{\partial \alpha}$. As α is common across time blocks, we keep the sum over t . Instead, all the neutral venue terms $P_t^{(u)}$ go to zero; furthermore, the derivatives of $P_t^{(h)}$ and

³Here $\odot$ is the Hadamard product (element-wise product). The Frobenius inner product would also work.

$P_t^{(a)}$ will differ by a sign.

$$\frac{\partial}{\partial \alpha} P_t^{(h)} = \begin{bmatrix} 0 & 1 - \pi_{1,2,t}^{(h)} & \cdots & 1 - \pi_{1,i,t}^{(h)} & \cdots & 1 - \pi_{1,I,t}^{(h)} \\ 1 - \pi_{2,1,t}^{(h)} & 0 & \cdots & 1 - \pi_{2,i,t}^{(h)} & \cdots & 1 - \pi_{2,I,t}^{(h)} \\ \vdots & \vdots & \ddots & \vdots & & \vdots \\ 1 - \pi_{i,1,t}^{(h)} & 1 - \pi_{i,2,t}^{(h)} & \cdots & 0 & \cdots & 1 - \pi_{i,I,t}^{(h)} \\ \vdots & \vdots & & \vdots & \ddots & \vdots \\ 1 - \pi_{I,1,t}^{(h)} & 1 - \pi_{I,2,t}^{(h)} & \cdots & 1 - \pi_{I,i,t}^{(h)} & \cdots & 0 \end{bmatrix}$$

Hence,

$$\begin{aligned} \frac{\partial l}{\partial \alpha} &= \sum_t \mathbf{1}^T \left[H_t \odot \frac{\partial P_t^{(h)}}{\partial \alpha} + A_t \odot \frac{\partial P_t^{(a)}}{\partial \alpha} \right] \mathbf{1} \\ &= \sum_{ijt} h_{ijt}(1 - \pi_{ijt}^{(h)}) - a_{ijt}(1 - \pi_{ijt}^{(a)}) \end{aligned}$$

The Hessian elements can follow using the same method.

$$\begin{aligned} \frac{\partial^2}{\partial \theta_{it} \partial \theta_{jt}} P_{i \bullet t}^{(x)} &= \left(0, \dots, \underbrace{\pi_{ijt}^{(x)}(1 - \pi_{ijt}^{(x)})}_{j\text{-th element}}, \dots, 0 \right)^T \\ \frac{\partial^2}{\partial \theta_{it} \partial \theta_{jt}} P_{\bullet it}^{(x)} &= \left(0, \dots, \underbrace{\pi_{jit}^{(x)}(1 - \pi_{jit}^{(x)})}_{j\text{-th element}}, \dots, 0 \right)^T \end{aligned}$$

Hence,

$$\begin{aligned} \frac{\partial^2 l}{\partial \theta_{it} \partial \theta_{jt}} &= \sum_x x_{ijt} \pi_{ijt}^{(x)}(1 - \pi_{ijt}^{(x)}) + x_{jit} \pi_{jit}^{(x)}(1 - \pi_{jit}^{(x)}) \\ &= (h_{ijt} + a_{jit}) \pi_{ijt}^{(h)}(1 - \pi_{ijt}^{(h)}) + (a_{ijt} + h_{jit}) \pi_{jit}^{(a)}(1 - \pi_{jit}^{(a)}) + (u_{ijt} + u_{jit}) \pi_{ijt}^{(u)}(1 - \pi_{ijt}^{(u)}) \end{aligned}$$

and so-on as necessary.

A.2 More relationship matrix examples

The relationship matrices are best understood visually. In Sections 2.5 and 2.6, we demonstrated some example relationship structures based on the NBA using only a subset of the teams. For completeness, here we provide the matrices for all $I = 30$ teams in the NBA.

		Vanilla Bradley-Terry																																	
Home	Celtics	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Nets	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Knicks	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	76ers	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Raptors	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Bulls	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Cavaliers	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Pistons	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Pacers	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Bucks	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Hawks	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Hornets	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Heat	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Magic	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Wizards	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Nuggets	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Timberwolves	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Thunder	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Blazers	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Jazz	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Warriors	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Clippers	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Lakers	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Suns	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Kings	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Mavericks	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Rockets	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Grizzlies	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Pelicans	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Spurs	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
		Celtics	Nets	Knicks	76ers	Raptors	Bulls	Cavaliers	Pistons	Pacers	Bucks	Hawks	Hornets	Heat	Magic	Wizards	Nuggets	Timberwolves	Thunder	Blazers	Jazz	Warriors	Clippers	Lakers	Suns	Kings	Mavericks	Rockets	Grizzlies	Pelicans	Spurs				
		Away																																	

		Common HGA																															
Home	Celtics	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Nets	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Knicks	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	76ers	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Raptors	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Bulls	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Cavaliers	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Pistons	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Pacers	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Bucks	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Hawks	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Hornets	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Heat	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Magic	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Wizards	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Nuggets	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Timberwolves	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Thunder	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Blazers	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1
	Jazz	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1
	Warriors	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1
	Clippers	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1
	Lakers	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1
	Suns	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1
	Kings	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1
	Mavericks	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1
	Rockets	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
	Grizzlies	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1
	Pelicans	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1
	Spurs	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1
		Away																															
		Celtics	Nets	Knicks	76ers	Raptors	Bulls	Cavaliers	Pistons	Pacers	Bucks	Hawks	Hornets	Heat	Magic	Wizards	Nuggets	Timberwolves	Thunder	Blazers	Jazz	Warriors	Clippers	Lakers	Suns	Kings	Mavericks	Rockets	Grizzlies	Pelicans	Spurs		

		Division-specific																													
Home	Celtics	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Nets	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Knicks	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	76ers	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Raptors	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Bulls	2	2	2	2	2	0	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
	Cavaliers	2	2	2	2	2	2	0	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
	Pistons	2	2	2	2	2	2	2	0	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
	Pacers	2	2	2	2	2	2	2	2	0	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
	Bucks	2	2	2	2	2	2	2	2	2	0	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
	Hawks	5	5	5	5	5	5	5	5	5	0	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
	Hornets	5	5	5	5	5	5	5	5	5	5	0	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
	Heat	5	5	5	5	5	5	5	5	5	5	5	0	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
	Magic	5	5	5	5	5	5	5	5	5	5	5	5	0	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
	Wizards	5	5	5	5	5	5	5	5	5	5	5	5	5	0	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
	Nuggets	3	3	3	3	3	3	3	3	3	3	3	3	3	0	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
	Timberwolves	3	3	3	3	3	3	3	3	3	3	3	3	3	3	0	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
	Thunder	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	0	3	3	3	3	3	3	3	3	3	3	3	3	3	3
	Blazers	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	0	3	3	3	3	3	3	3	3	3	3	3	3	3
	Jazz	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	0	3	3	3	3	3	3	3	3	3	3	3	3
	Warriors	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	0	4	4	4	4	4	4	4	4	4	4	4
	Clippers	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	0	4	4	4	4	4	4	4	4	4	4
	Lakers	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	0	4	4	4	4	4	4	4	4	4
	Suns	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	0	4	4	4	4	4	4	4	4
	Kings	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	0	4	4	4	4	4	4	4
	Mavericks	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	0	6	6	6	6	6	6	6
	Rockets	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	0	6	6	6	6	6	6
	Grizzlies	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	0	6	6	6	6	6
	Pelicans	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	0	6	6	6	6
	Spurs	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	0
		Away																													

		Division-specific hierarchical																														
Home	Celtics	0	1	1	1	1	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
	Nets	1	0	1	1	1	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
	Knicks	1	1	0	1	1	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
	76ers	1	1	1	0	1	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
	Raptors	1	1	1	1	0	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
	Bulls	4	4	4	4	4	0	5	5	5	5	4	4	4	4	4	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
	Cavaliers	4	4	4	4	4	5	0	5	5	5	4	4	4	4	4	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
	Pistons	4	4	4	4	4	5	5	0	5	5	4	4	4	4	4	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
	Pacers	4	4	4	4	4	5	5	5	0	5	4	4	4	4	4	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
	Bucks	4	4	4	4	4	5	5	5	5	0	4	4	4	4	4	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
	Hawks	7	7	7	7	7	7	7	7	7	7	0	8	8	8	8	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9
	Hornets	7	7	7	7	7	7	7	7	7	7	8	0	8	8	8	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9
	Heat	7	7	7	7	7	7	7	7	7	7	8	8	0	8	8	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9
	Magic	7	7	7	7	7	7	7	7	7	7	8	8	8	0	8	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9
	Wizards	7	7	7	7	7	7	7	7	7	7	8	8	8	8	0	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9
	Nuggets	10	10	10	10	10	10	10	10	10	10	10	10	10	10	0	11	11	11	11	11	12	12	12	12	12	12	12	12	12	12	12
	Timberwolves	10	10	10	10	10	10	10	10	10	10	10	10	10	10	11	0	11	11	11	11	12	12	12	12	12	12	12	12	12	12	12
	Thunder	10	10	10	10	10	10	10	10	10	10	10	10	10	10	11	11	0	11	11	12	12	12	12	12	12	12	12	12	12	12	12
	Blazers	10	10	10	10	10	10	10	10	10	10	10	10	10	10	11	11	11	0	11	12	12	12	12	12	12	12	12	12	12	12	12
	Jazz	10	10	10	10	10	10	10	10	10	10	10	10	10	10	11	11	11	11	0	12	12	12	12	12	12	12	12	12	12	12	12
	Warriors	13	13	13	13	13	13	13	13	13	13	13	13	13	13	14	14	14	14	14	0	15	15	15	15	14	14	14	14	14	14	14
	Clippers	13	13	13	13	13	13	13	13	13	13	13	13	13	13	14	14	14	14	14	15	0	15	15	15	14	14	14	14	14	14	14
	Lakers	13	13	13	13	13	13	13	13	13	13	13	13	13	13	14	14	14	14	14	15	15	0	15	15	14	14	14	14	14	14	14
	Suns	13	13	13	13	13	13	13	13	13	13	13	13	13	13	14	14	14	14	14	15	15	15	0	15	14	14	14	14	14	14	14
	Kings	13	13	13	13	13	13	13	13	13	13	13	13	13	13	14	14	14	14	14	15	15	15	15	0	14	14	14	14	14	14	14
	Mavericks	16	16	16	16	16	16	16	16	16	16	16	16	16	16	17	17	17	17	17	17	17	17	17	0	18	18	18	18	18	18	18
	Rockets	16	16	16	16	16	16	16	16	16	16	16	16	16	16	17	17	17	17	17	17	17	17	17	18	0	18	18	18	18	18	18
	Grizzlies	16	16	16	16	16	16	16	16	16	16	16	16	16	16	17	17	17	17	17	17	17	17	17	18	18	0	18	18	18	18	18
	Pelicans	16	16	16	16	16	16	16	16	16	16	16	16	16	16	17	17	17	17	17	17	17	17	17	18	18	18	0	18	18	18	18
	Spurs	16	16	16	16	16	16	16	16	16	16	16	16	16	16	17	17	17	17	17	17	17	17	17	18	18	18	18	0	18	18	18
		Away																														

		Team-specific HGA																													
Home	Celtics	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Nets	2	0	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
	Knicks	3	3	0	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
	76ers	4	4	4	0	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
	Raptors	5	5	5	5	0	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
	Bulls	6	6	6	6	6	0	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
	Cavaliers	7	7	7	7	7	7	0	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7
	Pistons	8	8	8	8	8	8	8	0	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8
	Pacers	9	9	9	9	9	9	9	9	0	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9
	Bucks	10	10	10	10	10	10	10	10	10	0	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10
	Hawks	11	11	11	11	11	11	11	11	11	11	0	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11
	Hornets	12	12	12	12	12	12	12	12	12	12	12	0	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12
	Heat	13	13	13	13	13	13	13	13	13	13	13	13	0	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13
	Magic	14	14	14	14	14	14	14	14	14	14	14	14	14	0	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14
	Wizards	15	15	15	15	15	15	15	15	15	15	15	15	15	15	0	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15
	Nuggets	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	0	16	16	16	16	16	16	16	16	16	16	16	16	16	16
	Timberwolves	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	0	17	17	17	17	17	17	17	17	17	17	17	17	17
	Thunder	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	0	18	18	18	18	18	18	18	18	18	18	18	18
	Blazers	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	0	19	19	19	19	19	19	19	19	19	19	19
	Jazz	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	0	20	20	20	20	20	20	20	20	20	20
	Warriors	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	0	21	21	21	21	21	21	21	21	21
	Clippers	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	0	22	22	22	22	22	22	22	22
	Lakers	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	0	23	23	23	23	23	23	23
	Suns	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	0	24	24	24	24	24	24
	Kings	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	0	25	25	25	25	25
	Mavericks	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	0	26	26	26	26
	Rockets	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	0	27	27	27
	Grizzlies	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	0	28	28
	Pelicans	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	0	29
	Spurs	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	0
		Away																													

		Team-specific hierarchical																																											
Home	Celtics	0	1	1	1	1	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3								
	Nets	4	0	4	4	4	5	5	5	5	5	5	5	5	5	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6								
	Knicks	7	7	0	7	7	8	8	8	8	8	8	8	8	8	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9							
	76ers	10	10	10	0	10	11	11	11	11	11	11	11	11	11	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12							
	Raptors	13	13	13	13	0	14	14	14	14	14	14	14	14	14	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15							
	Bulls	16	16	16	16	16	0	17	17	17	17	16	16	16	16	16	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18						
	Cavaliers	19	19	19	19	19	20	0	20	20	20	19	19	19	19	19	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21						
	Pistons	22	22	22	22	22	23	23	0	23	23	22	22	22	22	22	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24						
	Pacers	25	25	25	25	25	26	26	26	0	26	25	25	25	25	25	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27						
	Bucks	28	28	28	28	28	29	29	29	29	0	28	28	28	28	28	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30						
	Hawks	31	31	31	31	31	31	31	31	31	31	0	32	32	32	32	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33						
	Hornets	34	34	34	34	34	34	34	34	34	34	35	0	35	35	35	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36						
	Heat	37	37	37	37	37	37	37	37	37	37	38	38	0	38	38	39	39	39	39	39	39	39	39	39	39	39	39	39	39	39	39	39	39	39	39	39	39	39						
	Magic	40	40	40	40	40	40	40	40	40	40	41	41	41	0	41	42	42	42	42	42	42	42	42	42	42	42	42	42	42	42	42	42	42	42	42	42	42	42						
	Wizards	43	43	43	43	43	43	43	43	43	43	44	44	44	44	0	45	45	45	45	45	45	45	45	45	45	45	45	45	45	45	45	45	45	45	45	45	45	45						
	Nuggets	46	46	46	46	46	46	46	46	46	46	46	46	46	46	46	0	47	47	47	47	47	48	48	48	48	48	48	48	48	48	48	48	48	48	48	48	48	48						
	Timberwolves	49	49	49	49	49	49	49	49	49	49	49	49	49	49	49	50	0	50	50	50	50	51	51	51	51	51	51	51	51	51	51	51	51	51	51	51	51	51						
	Thunder	52	52	52	52	52	52	52	52	52	52	52	52	52	52	52	53	53	0	53	53	54	54	54	54	54	54	54	54	54	54	54	54	54	54	54	54	54	54						
	Blazers	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	56	56	56	0	56	57	57	57	57	57	57	57	57	57	57	57	57	57	57	57	57	57	57						
	Jazz	58	58	58	58	58	58	58	58	58	58	58	58	58	58	58	59	59	59	59	0	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60						
	Warriors	61	61	61	61	61	61	61	61	61	61	61	61	61	61	61	62	62	62	62	62	0	63	63	63	63	63	62	62	62	62	62	62	62	62	62	62	62	62						
	Clippers	64	64	64	64	64	64	64	64	64	64	64	64	64	64	64	65	65	65	65	65	66	0	66	66	66	66	65	65	65	65	65	65	65	65	65	65	65	65						
	Lakers	67	67	67	67	67	67	67	67	67	67	67	67	67	67	67	68	68	68	68	68	69	69	0	69	69	68	68	68	68	68	68	68	68	68	68	68	68	68						
	Suns	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	71	71	71	71	71	72	72	72	0	72	71	71	71	71	71	71	71	71	71	71	71	71	71						
	Kings	73	73	73	73	73	73	73	73	73	73	73	73	73	73	73	74	74	74	74	74	75	75	75	75	0	74	74	74	74	74	74	74	74	74	74	74	74	74						
	Mavericks	76	76	76	76	76	76	76	76	76	76	76	76	76	76	76	77	77	77	77	77	77	77	77	77	77	0	78	78	78	78	78	78	78	78	78	78	78	78						
	Rockets	79	79	79	79	79	79	79	79	79	79	79	79	79	79	79	80	80	80	80	80	80	80	80	80	80	81	0	81	81	81	81	81	81	81	81	81	81	81						
	Grizzlies	82	82	82	82	82	82	82	82	82	82	82	82	82	82	82	83	83	83	83	83	83	83	83	83	83	84	84	0	84	84	84	84	84	84	84	84	84	84						
	Pelicans	85	85	85	85	85	85	85	85	85	85	85	85	85	85	85	86	86	86	86	86	86	86	86	86	86	87	87	87	87	87	87	87	87	87	87	87	87	87						
	Spurs	88	88	88	88	88	88	88	88	88	88	88	88	88	88	88	89	89	89	89	89	89	89	89	89	89	90	90	90	90	90	90	90	90	90	90	90	90	0						
		Celtics	Nets	Knicks	76ers	Raptors	Bulls	Cavaliers	Pistons	Pacers	Bucks	Hawks	Hornets	Heat	Magic	Wizards	Nuggets	Timberwolves	Thunder	Blazers	Jazz	Warriors	Clippers	Lakers	Suns	Kings	Mavericks	Rockets	Grizzlies	Pelicans	Spurs														
		Away																																											

		Pairwise unique HGA																													
Home	Celtics	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29
	Nets	30	0	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58
	Knicks	59	60	0	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87
	76ers	88	89	90	0	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116
	Raptors	117	118	119	120	0	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145
	Bulls	146	147	148	149	150	0	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174
	Cavaliers	175	176	177	178	179	180	0	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203
	Pistons	204	205	206	207	208	209	210	0	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232
	Pacers	233	234	235	236	237	238	239	240	0	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	258	259	260	261
	Bucks	262	263	264	265	266	267	268	269	270	0	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285	286	287	288	289	290
	Hawks	291	292	293	294	295	296	297	298	299	300	0	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319
	Hornets	320	321	322	323	324	325	326	327	328	329	330	0	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345	346	347	348
	Heat	349	350	351	352	353	354	355	356	357	358	359	360	0	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375	376	377
	Magic	378	379	380	381	382	383	384	385	386	387	388	389	390	0	391	392	393	394	395	396	397	398	399	400	401	402	403	404	405	406
	Wizards	407	408	409	410	411	412	413	414	415	416	417	418	419	420	0	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435
	Nuggets	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450	0	451	452	453	454	455	456	457	458	459	460	461	462	463	464
	Timberwolves	465	466	467	468	469	470	471	472	473	474	475	476	477	478	479	480	0	481	482	483	484	485	486	487	488	489	490	491	492	493
	Thunder	494	495	496	497	498	499	500	501	502	503	504	505	506	507	508	509	510	0	511	512	513	514	515	516	517	518	519	520	521	522
	Blazers	523	524	525	526	527	528	529	530	531	532	533	534	535	536	537	538	539	540	0	541	542	543	544	545	546	547	548	549	550	551
	Jazz	552	553	554	555	556	557	558	559	560	561	562	563	564	565	566	567	568	569	570	0	571	572	573	574	575	576	577	578	579	580
	Warriors	581	582	583	584	585	586	587	588	589	590	591	592	593	594	595	596	597	598	599	600	0	601	602	603	604	605	606	607	608	609
	Clippers	610	611	612	613	614	615	616	617	618	619	620	621	622	623	624	625	626	627	628	629	630	0	631	632	633	634	635	636	637	638
	Lakers	639	640	641	642	643	644	645	646	647	648	649	650	651	652	653	654	655	656	657	658	659	660	0	661	662	663	664	665	666	667
	Suns	668	669	670	671	672	673	674	675	676	677	678	679	680	681	682	683	684	685	686	687	688	689	690	0	691	692	693	694	695	696
	Kings	697	698	699	700	701	702	703	704	705	706	707	708	709	710	711	712	713	714	715	716	717	718	719	720	0	721	722	723	724	725
	Mavericks	726	727	728	729	730	731	732	733	734	735	736	737	738	739	740	741	742	743	744	745	746	747	748	749	750	0	751	752	753	754
	Rockets	755	756	757	758	759	760	761	762	763	764	765	766	767	768	769	770	771	772	773	774	775	776	777	778	779	780	0	781	782	783
	Grizzlies	784	785	786	787	788	789	790	791	792	793	794	795	796	797	798	799	800	801	802	803	804	805	806	807	808	809	810	0	811	812
	Pelicans	813	814	815	816	817	818	819	820	821	822	823	824	825	826	827	828	829	830	831	832	833	834	835	836	837	838	839	840	0	841
	Spurs	842	843	844	845	846	847	848	849	850	851	852	853	854	855	856	857	858	859	860	861	862	863	864	865	866	867	868	869	870	0
		Away																													
		Celtics	Nets	Knicks	76ers	Raptors	Bulls	Cavaliers	Pistons	Pacers	Bucks	Hawks	Hornets	Heat	Magic	Wizards	Nuggets	Timberwolves	Thunder	Blazers	Jazz	Warriors	Clippers	Lakers	Suns	Kings	Mavericks	Rockets	Grizzlies	Pelicans	Spurs

Chapter B: Implementation Details

B.1 Attributes of Base Model Class

The BaseBradleyTerry abstract base class has the following variables.

Variable Name	Type	Description
<code>data</code>	Dictionary of DataFrames	A data structure containing the time-indexed win matrices, as described in 3.2.
<code>n_obs</code>	Integer	Total number of observations in the data.
<code>teams</code>	Array	List of team names.
<code>n_teams</code>	Integer	Number of teams.
<code>times</code>	Array	List of time block names.
<code>n_times</code>	Integer	Number of time blocks.
<code>_constraint_team_idx</code>	Integer	The index of the constraint team, which must be a global reference team.
<code>inactive_teams</code>	Dictionary of arrays	Dictionary mapping team names to a list of time blocks they have no data in.
<code>global_teams</code>	Array	List of global reference teams.
<code>params</code>	Array	List containing the vectorised parameters of the model.
<code>n_params</code>	Integer	Number of parameters in the model, including inactive ones.
<code>n_params_active</code>	Integer	Number of parameters in the model, excluding inactive ones.
<code>_fit_summary</code>	OptimizeResult object	scipy optimisation output containing attributes like the minimised parameter vector, success Boolean, number of iterations, estimated Hessian, and more.
<code>_hess_inv</code>	Array (2D)	Numerical Hessian estimated during optimisation.
<code>errors</code>	Array (2D)	List of calculated standard errors corresponding to each parameter in <code>params</code> .

The BaseHierarchicalBT base class has all the previous variables as well as the following.

Variable Name	Type	Description
<code>rel_mat</code>	<code>DataFrame</code>	The relationship matrix.
<code>_mask</code>	Array (2D)	A Boolean mask of the relationship matrix used to remove diagonal elements.
<code>levels</code>	Array	List of names of the levels in the <code>rel_mat</code> .
<code>n_levels</code>	Integer	Number of distinct levels in the <code>rel_mat</code> .

Chapter C: Additional Results

In Chapter 4, we fit several models to two major Australian league sports: the AFL and NRL. Of course, the number of model parameters can grow quite large as we take larger time periods, so we only discussed the key results for the home-ground advantage parameters in the main text. Here we provide some of the additional modelling results and plots from our analyses.

For more outputs and plots from the models fit in this paper, please refer to the Jupyter notebook files in the GitHub repository.¹

C.1 Team-specific HGA Plots

In the general AFL analysis (4.1) and NRL analysis (4.2), we first applied the VAN, CHA, and TSA models. For the former two models, we provided plots (e.g., Figures 8–11) of the (many) team strengths over the 2005–2024 period. The dynamic plots attempt to show the full set of parameters and the static plots simply average strengths for each team, sacrificing the time axis for some visual clarity. The results for the team-specific model were not included for brevity sake — only the HGA estimates — so we provide some extra plots here instead (under the nominal HGA assumption). Again, one can see (at least qualitatively) how similar the strength estimates are compared to the VAN and CHA models.

C.2 Team-specific HGA Results for COVID-19 Case Study

In Section 4.3, we looked for changes in the home advantage in the AFL and NRL by fitting our models to windows before, during, and after the COVID-19 pandemic. The following plots show the results of the team-specific HGA models, which are again quite cluttered due to the number of teams.

¹<https://github.com/booms14nger/uqhons-bt>

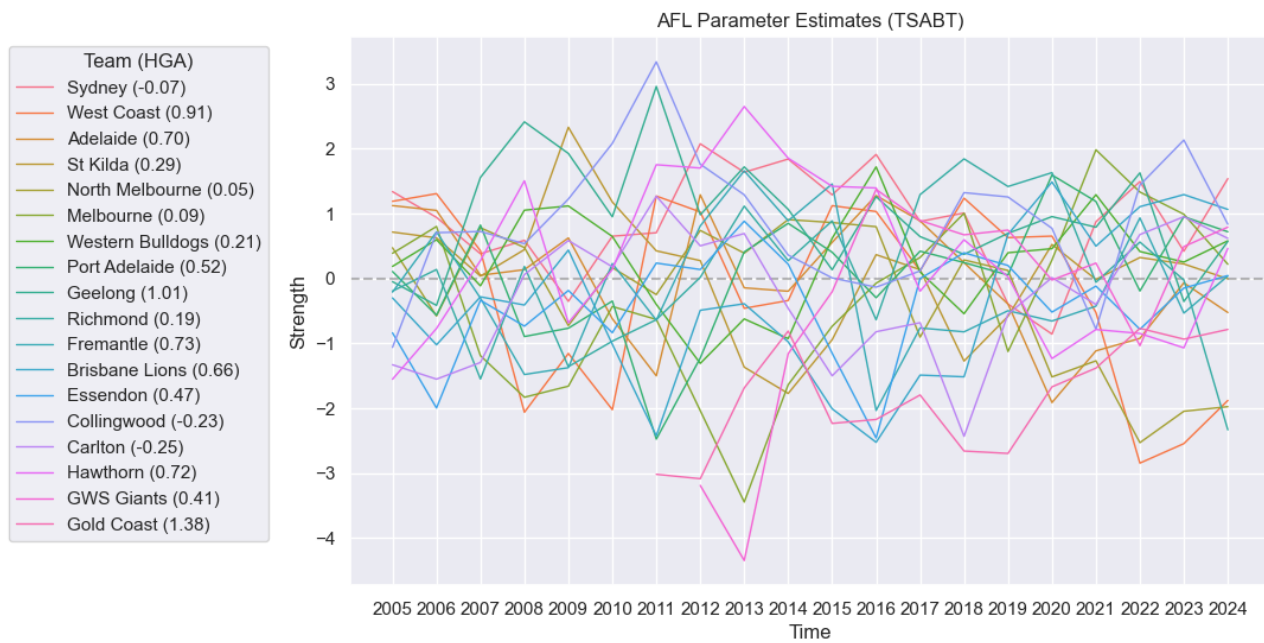


Figure 25: Parameter estimates for the AFL (2005–2024), team-specific HGA.

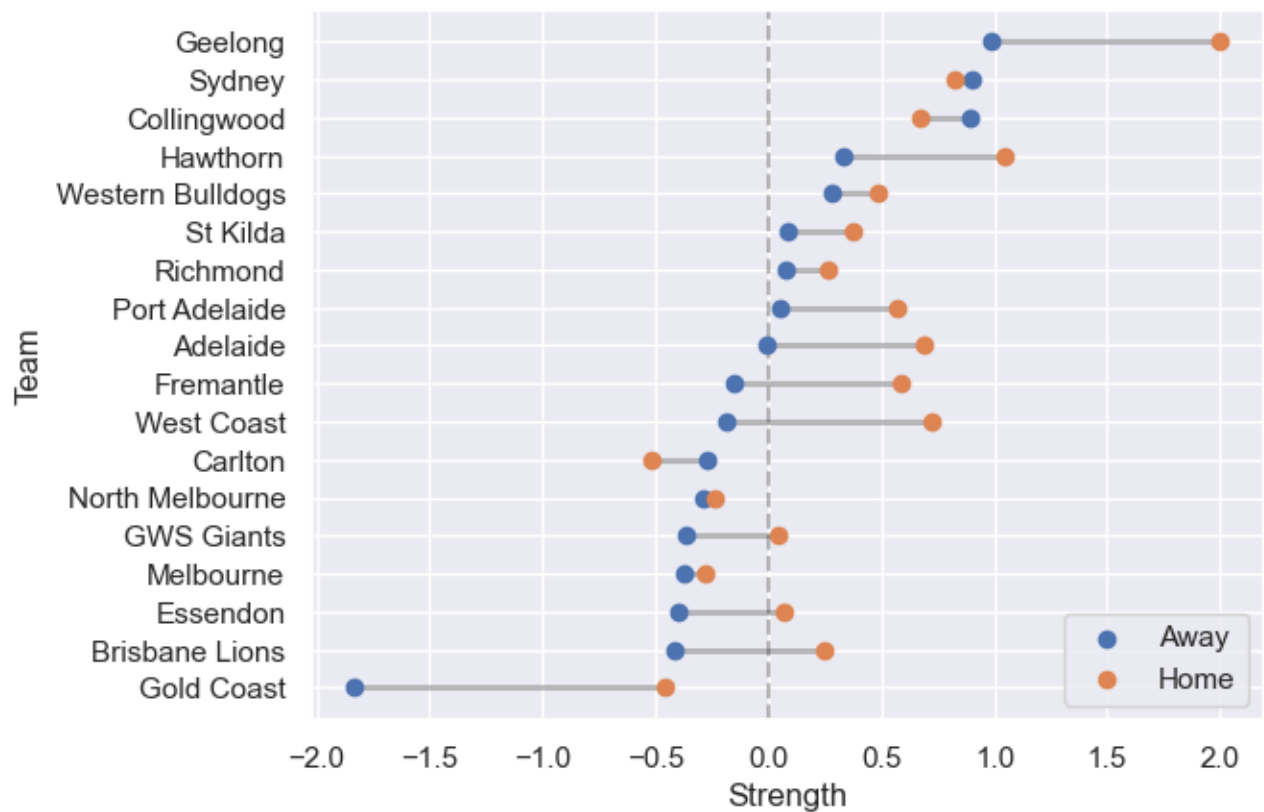


Figure 26: Average estimated team strengths for the AFL (2005–2024), team-specific HGA.

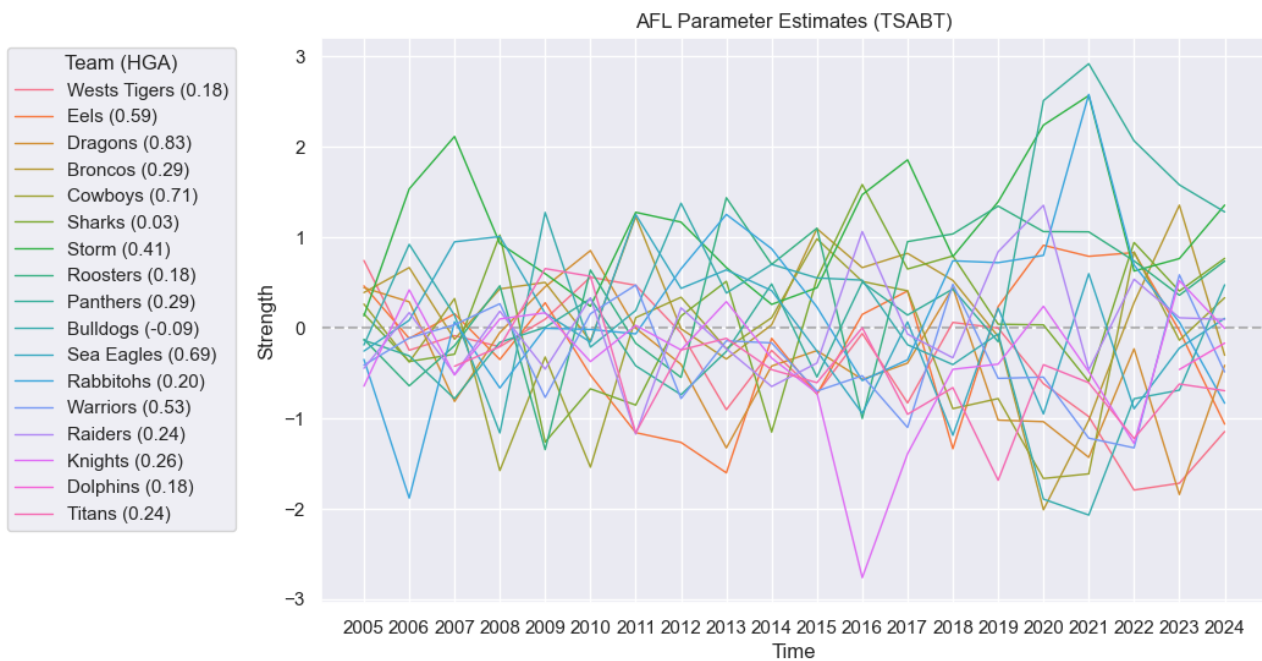


Figure 27: Parameter estimates for the NRL (2005–2024), team-specific HGA.

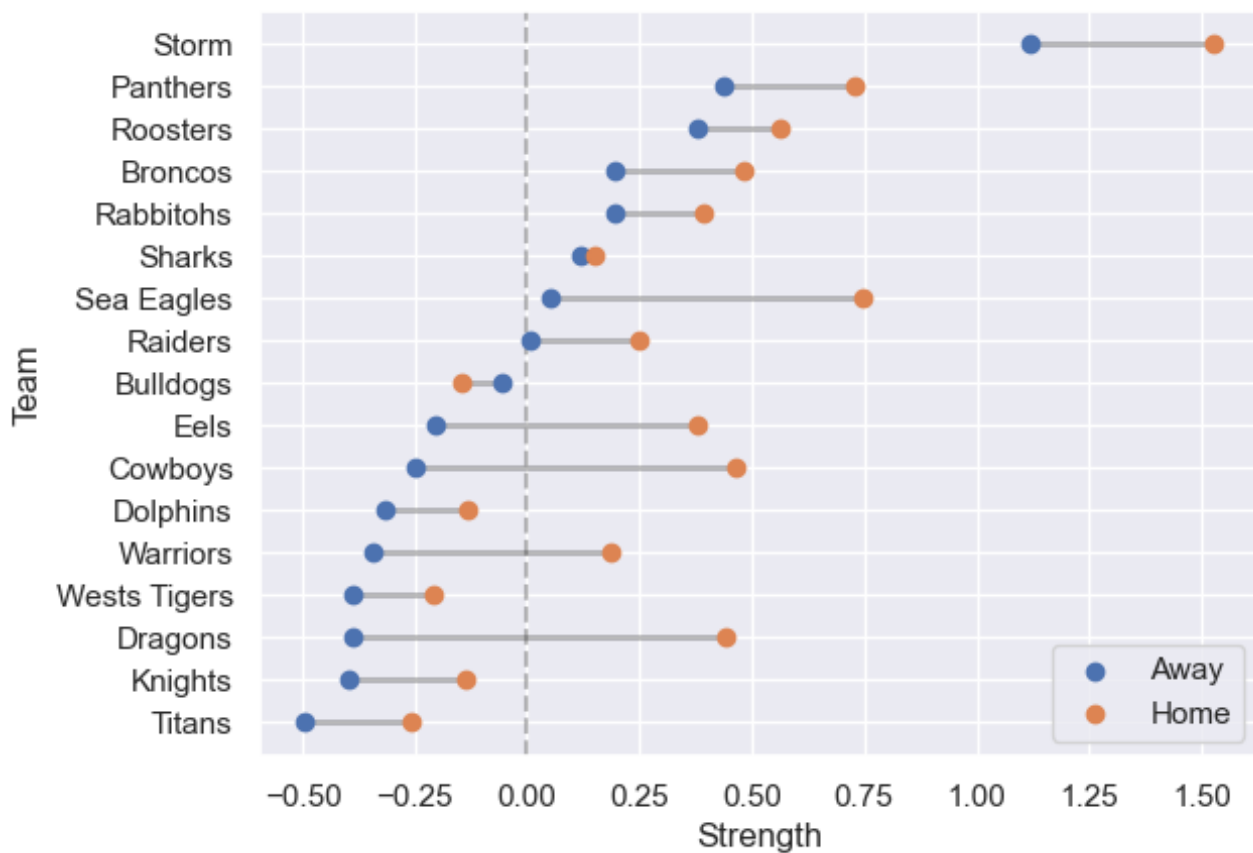


Figure 28: Average estimated team strengths for the NRL (2005–2024), team-specific HGA.

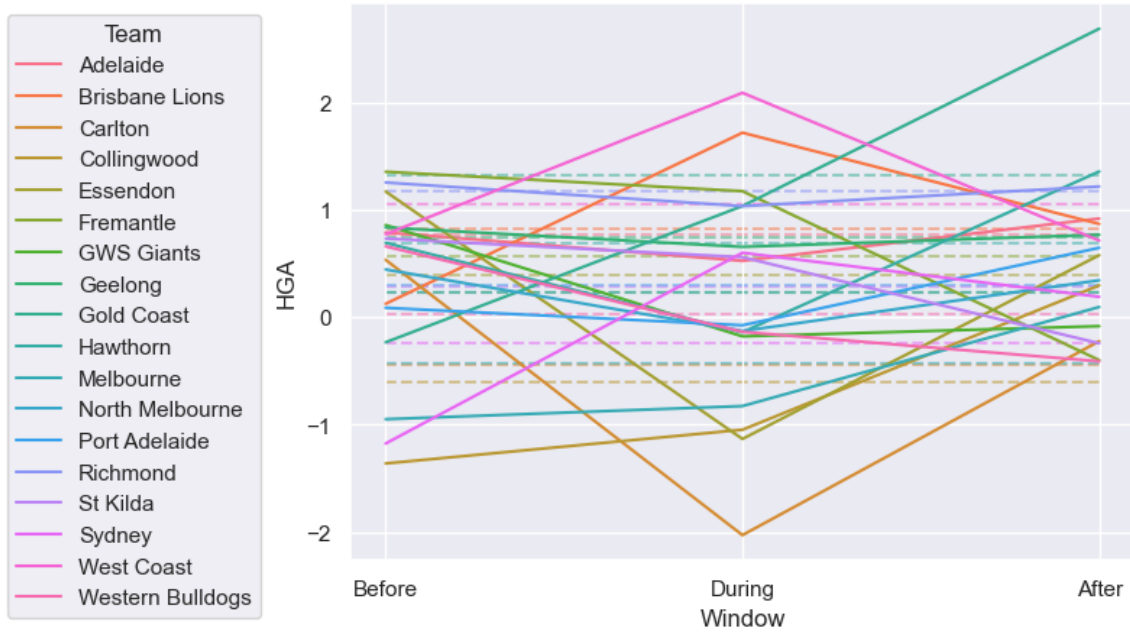


Figure 29: Estimated HGAs for the fully-dynamic (solid) and constant HGA (dashed) TSA models, fit to the AFL (2010–2019).

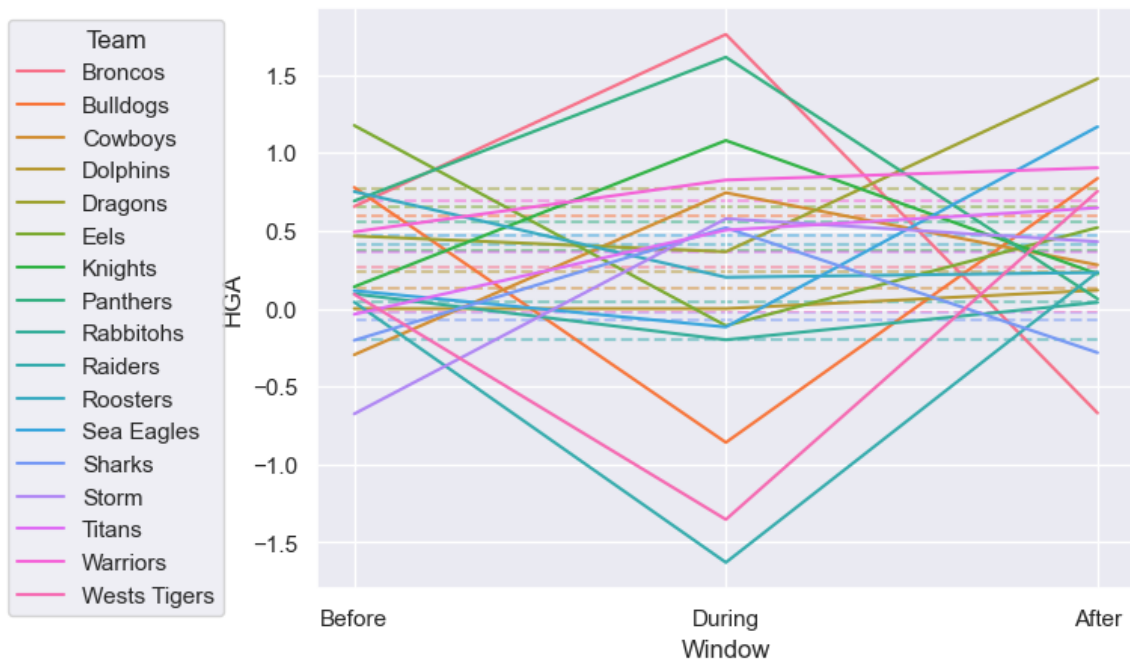


Figure 30: Estimated HGAs for the fully-dynamic (solid) and constant HGA (dashed) TSA models, fit to the NRL (2010–2019).